

The Compass



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Community Mathematics Centre

Schools of the Krishnamurti Foundation India
(KFI)

Statement of intent

The magazine is published by the Community Mathematics Centre. It is a collaborative initiative of the schools of the Krishnamurti Foundation India (KFI).

Its intention is:

- To provide material for mathematics and science teachers who teach at the high school or middle school levels, and for students who study at these levels.
- To serve as a platform where teachers and students can share their findings and discoveries with others who share their love for the subject.

In addition to mathematical content, in each issue we shall share a few ‘snippets’ from the world of science — short extracts abstracted from articles in online magazines, selected on the basis of intrinsic interest.

The Compass is published as an online magazine free of cost. Articles may be downloaded and used freely by teachers, but with proper attribution.

Edited by Shailesh Shirali & Padmapriya Shirali & Shashidhar Jagadeeshan on behalf of schools of the KFI

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Colour coding

The magazine has articles addressed to teachers who teach at different levels: Lower Primary level; Upper Primary and High School levels; and Senior Secondary and Junior College levels. There is also a Problem Corner. In addition we have the Snippets.

For the convenience of the reader, we use the following colour coding to distinguish between the various levels. Of course, such a division can never be exact. So, our apologies in advance for errors of judgement!

- Primary & Upper Primary (Classes 4, 5, 6, 7)
- Upper Primary & High School (Classes 6, 7, 8, 9, 10)
- Senior Secondary & Junior College (Classes 10, 11, 12)
- Problem Corner
- Snippets

Fast Calculations – 3

We continue our study of computational shortcuts (or, “the art of doing fast calculations and impressing an audience”). In this issue we shall look closely at some popular computing shortcuts. We shall also explain the logic behind the shortcuts given as problems in the October 2024 issue.

Multiplying a number by 5

The task is to compute the product $n \times 5$. Here are the steps.

1. If n is even, write n as $2m$, so $m = n/2$. The required answer is then $m0$ (that is, we append a 0 at the units end of m).

Example. To compute 142×5 . Since $142/2 = 71$, the required answer is 710.

2. If n is odd, write n as $2m + 1$, so $m = (n - 1)/2$. The required answer is then $m5$ (that is, we append a 5 at the units end of m).

Example. To compute 141×5 . Since $(141 - 1)/2 = 70$, the required answer is 705.

Questions for you to ponder

1. Why does this shortcut work?
2. Does this procedure work if the number of digits is much larger?

Squaring a number whose units digit is 5

This is a well-known shortcut that works for finding the square of any number whose units digit is 5, no matter how large the number is. It is particularly easy to implement for two-digit numbers. Here are the steps.

1. Write the number as $A5$. (Here, 5 is the units digit of the number and A is the 'rest' of the number; we get A by deleting the units digit. For example, if the number to be squared is 75, then $A = 7$.)
2. Compute the product $A(A + 1)$.
3. Append the digits '25' to the answer obtained in Step 2. This is the required answer.

We illustrate this by computing 75^2 , 85^2 and 135^2 .

- For 75^2 we have $A = 7$ and $A(A + 1) = 7 \times 8 = 56$.
So the required answer is 5625.
- For 85^2 we have $A = 8$ and $A(A + 1) = 8 \times 9 = 72$.
So the required answer is 7225.
- For 135^2 we have $A = 13$ and $A(A + 1) = 13 \times 14 = 182$.
So the required answer is 18225.

Please check that all three answers are correct.

Questions for you to ponder

1. Why does this shortcut work?
2. A challenge for you! How would you modify this shortcut to find the product of two numbers whose units digits are 5, i.e., to find the product $A5 \times B5$? For example, to compute 35×45 or 55×95 . You could study the following data and see whether they suggest any shortcut:

$$35 \times 45 = 1575, \quad 35 \times 55 = 1925, \quad 35 \times 65 = 2275,$$

$$45 \times 55 = 2475, \quad 45 \times 65 = 2925, \quad 45 \times 75 = 3375.$$

Analysis of two problems from the October 2024 issue

In the October 2024 issue we had asked you to find a shortcut for working out the squares of two-digit numbers whose tens digit is 3. That is, the squares of numbers of the form $n = 30 + a$ where a is a digit.

Here is a possible shortcut. We have:

$$\begin{aligned} n^2 &= (30 + a)^2 \\ &= 900 + 60a + a^2 \\ &= 30(30 + a + a) + a^2. \end{aligned}$$

This leads to the following procedure.

- $30 + a$ is the given number, and $30 + a + a$ means that we **add** the units digit a to the given number.

- $30(30 + a + a)$ means that we **multiply** the result from (A) by 30. (This is the same as multiplying by 3 and inserting a 0 at the units end.)
- To this we add a^2 , taking care of the carry-over as needed.

Let's try this out!

To find the square of 34

- $34 + 4 = 38$
- $30 \times 38 = 1140$ (we can also write this as 114_)
- $4^2 = 16$
- Hence $34^2 = 1140 + 16 = 1156$

To find the square of 37

- $37 + 7 = 44$
- $30 \times 44 = 1320$ (we can also write this as 132_)
- $7^2 = 49$
- Hence $37^2 = 1320 + 49 = 1369$

Try out the shortcut with a few other two-digit numbers starting with 3.

We shall leave you with the task of devising a similar shortcut for working out the squares of two-digit numbers whose tens digit is 4.

Pretty Number Patterns – 3

We continue our exploration of pretty patterns. We had started on the study of arithmetic sequences and we listed a few multiplicative properties of such sequences. We will continue that study now. As earlier, we will focus on the sequence of natural numbers $1, 2, 3, 4, 5, 6, 7, 8, \dots$

Last time we asked you to carry out two explorations:

- Look for some more multiplicative properties that quadruples of consecutive integers have.
- Take any five consecutive numbers a, b, c, d, e in the sequence. What interesting multiplicative properties can you spot that are *always* true?
- Take any six consecutive numbers a, b, c, d, e, f in the sequence. What interesting multiplicative properties can you spot that are *always* true?

Where did your exploration lead you? Here are some properties that you *may* have found.

Quadruples of consecutive numbers

Let (a, b, c, d) be any quadruple of consecutive numbers. Compute the quantity

$$(a^2 + d^2) - (b^2 + c^2).$$

You will find that you get the same answer each time.

For example, take the quadruple $(10, 11, 12, 13)$. We get:

$$10^2 + 13^2 = 100 + 169 = 269,$$

$$11^2 + 12^2 = 121 + 144 = 265.$$

The difference is 4.

Let's try now with the quadruple $(14, 15, 16, 17)$. We have:

$$14^2 + 17^2 = 196 + 289 = 485,$$

$$15^2 + 16^2 = 225 + 256 = 481.$$

The difference is 4, again.

You will always get 4 as the answer.

Quintuples or 5-tuples of consecutive numbers

We obtain a similar result when we consider quintuples or 5-tuples of consecutive numbers. Let (a, b, c, d, e) be a 5-tuple of consecutive numbers. Leave out the middle number and compute the quantity

$$(a^2 + e^2) - (b^2 + d^2).$$

You will find that you get the same answer each time.

For example, take the quadruple (10, 11, 12, 13, 14). We get:

$$10^2 + 14^2 = 100 + 196 = 296,$$

$$11^2 + 13^2 = 121 + 169 = 290.$$

The difference is 6.

Let's try now with the quadruple (14, 15, 16, 17, 18). We have:

$$14^2 + 18^2 = 196 + 324 = 520,$$

$$15^2 + 17^2 = 225 + 289 = 514.$$

The difference is 6, again.

You will always get 6 as the answer.

.....

Try to find the explanation for this phenomenon! Then look for similar properties of sextuples or 6-tuples of consecutive numbers.

Words of Wisdom from Prof. X

Article contributed by A. Ramachandran

Most unfortunately, the professor's "words of wisdom" crumble when examined closely! Here's why.

1. Above average

Prof. X. claims that all his students scored above-average marks in a test.

Comment ... For any student who scored above-average marks, there must be at least one student with below-average marks, and vice versa. The professor's statement is a fallacy as would be the statement that "all students scored below-average marks."

2. World's population in the distant past

Prof. X. muses, "I have two parents, four grandparents, eight great-grandparents, sixteen great-great-grandparents, and so on. So, the world's population must have been higher in the olden days."

Comment ... The professor's ancestors would also be the ancestors of many others — siblings, first cousins, second cousins, etc. No one has an exclusive set of ancestors. We all share our ancestors with a whole lot of others.

3. Proof of the Pythagorean theorem

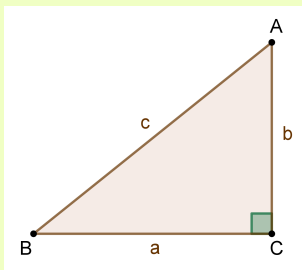


Figure 1

Prof. X. offers the following ‘proof’ of the Pythagorean theorem. In $\triangle ABC$ (Figure 1), angle C is a right angle. We know that $\sin^2 A + \cos^2 A = 1$. Now,

$$\sin A = \frac{BC}{AB} \quad \text{and} \quad \cos A = \frac{AC}{AB},$$

$$\therefore \frac{BC^2}{AB^2} + \frac{AC^2}{AB^2} = 1,$$

therefore $AB^2 = BC^2 + AC^2$. Hence proved!

Comment ... The trigonometric relation $\sin^2 A + \cos^2 A = 1$ is itself a consequence of the Pythagorean theorem. It cannot be invoked to prove

the latter. The professor's statement is an example of circular reasoning which is a fallacy.

About the author ...

A Ramachandran has had a long-standing interest in the teaching of mathematics and science. He studied physical science and mathematics at the undergraduate level and shifted to life science at the postgraduate level. He taught science, mathematics and geography to middle school students at Rishi Valley School for two decades. His other interests include the English language and Indian music. He may be contacted at archandran.53@gmail.com

Playing with Primitive Pythagorean Triples – 3

We continue our exploration of Primitive Pythagorean Triples (PPTs). We had discovered in the previous article that there were certain values of k for which PPTs of the form $(a, b, b + k)$ are just not possible; e.g., $k = 3, 4, 5, 6$. On the other hand, we found ways of listing infinitely many PPTs having $k = 1$ or 2 . In this article, we shall find a way of listing all values of k which are possible, and all values which are not possible.

But we shall approach the problem in a very different way from the route taken earlier. Our approach now will be to find a single formula that can generate all possible PPTs. The answer to the above question will then follow from that formula as a simple corollary.

There are many ways of arriving at this formula. We follow an approach that uses trigonometry. The half-angle identities for the trigonometric functions are going to play a key role in this approach!

Arriving at a formula to generate PPTs

Let (a, b, c) be a PPT; then a, b, c are coprime positive integers obeying the Pythagorean relation, $a^2 + b^2 = c^2$. Let ABC be a triangle with sides a, b, c and $\angle C = 90^\circ$; let $\theta = \angle ABC$. Let AD be the bisector of $\angle ABC$, with D on AC (see Figure 2).

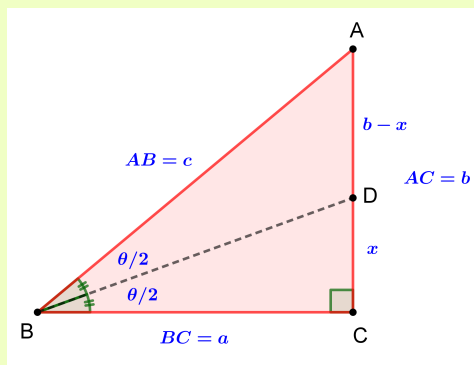


Figure 2

Obviously, $\tan \theta = b/a$ is a rational number. We shall show that $\tan \frac{1}{2}\theta$ too is a rational number.

It is known that *the internal bisector of an angle of a triangle divides the opposite side in the ratio of the adjacent sides*. With reference to Figure 2, this means that

$$\frac{AD}{DC} = \frac{BA}{BC},$$

or, writing $x = DC$,

$$\frac{b-x}{x} = \frac{c}{a}.$$

This yields

$$x = \frac{ab}{a+c}, \quad (1)$$

and so,

$$\tan \frac{\theta}{2} = \frac{b}{a+c}. \quad (2)$$

Since a, b, c are positive integers, it follows that $\tan \frac{1}{2}\theta$ is a rational number.

Let $t = \tan \frac{1}{2}\theta$. Note that $0 < t < 1$. We have,

$$t(a+c) = b. \quad (3)$$

We also have,

$$c^2 - a^2 = b^2. \quad (4)$$

Division yields,

$$\frac{c-a}{t} = b, \quad \therefore c-a = tb.$$

So we have,

$$c+a = \frac{b}{t}, \quad c-a = tb.$$

We treat these as a pair of simultaneous equations in the unknowns c and a , and we solve for the unknowns in terms of t and b . By addition we get

$$2c = b \left(t + \frac{1}{t} \right),$$

and by subtraction we get,

$$2a = b \left(\frac{1}{t} - t \right).$$

Hence we have:

$$a : b : c = \frac{1}{2} \left(\frac{1}{t} - t \right) : 1 : \frac{1}{2} \left(t + \frac{1}{t} \right), \quad (5)$$

which may be written more conveniently as

$$a : b : c = 1 - t^2 : 2t : 1 + t^2. \quad (6)$$

Consider for a moment where we have reached. Corresponding to any PPT (a, b, c) , we have a definite number t given by $t = b / (a + c)$; it lies between 0 and 1, and it is rational. The above relation now shows that we can *reverse* this relation: from t we can obtain the ratios $a : b : c$ and hence the actual values of a, b, c (keeping in mind that a, b, c are coprime, positive integers). This means that we have found a formula that can generate every possible PPT!

Illustrative examples

A few examples will illustrate this finding.

- Let $t = \frac{1}{2}$. Then we get

$$a : b : c = 1 - \frac{1}{4} : 1 : 1 + \frac{1}{4},$$

i.e., $a : b : c = 3 : 4 : 5$. This yields the PPT (3,4,5).

- Let $t = \frac{2}{3}$. Then we get

$$a : b : c = 1 - \frac{4}{9} : \frac{4}{3} : 1 + \frac{4}{9} = \frac{5}{9} : \frac{12}{9} : \frac{13}{9},$$

i.e., $a : b : c = 5 : 12 : 13$. This yields the PPT (5,12,13).

- Let $t = \frac{3}{4}$. Then we get

$$a : b : c = 1 - \frac{9}{16} : \frac{3}{2} : 1 + \frac{9}{16} = \frac{7}{16} : \frac{24}{16} : \frac{25}{16},$$

i.e., $a : b : c = 7 : 24 : 25$. This yields the PPT (7,24,25).

- Let $t = \frac{2}{5}$. Then we get

$$a : b : c = 1 - \frac{4}{25} : \frac{4}{5} : 1 + \frac{4}{25} = \frac{21}{25} : \frac{20}{25} : \frac{29}{25},$$

i.e., $a : b : c = 21 : 20 : 29$. This yields the PPT (21,20,29).

The above four examples worked out perfectly. However, the following two examples may bring up some doubts in your mind.

- Let $t = \frac{1}{3}$. Then we get

$$a : b : c = 1 - \frac{1}{9} : \frac{2}{3} : 1 + \frac{1}{9} = \frac{8}{9} : \frac{6}{9} : \frac{10}{9},$$

i.e., $a : b : c = 8 : 6 : 10$. This yields the triple (8,6,10).

- Let $t = \frac{3}{5}$. Then we get

$$a : b : c = 1 - \frac{9}{25} : \frac{6}{5} : 1 + \frac{9}{25} = \frac{16}{25} : \frac{30}{25} : \frac{34}{25},$$

i.e., $a : b : c = 16 : 30 : 34$. This yields the triple (16,30,34).

In the last two cases we obtain Pythagorean triples that are not primitive (so they are not PPTs), as the numbers in the triple carry a common factor greater than 1. If we divide the numbers in the triple by the respective GCDs, then we do obtain PPTs: (4,3,5) and (8,15,17), respectively.

For which values of t do we obtain PPTs, and for which values of t do we obtain Pythagorean triples that are not PPTs? The puzzle is not difficult to unravel. Try to find an answer on your own before reading on!

Resolution of the puzzle

Let $t = m/n$ where m, n are integers with $0 < m < n$ and $\text{GCD}(m, n) = 1$. Then we have:

$$a : b : c = 1 - \frac{m^2}{n^2} : \frac{2m}{n} : 1 + \frac{m^2}{n^2},$$

$$\text{i.e., } a : b : c = n^2 - m^2 : 2mn : n^2 + m^2.$$

So we obtain the triple $(n^2 - m^2, 2mn, n^2 + m^2)$.

If m, n are both odd then $n^2 - m^2, 2mn, n^2 + m^2$ are all even and so have a common factor of 2. Can there be any other prime factor shared by all three numbers? We shall show that there cannot be any such prime factor. The assumption made at the start that m, n are coprime plays a crucial role in the proof.

Suppose that p is an odd prime number that is a common factor of $n^2 - m^2, 2mn$ and $n^2 + m^2$. Since p divides both $n^2 - m^2$ and $n^2 + m^2$, it divides their sum as well as their difference. So, p divides both $2m^2$ and $2n^2$. Since p is an odd prime number, it must be that p divides both m^2 and n^2 . From this we infer that p divides both m and n . But m, n are coprime. It follows that there can be no such prime number. Therefore if m, n are both odd, 2 is the only prime factor that divides all three of $n^2 - m^2, 2mn$ and $n^2 + m^2$. Please write out the argument on your own.

Exactly the same argument can be used to show that if m, n are coprime and one of m, n is odd while the other is even, then there is no prime factor that divides all three of $n^2 - m^2, 2mn$ and $n^2 + m^2$. In other words, if m, n are coprime and one of m, n is odd while the other is even, then $n^2 - m^2, 2mn$ and $n^2 + m^2$ are coprime.

From the above we deduce the following. Let m, n be coprime integers with $0 < m < n$. Then:

- (1) If one of m, n is odd while the other is even, then the triple

$$\left(n^2 - m^2, 2mn, n^2 + m^2 \right) \quad (7)$$

is a PPT. Example: $m = 2, n = 5$ yields the PPT $(21, 20, 29)$.

- (2) If both m, n are odd, then the triple

$$\left(\frac{n^2 - m^2}{2}, mn, \frac{n^2 + m^2}{2} \right) \quad (8)$$

is a PPT. Example: $m = 5, n = 7$ yields the PPT $(12, 35, 37)$.

- (3) Moreover, every possible PPT can be generated in this manner.

A surprising finding

The above listing contains an unexpected puzzle.

- Let $m = 5, n = 9$. (Check that the numbers are coprime; they are.) Both the numbers are odd, so formula (8) applies. We get the following PPT: $(28, 45, 53)$.
- Let $m = 2, n = 7$. (Check that the numbers are coprime; they are.) One number is odd and the other number is even, so formula (7) applies. We get the following PPT: $(45, 28, 53)$.

How curious! The two triples have exactly the same numbers but in a different order.

It turns out that the following is true:

Formula 7 and Formula 8 generate exactly the same collection of PPTs, except that the numbers in the triples occur in a different order.

Try to find an explanation for this fact!

Resolution of an earlier puzzle

We are now in a position to answer a question posed earlier: **For arbitrary PPTs (a, b, c) , what values are possible for the differences $c - a$ and $c - b$?**

Consider the triple in formula 7:

$$(a, b, c) = (n^2 - m^2, 2mn, n^2 + m^2).$$

We observe that

- $c - a = 2m^2$, which is twice a square;
- $c - b = n^2 - 2mn + m^2 = (n - m)^2$, which is an odd square.

We see from this that each difference is either an odd square or twice a square.

Now consider the triple in formula 8:

$$(a, b, c) = \left(\frac{n^2 - m^2}{2}, mn, \frac{n^2 + m^2}{2} \right).$$

Here we observe that

- $c - a = m^2$, which is an odd square;
- $c - b = \frac{1}{2}(n^2 - 2mn + m^2)$, which is half a square.

We see from this that each difference is either an odd square or half a square. But half a square is also twice some other square! So the two kinds of triples yield the same sets of differences.

We have found the answer to the question. We express our finding in the following theorem.

Theorem. *For an arbitrary PPT (a, b, c) , let k represent either of the differences $c - a, c - b$. Then the set $\mathcal{K}$ of possible values of k is given by*

$$\mathcal{K} = \left\{ r^2 : r \in \mathbb{N}, r \text{ odd} \right\} \cup \left\{ 2r^2 : r \in \mathbb{N} \right\}.$$

Now we can understand our earlier findings:

- $1 \in \mathcal{K}$ and $2 \in \mathcal{K}$; the next number in $\mathcal{K}$ is 8;
- $4 \notin \mathcal{K}$ (as 4 is an even square, and only odd squares are permitted);
- $3 \notin \mathcal{K}, 5 \notin \mathcal{K}, 6 \notin \mathcal{K}, 7 \notin \mathcal{K}$.

An odd way to generate PPTs!

In closing, we offer this unusual way to generate PPTs using the sequence of odd numbers.

- Select any two consecutive odd numbers, e.g., 3 & 5, or 5 & 7, or 7 & 9, etc.
- Add the reciprocals of the two numbers. Let the sum be written as $\frac{a}{b}$ where a, b are coprime.
- Let $c = \sqrt{a^2 + b^2}$. It will be found that c is an integer, and (a, b, c) is a PPT.

Examples

- From the consecutive odd numbers 3, 5, we get the sum $\frac{8}{15}$ and hence the PPT (8, 15, 17).

- From the consecutive odd numbers 5,7, we get the sum $\frac{12}{35}$ and hence the PPT (12,35,37).
- From the consecutive odd numbers 7,9, we get the sum $\frac{16}{63}$ and hence the PPT (16,63,65).

And so on. Justifying why this always yields a PPT should be an easy exercise for you.

Can the sequence of consecutive even numbers also yield PPTs? We leave this too as an exercise for you.

Article contributed by Shailesh Shirali

The Big Wide World of Data

Visualisation – 3

As stated in earlier issues, this column aims to present a running series of short articles on the theme of data visualisation. We shall present a few striking examples of this in each issue (with little or no commentary).

Air pollution

Here are two data visualizations featuring air pollution in our country. They should be of great concern to us.

What are we going to do about this situation? What kind of initiatives can we take to bring about some change?

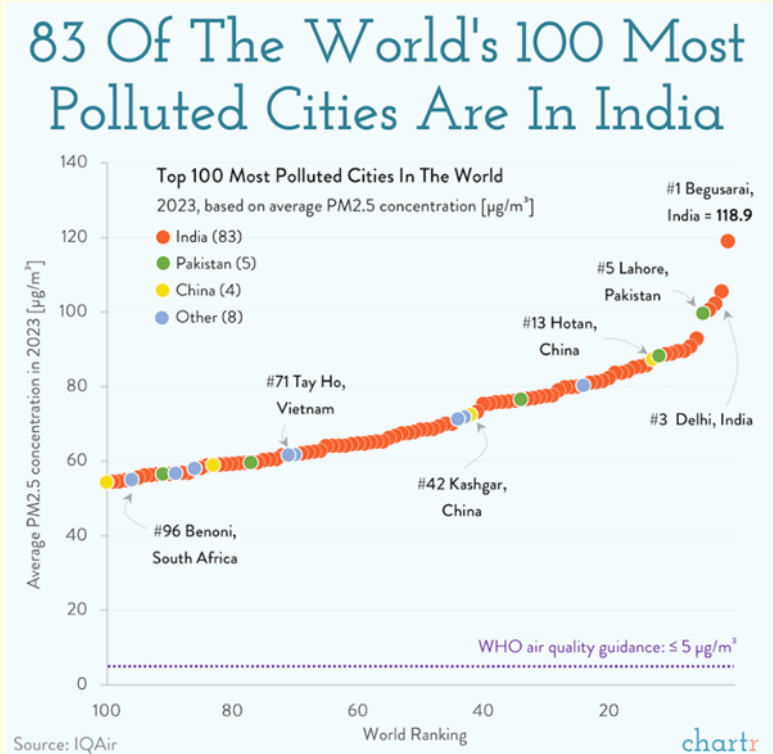


Figure 3: Polluted cities. Source: <https://www.chartr.co/newsletters/2024-03-24>

From the article in *Nature*, 17 December 24, available here: <https://www.nature.com/articles/d41586-024-04118-2> ...

By mid- to late November every year, a pall of noxious air settles over the entire Indo-Gangetic Plain, stretching from

eastern Pakistan to Bangladesh. This year, the haze was visible in satellite images and disrupted daily life in large cities such as Lahore and Delhi, leading to school closures and weekend curfews reminiscent of COVID-19 lockdowns.

The causes of this phenomenon are well known. Pollution levels typically begin to surge during the autumn festival of Diwali because of the widespread use of fireworks. The situation worsens as farmers burn leftover rice-paddy stubble to prepare fields for the winter (rabi) crop, following the harvest of the summer (kharif) crop. These seasonal factors, combined with background pollution from vehicular and industrial sources, create a public-health emergency in the region every year.

On cue, on 18 November this year, Delhi's Air Quality Index soared to 1,700 — far exceeding the safe limit of 50 set by the World Health Organization (WHO). Lahore in Pakistan had recorded a value of 1,100 a few days earlier. Because the Indo-Gangetic Plain is one of the most densely populated regions globally, the number of people affected would be in the millions — making the north Indian pollution plume one of the world's biggest public-health challenges. Yet it is solvable.

Despite pollution being a persistent, year-round problem, the media and policymakers often prioritize temporary fixes targeting the October–November spikes (see 'Short attention span'). This needs to change. ...

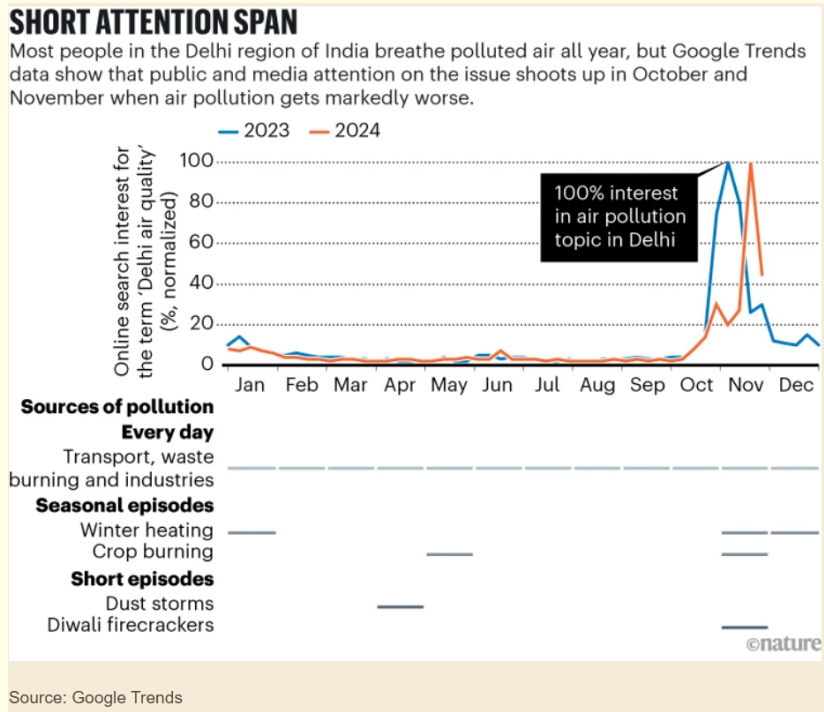


Figure 4: Pollution in northern India. Source: <https://www.nature.com/articles/d41586-024-04118-2>

Electric vehicle slowdown

From <https://sherwood.news/business/porsche-gm-and-tesla-confirm-continued-ev-slowdown/> ...

[It appears that the electric vehicle (EV) transition is decelerating ...] ... Porsche poured cold water on its transition to EVs, saying in a statement that the “transition to electric cars is taking longer than we thought five years ago”. ... General Motors said it would delay plans for its Buick EV and postpone a new electric factory in Michigan, before Tesla revealed a 45% slump in profits, facing slower demand (although the company did report significant growth in its energy generation and storage business). ... These latest delays follow similar decisions at Ford, Nissan, Volkswagen, and Mercedes-Benz, who have also eased off the accelerator in their race to boost EV production.

Despite EV sales skyrocketing from 3 million in 2021 to almost 14 million last year, both established carmakers and startups are seeing softer demand. As a result, dealerships are now grappling with parking lots full of EVs. ... Although the steep price tags that once came with going all-electric have dropped, ... concerns about range anxiety, the availability of public chargers, and battery performance at low temperatures have all persisted. ... Instead, many car buyers who might have considered ditching the combustion engine altogether now seem to be opting for hybrids. Toyota’s decision to focus on hybrids appears to be paying off, and even Lamborghini’s CEO expects hybrids to be a “success story”.

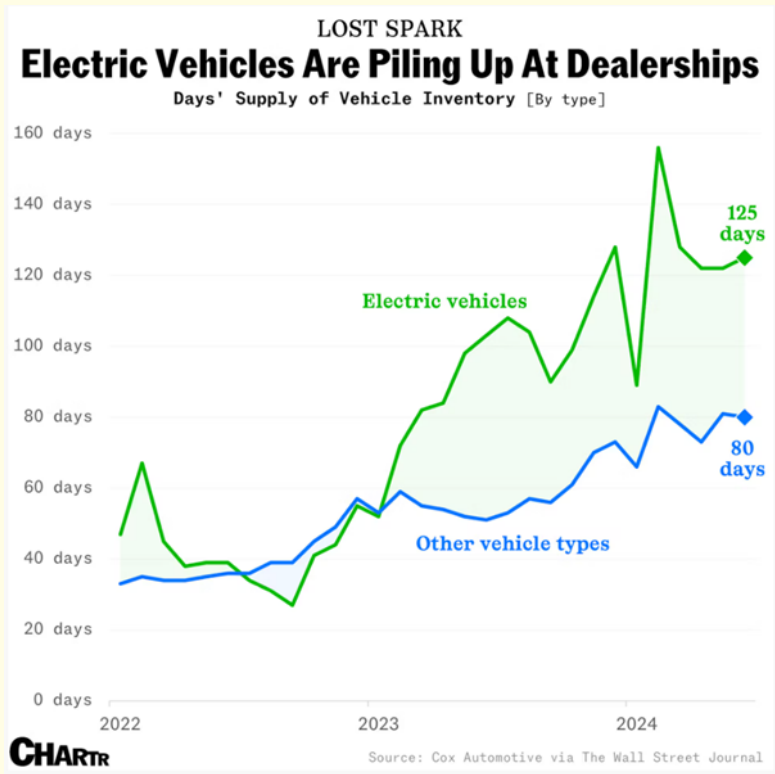


Figure 5: Electric vehicle slow down. Source: <https://sherwood.news/business/porsche-gm-and-tesla-confirm-continued-ev-slowdown/>

The Baby Bust

From <https://sherwood.news/world/five-major-themes-to-watch-in-2025> ...

While kids movies are having a moment – there might not be quite the same audience for them in the next century, as the widespread slump in fertility rates continues. Indeed, the global total fertility rate has more than halved since 1970 to just 2.25 in 2023.

The worldwide total still lands above the rate required for a population to replace itself from one generation to the next (roughly 2.1 births per woman), meaning that the global population is growing slightly (for now). But it's estimated that, by 2050, more than three-quarters of countries won't be able to sustain their population size.

Conti-natal drift

America's fertility rate has been below replacement level since 2007. In Europe, the brief baby bump observed in the Nordic countries in the '90s has all but vanished. In Asia, South Korea's fertility rate remains the world's lowest at 0.72, and China now has a rate less than one-seventh of its 1963 peak. With the exception of Africa (the continent's population is set to double by 2050), what's so startling is how universal the reproductive trend is – though it is, in a sense, unsurprising.

Improved education and contraceptive access have meant that there's a growing female workforce; but, perhaps most impor-

tantly, many people just don't seem to want kids any more, for various social, financial, and political reasons. Despite governments pressing on with population-boosting policies, the global fertility crisis shows no signs of slowing.

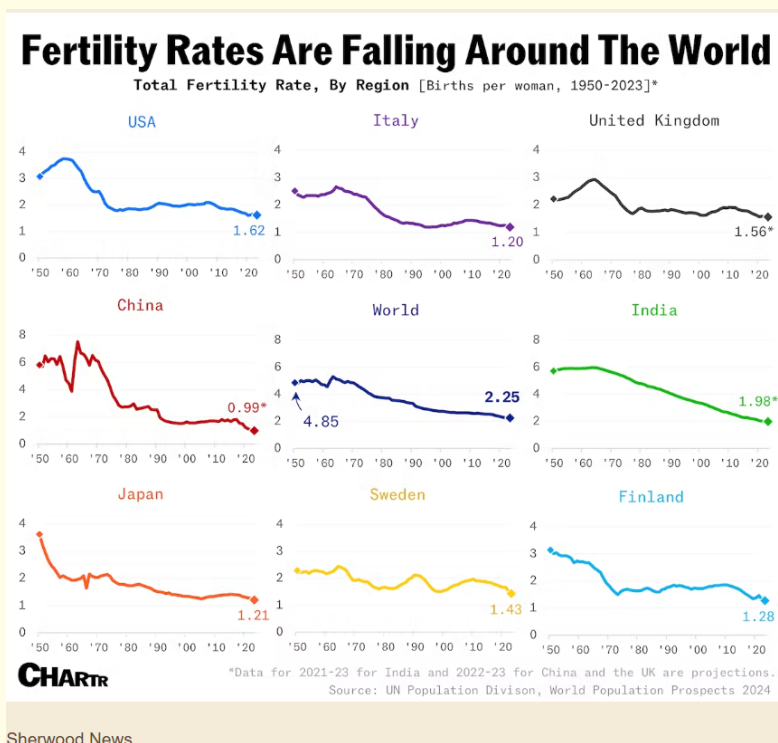


Figure 6: Fertility rates are coming down everywhere. Source: <https://sherwood.news/world/five-major-themes-to-watch-in-2025>

How To Prove It – 3

We continue our series of articles on the theme of proof. In the last two issues we dwelt exclusively on the theme of proof by induction. Now we shift direction and consider another technique that is very widely used in mathematics: *proof by enumeration of all possibilities*. It is known more informally as *brute force enumeration*.

It is also known as *proof by exhaustion*. But, as you can readily imagine, this phrase can have multiple meanings! (See Figure 7.)

Wikipedia [1] describes the approach as follows: *Proof by exhaustion, also known as ... proof by cases ... or the brute force method, is a method of mathematical proof in which the statement to be proved is split into a finite number of cases ... and where each type of case is checked to see if the proposition ... holds. This is a method of direct proof.*

The easy availability these days of high-speed electronic computers has hugely increased the ease of using this approach.

We will explain this technique by describing a few examples. You will surely understand the approach by studying the examples.

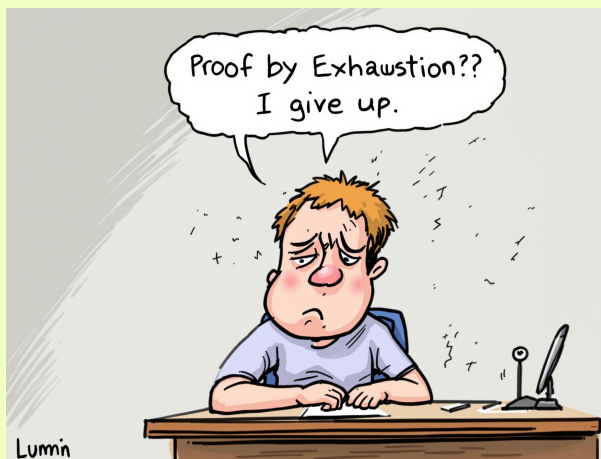


Figure 7: Sometimes, proof by exhaustion does not work out!

Example 1. Squares modulo 4

Claim: *There are no squares that leave remainder 2 or remainder 3 when divided by 4.*

Proof of claim: We consider all possible forms of an arbitrary integer n in terms of its remainder when divided by 4. Clearly, n must be of one of the following forms (here, k represents an arbitrary integer):

- $n = 4k$ (when n is divisible by 4);
- $n = 4k + 1$ (when n leaves remainder 1 when divided by 4);
- $n = 4k + 2$ (when n leaves remainder 2 when divided by 4);
- $n = 4k + 3$ (when n leaves remainder 3 when divided by 4).

We find the squares of each of these forms and examine what happens.

- If $n = 4k$, then $n^2 = 16k^2$, which is a multiple of 4;
- If $n = 4k + 1$, then $n^2 = 16k^2 + 8k + 1 = 4(4k^2 + 2k) + 1$, which is 1 more than a multiple of 4;
- If $n = 4k + 2$, then $n^2 = 16k^2 + 16k + 4 = 4(4k^2 + 4k + 1)$, which is a multiple of 4;
- If $n = 4k + 3$, then $n^2 = 16k^2 + 24k + 9 = 4(4k^2 + 6k) + 1$, which is 1 more than a multiple of 4.

We see that in all four cases, n^2 is either a multiple of 4, or 1 more than a multiple of 4. Expressed differently: every square is either divisible by 4, or leaves remainder 1 when divided by 4. The stated claim thus follows. ■

Comment 1

The argument presented above can be written more cleanly and neatly using the 'mod' symbol, thus:

$$\text{If } n \equiv 0 \pmod{4}, \text{ then } n^2 \equiv 0 \pmod{4};$$

$$\text{If } n \equiv 1 \pmod{4}, \text{ then } n^2 \equiv 1 \pmod{4};$$

$$\text{If } n \equiv 2 \pmod{4}, \text{ then } n^2 \equiv 0 \pmod{4};$$

$$\text{If } n \equiv 3 \pmod{4}, \text{ then } n^2 \equiv 1 \pmod{4},$$

hence $n^2 \equiv 0 \pmod{4}$ or $1 \pmod{4}$ in all cases. ■

We urge all students to become thoroughly familiar with the use of this symbol. It makes mathematical writing much cleaner and more compact. In follow-up articles we shall frequently use this notation.

Comment 2

The reader may have noticed that it is possible to shorten this proof. We only need to consider two cases: n even, and n odd.

- If n is even, then $n = 2k$ for some integer k , giving $n^2 = 4k^2$, which is of the form $0 \pmod{4}$.
- If n is odd, then $n = 2k + 1$ for some integer k , giving $n^2 = 4k^2 + 4k + 1$, which is of the form $1 \pmod{4}$.

Hence $n^2 \equiv 0 \pmod{4}$ or $1 \pmod{4}$ in all cases. ■

This is clearly a much better proof!

Example 2. Odd squares modulo 8

Claim: *All odd squares are of the form $1 \pmod{8}$.*

Proof of claim: We consider all possibilities for an arbitrary odd integer n modulo 4. Since n is known to be odd, it must be of one of the forms $4k \pm 1$. We consider both possibilities.

- If $n = 4k + 1$, then $n^2 = 16k^2 + 8k + 1 = 8k(2k + 1) + 1$, which is of the form $1 \pmod{8}$.
- If $n = 4k - 1$, then $n^2 = 16k^2 - 8k + 1 = 8k(2k - 1) + 1$, which too is of the form $1 \pmod{8}$.

The claim thus follows. ■

Here is a proof of the same claim that avoids enumeration of all possibilities.

Since n is odd, we have $n = 2k + 1$ where k is some integer. Hence,

$$\begin{aligned}n^2 &= (2k + 1)^2 = 4k^2 + 4k + 1 \\&= 4k(k + 1) + 1.\end{aligned}$$

The quantity $k(k + 1)$ is the product of two consecutive integers of which one must be odd and the other even, hence the product is even, hence $4k(k + 1)$ is a multiple of 8. It follows that $n^2 \equiv 1 \pmod{8}$. ■

Example 3. Squares modulo 3

Claim: *All squares are of one of the forms $0 \pmod{3}$ or $1 \pmod{3}$. There are no squares of the form $2 \pmod{3}$.*

Proof of claim: Let n be an arbitrary integer. Consider the different possibilities modulo 3, namely: $n \equiv 0 \pmod{3}$ and $n \equiv \pm 1 \pmod{3}$, i.e., $n = 3k$ and $n = 3k \pm 1$.

- If $n = 3k$, then $n^2 = 9k^2 \equiv 0 \pmod{3}$.
- If $n = 3k + 1$, then $n^2 = 9k^2 + 6k + 1 \equiv 1 \pmod{3}$.
- If $n = 3k - 1$, then $n^2 = 9k^2 - 6k + 1 \equiv 1 \pmod{3}$.

The result in each case is in agreement with the claim. As all possibilities have been exhausted, the claim is thereby proved. ■

Example 4. Cubes modulo 9

Claim: *All cubes are of one of the forms $0 \pmod{9}$ or $\pm 1 \pmod{9}$.*

Proof of claim: Let n be an arbitrary integer. Consider the different possibilities modulo 3, namely: $n \equiv 0 \pmod{3}$ and $n \equiv \pm 1 \pmod{3}$, i.e., $n = 3k$ and $n = 3k \pm 1$.

- If $n = 3k$, then $n^3 = 27k^3 \equiv 0 \pmod{9}$.
- If $n = 3k + 1$, then $n^3 = 27k^3 + 27k^2 + 9k + 1 \equiv 1 \pmod{9}$.
- If $n = 3k - 1$, then $n^3 = 27k^3 - 27k^2 + 9k - 1 \equiv -1 \pmod{9}$.

The result in each case is in agreement with the claim. As all possibilities have been exhausted, the claim is thereby proved. ■

Example 5. Tens and units digits of squares

Claim: *There is no square number whose units and tens digits are both odd.*

Here, the reference to “units and tens digits” indicates that we are working in base ten (i.e., we are using decimal notation).

Proof of claim: How would we prove this using brute force enumeration? We start by noting that for any integer n , the last two digits of n^2 depend only on the last two digits of n . This is so because for any integer a ,

$$(100 + a)^2 \equiv a^2 \pmod{100}.$$

So we only need to verify the stated claim for odd integers between 0 and 99. But we can do better than that by observing that this too is true:

$$(100 - a)^2 \equiv a^2 \pmod{100}.$$

So, if $a + b = 100$, then b^2 has the same last two digits as a^2 . Therefore we only need to verify the stated claim for odd integers between 0 and 50, i.e.,

for the twenty-five integers $1, 3, 5, 7, 9, 11, \dots, 41, 43, 45, 47, 49$. Here are the last two digits of the squares of these numbers:

01, 09, 25, 49, 81, 21, 69, 25, 89, 61,
41, 29, 25, 29, 41, 61, 89, 25, 69, 21,
81, 49, 25, 09, 01

We see that *the tens digits of these numbers are all even*. The stated conclusion thus follows: *there is no square whose last two digits are both odd*. ■

Question for the reader

Is there a proof that avoids brute force enumeration? We leave this question for the reader to unravel.

Example 6. Squares of prime numbers modulo 24

Claim: *If p is a prime number greater than 3, then $p^2 \equiv 1 \pmod{24}$.*

Here are some examples of this:

$$5^2 = 25 \equiv 1 \pmod{24}, \quad 7^2 = 49 \equiv 1 \pmod{24}, \quad 11^2 = 121 \equiv 1 \pmod{24}.$$

Proof of claim: Since p is a prime number greater than 3, it is not divisible by either 2 or 3. Therefore, it cannot be of any of the following forms: $6k, 6k + 2, 6k + 3, 6k + 4$; here, k represents an arbitrary positive integer. Otherwise put, it cannot be of any of these forms: $0 \pmod{6}, 2 \pmod{6}, 3 \pmod{6}, 4 \pmod{6}$.

Hence p is of one of these forms: $6k + 1, 6k + 5$.

Let us examine the squares of these expressions:

$$\begin{aligned}(6k+1)^2 &= 36k^2 + 12k + 1 = 12(3k^2 + k) + 1, \\ (6k+5)^2 &= 36k^2 + 60k + 25 = 12(3k^2 + 5k) + 25.\end{aligned}$$

Consider the quantity $3k^2 + k$. If k is even, it is the sum of two even numbers and is therefore even; and if k is odd, then it is the sum of two odd numbers and is therefore even. So, $3k^2 + k$ is an even number independent of the value of k . Hence $12(3k^2 + k)$ is a multiple of 24 for all k . This means that $12(3k^2 + k) + 1$ is of the form $1 \pmod{24}$ for all k .

Next, consider the quantity $3k^2 + 5k$. If k is even, it is the sum of two even numbers and is therefore even; and if k is odd, then it is the sum of two odd numbers and is therefore even. So, $3k^2 + 5k$ is an even number independent of the value of k . Hence $12(3k^2 + 5k)$ is a multiple of 24 for all k . This means that $12(3k^2 + 5k) + 1$ is of the form $1 \pmod{24}$ for all k .

The stated claim follows. ■

Comment

Did you notice that we have proved more than what was required? The claim was, “If p is a prime number greater than 3, then ...,” but to what extent did we make use of the fact that p is a *prime* number? All that we inferred was that p is not divisible by either 2 or 3, and therefore that p must be of the form $\pm 1 \pmod{6}$. Beyond that we made no use of the primality nature of p . This means that we have actually proved the following:

If n is an integer not divisible by either 2 or 3, then $n^2 \equiv 1 \pmod{24}$.

Problems for you to solve

Try your hand at the following problems. Prove them using the approach described above (brute force enumeration). Then see if you can find some other way of proving the claim without resorting to complete enumeration.

1. Show that for all integers n , the quantity $n(n+1)(2n+1)$ is divisible by 6.
2. Show that for all integers n , the quantity $n^3 - n$ is divisible by 3.
3. Show that for all integers n , the quantity $n^5 - n$ is divisible by 5.
4. Show that for all integers n , the quantity $n^7 - n$ is divisible by 7.

Further reading

[1] Wikipedia, “Proof by exhaustion” from

https://en.wikipedia.org/wiki/Proof_by_exhaustion

Article contributed by Shailesh Shirali

Cutting Numbers Down to Size – an Imaginary Conversation

Article contributed by A.Ramachandran

Mr. Sharma: Rahul, I see that you have been rather busy for some time.

Rahul: Yes, Dad. I am working on a math assignment. I am to make a bar graph showing the areas of the countries of Asia. I have got the data from the Atlas. But I am finding it difficult to make the bar graph.

Mr. Sharma: What is the problem?

Rahul: I don't know what scale to choose for the Y-axis. There is a wide spread of values – from about 300 km^2 for Maldives to more than 15 million km^2 for Russia. If I try to fit in Russia, Maldives is hardly noticeable; if I

expand the scale to make Maldives fairly visible, then Russia goes out of the paper.

Mr. Sharma: Hm! I see your difficulty now. You are using an additive scale for the Y-axis. Try using a multiplicative scale.

Rahul: I don't understand you, Dad.

Mr. Sharma: You see, in your graph, the values increase by a constant amount as you move up notch by notch. That is, if one step counts for 1000 km², three steps would count for 3000 km². You could also have a graph where the values get multiplied by a constant factor every step you go up. If one marking stands for 100 km², the next could stand for 1000 km², the next for 10000 km², and so on.

Rahul: Then the base line cannot be zero.

Mr. Sharma: That's right. We could start with 1(= 10⁰), and have 10(= 10¹), 100(= 10²), 1000(= 10³), etc., at equal spacings, as we go up. Moving down from 1, we would have 0.1(= 10⁻¹), 0.01(= 10⁻²), etc., again at equal spacings.

Rahul: Oh, that's an interesting idea.

Mr. Sharma: We call such a scale a *logarithmic scale*. Instead of having to take the logarithms of the data values and plotting them on a linear scale, the Y-axis is marked in such a way that you directly plot the data.

Rahul: Now, between two markings, say 1 and 10, will the markings for 2, 3, 4, etc., be equally spaced?

Mr. Sharma: No. The logarithms of equally spaced numbers will not be equally spaced. Rather they will look like what you see below (Figure 8).

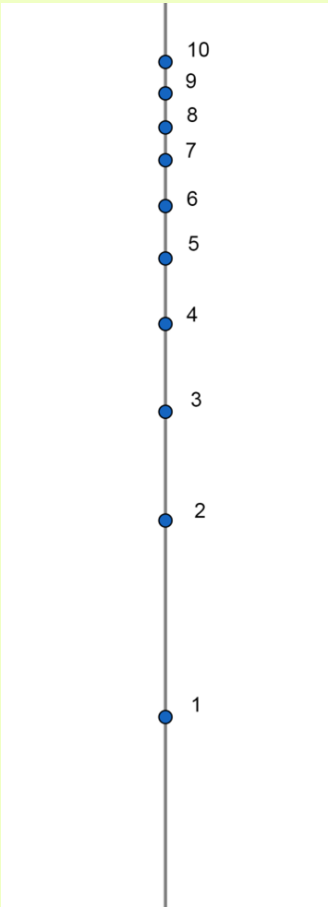


Figure 8

The same pattern continues up to 100 (now indicating 20, 30, 40, etc.), 1000 (indicating 200, 300, 400, etc.), and so on, and similarly down to 0.1, 0.01,

and so on. The spacings are according to the log to base 10 values, as you can see from this table, with values correct to four decimal places.

Number	Logarithm (base 10)
1	0.0000
2	0.3010
3	0.4771
4	0.6020
5	0.6989
6	0.7781
7	0.8450
8	0.9030
9	0.9542
10	1.0000

Rahul: How do I prepare such a graph sheet?

Mr. Sharma: You can download or buy such sheets but you may not find any numbers indicated. You need to mark up the Y-axis as per your need – depending on the range of data values you are working with.

Rahul: I now see how such a scale can accommodate a wide range of data values. Is this sort of scale widely used?

Mr. Sharma: Yes. Data related to some physical and chemical processes yield curved graphs on normal (linear) graph paper but yield straight lines on logarithmic graph sheets. This is useful in some ways. Also, you may have heard of the *decibel scale*.

Rahul: It tells us how loud a sound is.

Mr. Sharma: Correct. Loudness is related to the amount of energy released per second. If a sound is stated to be 10 decibels more than another, then the former is 10 times as loud as the latter. A sound that is 20 decibels higher is 100 times as loud, and so on. The sound of a jet airplane taking off at close quarters is about 10^{12} or a trillion times as loud as a whisper. It would be difficult to express both these on a linear scale.

Rahul: I think the pH scale of acids and bases is similar to this.

Mr. Sharma: Yes. The acidic or basic nature of a solution depends on the concentration of hydrogen ions. This usually ranges from 10^{-1} to 10^{-14} moles/litre. So the logarithm of this to base 10 is taken and the negative sign is ignored. So, the lower the pH value the greater the acidic nature. The Richter scale used to express earthquake intensities is also similar. For every unit you go up on the Richter scale the wave amplitude increases by a factor of 10; however the energy released increases by a factor of about 31.6.

Rahul: So the base does not have to be 10 always?

Mr. Sharma: That's a point. The base could be different. There is a scale used to express the apparent brightness of stars, that is, the brightness as viewed from earth. The base used is about 2.5, chosen for historical reasons. Also, the higher or more positive values represent dimmer stars.

Rahul: Well, that's a lot of information to take in at one go. I shall look up these logarithmic scales later. Now I must complete my bar graph.

Mr. Sharma: You must do that. But I would like to point out just one thing before that. There are many connections among the log values in the above

table. I am giving them here:

$$\begin{aligned}\log 4 &= 2(\log 2), & \log 8 &= 3(\log 2), & \log 9 &= 2(\log 3), \\ \log 6 &= \log 2 + \log 3, & \log 10 &= \log 2 + \log 5.\end{aligned}$$

About the author ...

A Ramachandran has had a long-standing interest in the teaching of mathematics and science. He studied physical science and mathematics at the undergraduate level and shifted to life science at the postgraduate level. He taught science, mathematics and geography to middle school students at Rishi Valley School for two decades. His other interests include the English language and Indian music. He may be contacted at archandran.53@gmail.com

Proofs-without-words for the Pythagorean theorem – 2

Article contributed by Adrian Kujur

In Part 1 of this article (October 2024 issue), we described a few dissection schemes that serve as proofs-without-words (or PWWs) of the Pythagorean Theorem. These schemes describe how the two smaller squares on the legs of a right-angled triangle can be dissected and the pieces reassembled to make up the larger square on the hypotenuse of the triangle.

Working out why the pieces fit together perfectly in the manner suggested diagram is usually done geometrically (angle and side length comparisons). It is important that this detail is not omitted because it is possible to be deceived by a diagram; pieces may appear to fit together perfectly when they do not.

Such proofs are fundamentally different from proofs whose basis is an algebraic identity, e.g., the expansion identity for $(a + b)^2$ or $(a - b)^2$.

Now we describe some more such schemes. In each case we simply display

the scheme, with no commentary. We leave the task of understanding how the schemes work to you.

As noted in Part 1, these proofs work only for particular ratios of the legs. So, they are not meant to be general proofs.

When the ratio of the legs is 1 : 3

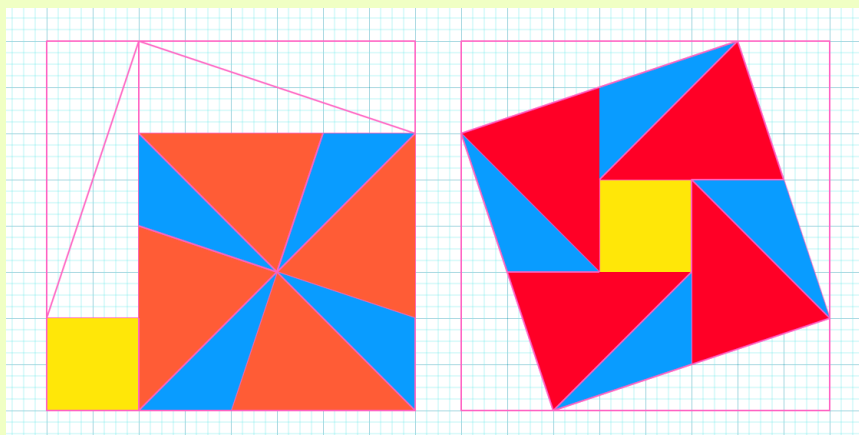


Figure 9: A dissection scheme when the ratio of the legs is 1 : 3

When the ratio of the legs is 1 : 4

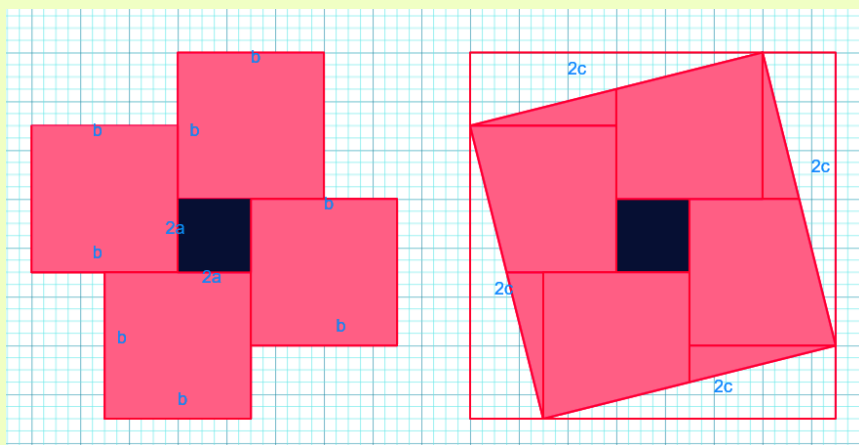


Figure 10: A dissection scheme when the ratio of the legs is 1 : 4

When the ratio of the legs is 3 : 4

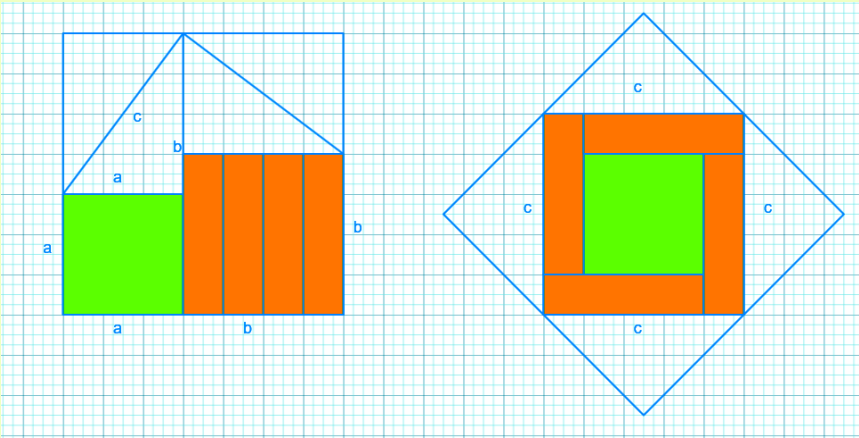


Figure 11: A dissection scheme when the ratio of the legs is 3 : 4

When the ratio of the legs is 3 : 5

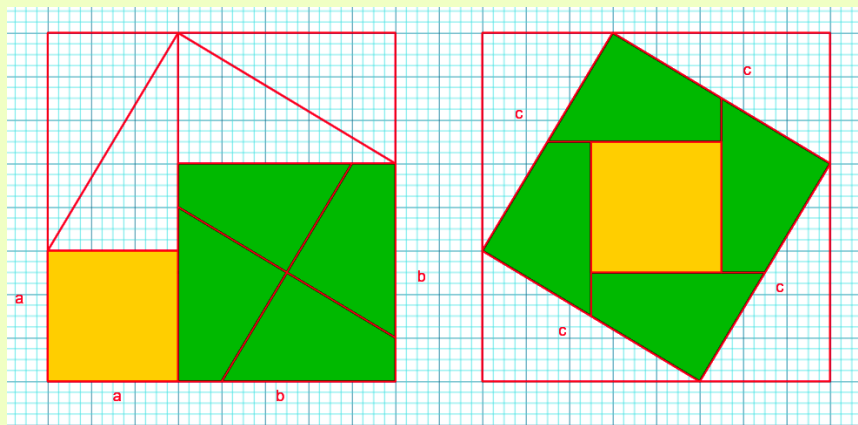


Figure 12: A dissection scheme when the ratio of the legs is 3 : 5

About the author ...

Adrian Kujur is a tribal teacher from Chhattisgarh. He teaches at the upper primary level in a government school, Kanya Shiksha Parisar Ambikapur, Chhattisgarh. He is a graduate from NIT Bhopal in Electrical Engineering.

He has been teaching Mathematics since 2010. At the same time he has been associated with the production of teaching-learning materials and running a teacher training programme in mathematics for many years. His teaching innovations have been recognised by Azim Premji Foundation in Ambikapur, Chhattisgarh.

He may be contacted at kujuradrian@gmail.com.

Gems From Geometry – 2

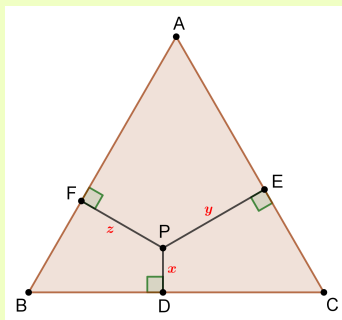
We continue on our series of articles on striking results (“gems”) from geometry. As noted last time, there is a vast collection of such geometric gems discovered over the past two millennia — so we will not run out of topics any time soon!

In this issue we study a theorem known today as *Viviani’s theorem*. It is not only the statement of the theorem that is striking and beautiful; the proof methods are equally beautiful.

Pedagogically, it is of particular interest as it is a result that can be found experimentally even by young children. For the mathematics practitioner too it is of great interest, because it can be proved in many different ways. It even gives us the opportunity for generalization!

The theorem is named after the Italian mathematician-scientist Vincenzo Viviani (1622–1703), a disciple of Galileo during the years when Galileo was under ‘house arrest’ for making politically unwise statements about the heliocentric theory of Copernicus. Among other things, Viviani helped Galileo in the compilation of the book, *Discourses and Mathematical Demonstrations Relating to Two New Sciences*.

Figure 13 depicts what Vivian's theorem states. Let ABC be an equilateral triangle, and let P be any interior point of the triangle.

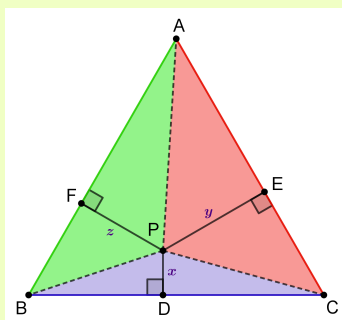


Let perpendiculars PD, PE, PF be dropped from P to the sides BC, CA, AB , as shown.

The theorem then states that the distance sum $PD + PE + PF$ is the same for all positions of P .

Figure 13: Vivian's theorem: $PD + PE + PF$ is the same for all P within the triangle

An easy way to prove the theorem is indicated in Figure 14.

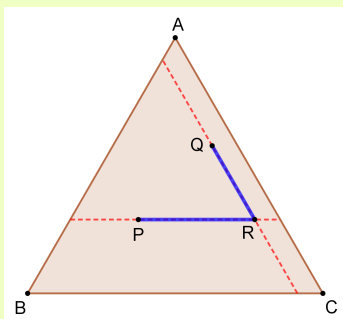


Join PA, PB, PC . Express the areas of triangles PBC, PCA, PAB in terms of x, y, z and the side a of the triangle. Now use the fact that the sum of the areas of PBC, PCA, PAB is equal to the area of ABC (and, therefore, $x + y + z =$ the altitude h of the triangle).

Figure 14: Proof of Viviani's theorem using areas

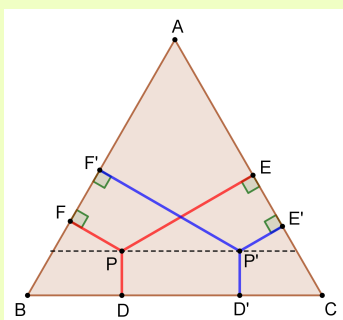
Is there a 'pure geometry' proof?

The proof described above (Figure 14) is compact and elegant. However, it requires some algebraic simplification at the end. This may prompt us to look for a purely geometric proof. Here is one such.



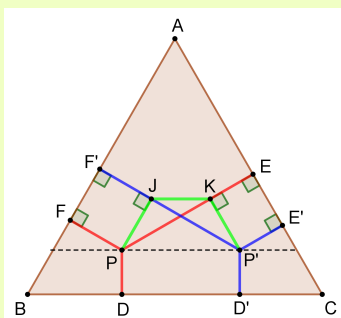
Step 1. We show that it is possible to move from any interior point P to any other interior point Q via movements that are parallel to the sides of the triangle. Just two such movements are needed, as the figure clearly indicates.

Figure 15: Step 1 of the 'pure geometry' proof



Step 2. Consider a pair of points P, P' such that PP' is parallel to BC . Drop perpendiculars from P to the sides as shown. We clearly have $P'D' = PD$. We must now show $P'E' + P'F' = PE + PF$.

Figure 16: Step 2 of the 'pure geometry' proof



Step 3. Drop perpendiculars PJ from P to $P'F'$ and $P'K$ from P' to PE , as shown.

Since $PF = JF'$ and $P'E' = KE$, we only need to show that $PK = P'J$.

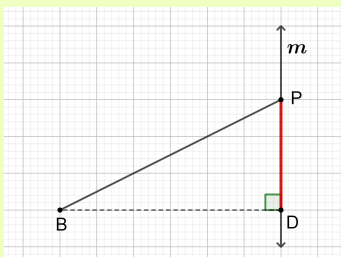
This follows when we see that $PJKP'$ is an isosceles trapezium.

It will help if we observe that quadrilateral $PJKP'$ is cyclic. Please try to fill in the details of the proof on your own. ■

Figure 17: Step 3 of the 'pure geometry' proof

A proof using vector algebra

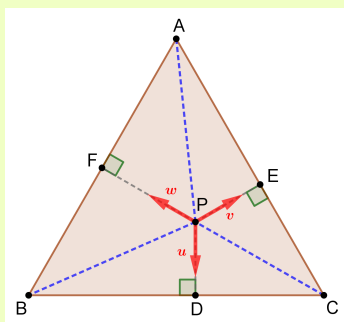
We round off this article by describing a compact and elegant proof using vectors. It draws on a simple observation.



Let PB be a segment; let m be a line through P (Figure 18). Let PD be the projection of PB on m . Then $PD = \vec{PB} \cdot \mathbf{u}$, where $\mathbf{u}$ is a unit vector along the line m . (We must orient the vector suitably.)

Figure 18: A simple observation about projections

We apply this observation to the proof of Viviani's theorem.



Let P be a point within $\triangle ABC$. Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be unit vectors perpendicular respectively to BC, CA, AB , as shown in Figure 19. From symmetry, we see that the vector sum $\mathbf{u} + \mathbf{v} + \mathbf{w}$ is zero.

Figure 19

Drawing on the observation made above we see that

$$PD = \overrightarrow{PB} \cdot \mathbf{u}, \quad PE = \overrightarrow{PC} \cdot \mathbf{v}, \quad PF = \overrightarrow{PA} \cdot \mathbf{w}.$$

For some other point Q within the triangle, we similarly have

$$QD = \overrightarrow{QB} \cdot \mathbf{u}, \quad QE = \overrightarrow{QC} \cdot \mathbf{v}, \quad QF = \overrightarrow{QA} \cdot \mathbf{w}.$$

Hence we have

$$PD + PE + PF = \overrightarrow{PB} \cdot \mathbf{u} + \overrightarrow{PC} \cdot \mathbf{v} + \overrightarrow{PA} \cdot \mathbf{w},$$

$$QD + QE + QF = \overrightarrow{QB} \cdot \mathbf{u} + \overrightarrow{QC} \cdot \mathbf{v} + \overrightarrow{QA} \cdot \mathbf{w}.$$

By subtraction we obtain

$$\begin{aligned} & (PD + PE + PF) - (QD + QE + QF) \\ &= \left(\overrightarrow{PB} \cdot \mathbf{u} + \overrightarrow{PC} \cdot \mathbf{v} + \overrightarrow{PA} \cdot \mathbf{w} \right) - \left(\overrightarrow{QB} \cdot \mathbf{u} + \overrightarrow{QC} \cdot \mathbf{v} + \overrightarrow{QA} \cdot \mathbf{w} \right). \end{aligned}$$

The above expression may be rewritten as

$$\begin{aligned}
 & (\overrightarrow{PB} - \overrightarrow{QB}) \cdot \mathbf{u} + (\overrightarrow{PC} - \overrightarrow{QC}) \cdot \mathbf{v} + (\overrightarrow{PA} - \overrightarrow{QA}) \cdot \mathbf{w} \\
 &= \overrightarrow{PQ} \cdot \mathbf{u} + \overrightarrow{PQ} \cdot \mathbf{v} + \overrightarrow{PQ} \cdot \mathbf{w} \\
 &= \overrightarrow{PQ} \cdot (\mathbf{u} + \mathbf{v} + \mathbf{w}) \\
 &= 0,
 \end{aligned}$$

since $\mathbf{u} + \mathbf{v} + \mathbf{w}$ is the zero vector. Hence

$$PD + PE + PF = QD + QE + QF.$$

Since P, Q are an arbitrary pair of points, it follows that $PD + PE + PF$ is a constant for all points P within the triangle. ■

Generalization

Do these proofs help us in generalizing Viviani's theorem? Yes. Let's see how. Consider the proof based on area computation (Figure 14). As a corollary we deduced that $x + y + z = h$.

Now suppose that $\triangle ABC$ is not equilateral. Let $BC = a$, $CA = b$, $AB = c$. Let the area of $\triangle ABC$ be K . Then we have,

$$K = \frac{1}{2}a \cdot PD + \frac{1}{2}b \cdot PE + \frac{1}{2}c \cdot PF.$$

Among the lengths a, b, c , there is a smallest and a largest length. Let us label the vertices in such a way that $a \leq b \leq c$. Then we have

$$\frac{a(PD + PE + PF)}{2} \leq K \leq \frac{c(PD + PE + PF)}{2}.$$

This implies that

$$\frac{2K}{c} \leq PD + PE + PF \leq \frac{2K}{a}.$$

Let the altitudes of $\triangle ABC$ be h_a (from A to BC), h_b (from B to CA), h_c (from C to AB). Then we have,

$$K = \frac{ah_a}{2} = \frac{bh_b}{2} = \frac{ch_c}{2},$$

so

$$h_c = \frac{2K}{c}, \quad h_b = \frac{2K}{b}, \quad h_a = \frac{2K}{a},$$

and since $a \leq b \leq c$, we have $h_c \leq h_b \leq h_a$.

So we have:

$$h_c \leq PD + PE + PF \leq h_a.$$

A nicer way of expressing this is

$$\min\{h_a, h_b, h_c\} \leq PD + PE + PF \leq \max\{h_a, h_b, h_c\}.$$

In words: $PD + PE + PF$ lies between the shortest altitude and the longest altitude.

This is clearly a genuine generalization of Viviani's theorem. ■

We will study another generalization in the April 2025 issue.

Article contributed by Shailesh Shirali

Memorable Sums – 2

In the previous article we played around with a few infinite series. We succeeded in finding their sums by using some clever strategies. We started with a straightforward summation; we showed that

$$a = \frac{1}{10} + \frac{1}{100} + \frac{1}{1000} + \frac{1}{10000} + \frac{1}{100000} + \cdots = \frac{1}{9}. \quad (9)$$

Then we showed the following using some ingenious bracketing of terms:

$$b = \frac{1}{10} + \frac{2}{100} + \frac{3}{1000} + \frac{4}{10000} + \frac{5}{100000} + \cdots = \frac{10}{81}. \quad (10)$$

Then we took on a more challenging summation but succeeded in finding its sum as well:

$$c = \frac{1^2}{10} + \frac{2^2}{100} + \frac{3^2}{1000} + \frac{4^2}{10000} + \frac{5^2}{100000} + \cdots = \frac{110}{729}. \quad (11)$$

We ended the article by asking you to find the sum of this series:

$$d = \frac{1^3}{10} + \frac{2^3}{100} + \frac{3^3}{1000} + \frac{4^3}{10000} + \frac{5^3}{100000} + \cdots. \quad (12)$$

We shall replicate the path we took earlier and find the value of d .

$$\begin{aligned}
 d &= \frac{1^3}{10} + \frac{2^3}{100} + \frac{3^3}{1000} + \frac{4^3}{10000} + \frac{5^3}{100000} + \cdots \\
 \therefore \frac{d}{10} &= \frac{1^3}{100} + \frac{2^3}{1000} + \frac{3^3}{10000} + \frac{4^3}{100000} + \cdots, \\
 \therefore \frac{9d}{10} &= \frac{1}{10} + \frac{7}{100} + \frac{19}{1000} + \frac{37}{10000} + \frac{61}{100000} + \cdots.
 \end{aligned}$$

Next,

$$\begin{aligned}
 \frac{9d}{10} &= \frac{1}{10} + \frac{7}{100} + \frac{19}{1000} + \frac{37}{10000} + \frac{61}{100000} + \cdots, \\
 \therefore \frac{9d}{100} &= \frac{1}{100} + \frac{7}{1000} + \frac{19}{10000} + \frac{37}{100000} + \cdots, \\
 \therefore \frac{81d}{100} - \frac{1}{10} &= \frac{6}{100} + \frac{12}{1000} + \frac{18}{10000} + \frac{24}{100000} + \cdots.
 \end{aligned}$$

We know how to find the sum of the last series:

$$\begin{aligned}
 &\frac{6}{100} + \frac{12}{1000} + \frac{18}{10000} + \frac{24}{100000} + \cdots \\
 &= \frac{6}{10} \left(\frac{1}{10} + \frac{2}{100} + \frac{3}{1000} + \frac{4}{10000} + \cdots \right) \\
 &= \frac{6}{10} \cdot \frac{10}{81} = \frac{2}{27}.
 \end{aligned}$$

Therefore we have:

$$\frac{81d}{100} - \frac{1}{10} = \frac{2}{27}, \quad \therefore d = \left(\frac{1}{10} + \frac{2}{27} \right) \frac{100}{81} = \frac{470}{2187}.$$

That was quite a journey!

Another approach — using calculus!

We now describe a completely different approach to the problem of finding these sums.

We start with the basic identity

$$1 + x + x^2 + x^3 + x^4 + \cdots = \frac{1}{1-x}, \quad \text{for } |x| < 1. \quad (13)$$

Value of b

Differentiating both sides of (13) with respect to x , we get

$$1 + 2x + 3x^2 + 4x^3 + \cdots = \frac{d}{dx} \left(\frac{1}{1-x} \right) = \frac{1}{(1-x)^2}, \quad \text{for } |x| < 1,$$

and so,

$$x + 2x^2 + 3x^3 + 4x^4 + \cdots = \frac{x}{(1-x)^2}, \quad \text{for } |x| < 1. \quad (14)$$

The substitution $x = \frac{1}{10}$ yields

$$b = \frac{1}{10} + \frac{2}{100} + \frac{3}{1000} + \frac{4}{10000} + \cdots = \frac{1/10}{81/100} = \frac{10}{81},$$

in agreement with the value obtained earlier.

Value of c

Differentiating both sides of (14) with respect to x , we get

$$1^2 + 2^2x + 3^2x^2 + 4^2x^3 + \cdots = \frac{d}{dx} \left(\frac{x}{(1-x)^2} \right) = \frac{1+x}{(1-x)^3}, \quad \text{for } |x| < 1,$$

and so,

$$1^2x + 2^2x^2 + 3^2x^3 + 4^2x^4 + \dots = \frac{x(1+x)}{(1-x)^3}, \quad \text{for } |x| < 1. \quad (15)$$

The substitution $x = \frac{1}{10}$ yields

$$c = \frac{1^2}{10} + \frac{2^2}{100} + \frac{3^2}{1000} + \frac{4^2}{10000} + \dots = \frac{11/100}{729/1000} = \frac{110}{729},$$

in agreement with the value obtained earlier.

Value of d

Differentiating both sides of (15) with respect to x , we get

$$1^3 + 2^3x + 3^3x^2 + 4^3x^3 + \dots = \frac{d}{dx} \left(\frac{x(1+x)}{(1-x)^3} \right) = \frac{x^2 + 4x + 1}{(1-x)^4}, \quad \text{for } |x| < 1,$$

and so,

$$1^3x + 2^3x^2 + 3^3x^3 + 4^3x^4 + \dots = \frac{x(x^2 + 4x + 1)}{(1-x)^4}, \quad \text{for } |x| < 1. \quad (16)$$

The substitution $x = \frac{1}{10}$ yields

$$d = \frac{1^3}{10} + \frac{2^3}{100} + \frac{3^3}{1000} + \frac{4^3}{10000} + \dots = \frac{4700}{2187},$$

in agreement with the value obtained earlier.

This method works like a well-oiled machine!

Yet one more summation

Let's close this article with yet one more challenging summation, just to show how easily this method works. We shall evaluate the sum

$$f = \frac{1^4}{10} + \frac{2^4}{100} + \frac{3^4}{1000} + \frac{4^4}{10000} + \dots$$

Differentiating both sides of (16) with respect to x , we get

$$\begin{aligned} 1^4 + 2^4x + 3^4x^2 + 4^4x^3 + \dots &= \frac{d}{dx} \left(\frac{x(x^2 + 4x + 1)}{(1-x)^4} \right) \\ &= \frac{x^3 + 11x^2 + 11x + 1}{(1-x)^5}, \quad \text{for } |x| < 1, \end{aligned}$$

and so,

$$1^4x + 2^4x^2 + 3^4x^3 + 4^4x^4 + \dots = \frac{x(x^3 + 11x^2 + 11x + 1)}{(1-x)^5}, \quad \text{for } |x| < 1. \quad (17)$$

The substitution $x = \frac{1}{10}$ yields

$$d = \frac{1^4}{10} + \frac{2^4}{100} + \frac{3^4}{1000} + \frac{4^4}{10000} + \dots = \frac{73700}{19683}.$$

We have obtained the required answer.

Closing remarks

- You will have noticed that we left out many algebraic details above. This was intentional! We leave the detailed steps for the student to check for himself or herself.
- We have made use only of differentiation in the exposition above. Why don't we make use of integration as well? This opens up fresh possibilities. We invite you to explore further on your own.

Proof-Based Grid-Puzzles – 2: The Grid-Diagonal Puzzle

Article contributed by Rossi D'Souza

Introduction

The Grid-Diagonal Puzzle is an activity based on a famous old mathematics problem that is often posed to students at talent-development programs. [3] states the problem as: “How many boxes are crossed by a diagonal in a rectangular table formed by 199×991 small squares?” (p. 35). [2] presents a similar problem: “A 200×3 rectangle is drawn on graph paper along the grid lines. How many grid squares will a diagonal of the rectangle cross?” (p. 15).

The solutions offered in these books are limited to grid-rectangles whose length and breadth were coprime. But generalizing the problem to finding

the number of boxes (or unit squares) crossed by a diagonal (d) of any grid rectangle of dimensions $l \times b$ turns out to be a fun classroom activity (that can serve as a stepping stone into Coordinate geometry for middle school students).

As a classroom activity, we provide students with graph paper, a pencil and a ruler, and ask them to do the following:

Step 1. Draw a rectangle with length 2 units and breadth 3 units on a sheet of graph paper.

Step 2. Draw the diagonal and count the number of unit squares it passes through.

Step 3. Increase the breadth by 1 unit and repeat Step 2.

Step 4. Record the value of d for different values of l and b in a table, and try to find the relationship between d , l and b .

In the course of the activity, the students will likely discover the relationship: $d = l + b - \text{GCD}(l, b)$.

We now ask our students to try to prove their observation.

Proof of the conjecture

To prove the formula $d = l + b - \text{GCD}(l, b)$, we consider two cases of grid-rectangles — rectangles whose length and breadth are coprime, and those whose length and breadth are not coprime.

Let's first look at the case of rectangles where l and b are coprime and observe if the diagonal passes through any grid points. We are looking at grid points as this offers an insight into the number of grid squares that our diagonal will pass through.

If l and b are coprime, it is easy to see that the diagonal will not pass through any grid point except the vertices of the rectangle. Why? Because (see Figure 20), if a diagonal passes through a grid point (x, y) , then we must have $\frac{l}{b} = \frac{y}{x}$. This means that the fraction $\frac{l}{b}$ can be “reduced” (by cancelling factors) to $\frac{y}{x}$. But this cannot happen if l and b are coprime; no such cancellation can take place.

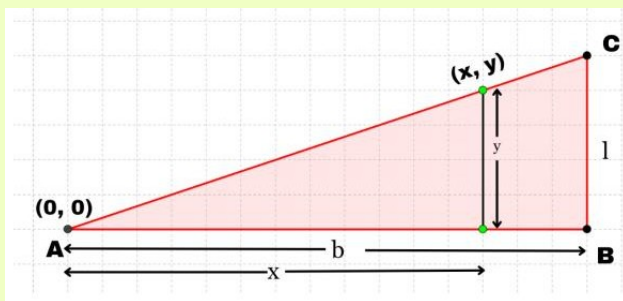


Figure 20: Case of grid-rectangle whose diagonal passes through a non-vertex grid point

In the classroom we discovered that if we regard the floor tiles as a grid and walk along the diagonal, in order to step into a new square, we must cross its border. Moreover, each border is part of either a grid-row or grid-column.

So, if we walk from one corner of the rectangle to the diagonally opposite corner, we must cross $b - 1$ columns and $l - 1$ rows (and not b columns and l rows, because we are excluding the borders of the rectangle).

Thus, we cross a total of $(l - 1) + (b - 1) = l + b - 2$ borders, which is the number of squares we enter.

Adding the one square from which we begin to the $l + b - 2$ squares that

we enter, we see that the diagonal passes through $l + b - 2 + 1 = l + b - 1$ grid squares.

Therefore, if l and b are coprime, $d = l + b - 1$.

Now for the second case. If l and b are not coprime, the diagonal must (as earlier) cross $l + b - 2$ borders. However, since l and b have a common factor greater than 1, the diagonal will pass through a number of in-between grid points (the intersection of a row and a column of the grid).

Thus, to get an accurate number of borders that we cross (or squares that we enter), we must subtract 1 for each such grid point that we pass through.

Let k be the number of grid-points that we pass through while walking along the diagonal. Then $d = (l + b - 1) - k$.

Let $p = \frac{b}{\text{GCD}(l,b)}$ and $q = \frac{l}{\text{GCD}(l,b)}$. Then $\frac{p}{q} = \frac{b}{l}$.

Since the line segment joining the 'starting vertex' $A(0,0)$ and the point (p,q) has the same 'slope' as the diagonal AC , the diagonal will pass through point (p,q) . (If we denote points (p,q) and $(p,0)$ as E and D respectively, as shown in Figure 21, we see that $\triangle ADE \sim \triangle ABC$ since $\frac{q}{p} = \frac{ED}{AD} = \frac{CB}{AB} = \frac{l}{b}$, and $\angle ADE = \angle ABC = 90^\circ$. Therefore $\angle EAD = \angle CAB$. In other words, AE and AC have the same slope).

Similarly, diagonal AC passes through $(2p, 2q)$ provided that $2p \leq b$ (or, equivalently, $2q \leq l$).

In general, diagonal AC passes through all points (np, nq) where $1 \leq n \leq \frac{b}{p}$.

In other words, the diagonal encounters $\frac{b}{p}$ (or $\frac{l}{q}$) grid points (including the final vertex) along the way.

But $\frac{b}{p}$ (or $\frac{l}{q}$) is the GCD of l and b .

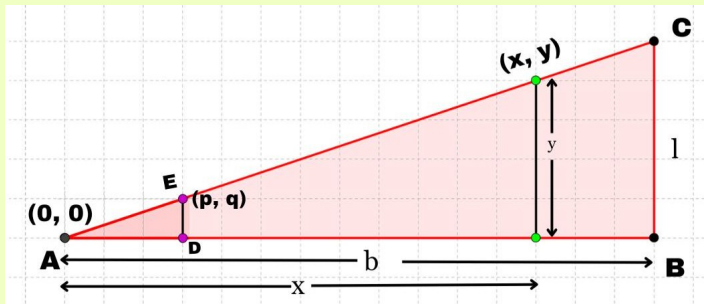


Figure 21: AC passes through point E(p, q)

This means that the number of grid points (including the final vertex) that the diagonal encounters is $\text{GCD}(l, b)$.

Thus the number of *non-vertex* grid points that the diagonal encounters is $k = \text{GCD}(l, b) - 1$. Thus $d = l + b - 1 - k = l + b - 1 - (\text{GCD}(l, b) - 1)$, i.e., $d = l + b - \text{GCD}(l, b)$.

So, if l and b are not coprime, $d = l + b - \text{GCD}(l, b)$.

But if l and b are coprime, then $\text{GCD}(l, b) = 1$.

Thus, the equation $d = l + b - \text{GCD}(l, b)$ holds true in all cases. And this proves the conjecture!

Concluding discussion

Mathematical puzzles serve as an effective pedagogical hook for stimulating active and collaborative participation in mathematics learning. As [1] notes, “puzzles at first generate bewilderment and confusion, at the same

time that they challenge our wits” (p. 17) ... “People simply cannot ignore a challenge, no matter what the costs are in time and energy”(p. 79).

Proof-based puzzles foster the development of students’ mathematical intuition and also helps sharpen their logical rigour in expressing mathematical observations. Proof-based puzzles thus also provide a window into students’ understanding of the different levels of the mathematics that is involved.

However, for various reasons, the use of proof-based puzzles as a pedagogical practice remains underutilized whether in school classrooms or in mathematical circles. I hope that this two-part article contributes to some change in this regard.

Acknowledgements

I thank Praneta Jha for the discussions that went into developing and presenting the puzzles shared in this two-part article.

Further reading

- [1] Danesi, M. (2004). *The Liar Paradox And The Towers of Hanoi: The Ten Greatest Math Puzzles Of All Time*. New Jersey: John Wiley & Sons, Inc.
- [2] Dorichenko, S. (2012). *A Moscow Math Circle*. Hyderabad: Universities Press.
- [3] Formin, D., S. Genkin, and I. Itenberg (1996). *Mathematical Circles (Russian Experience)*. Hyderabad: Universities Press.

About the author ...

Rossi D'Souza is an independent researcher in mathematics education with a wide range of experience in teaching at the college and university level. He has a PhD in Mathematics Education from Homi Bhabha Centre for Science Education, TIFR. His research interests range from developing mathematical puzzles and exploring their pedagogical potential and philosophical implications to uncovering the political economy of mainstream mathematics education research at the international level. He may be contacted at rossi.dsouza@gmail.com.

Primes in Arithmetic Progression – 2

In the previous issue we asked you to look for an 8-tuple of prime numbers in Arithmetic Progression (AP). Then we asked you to look for a 10-tuple of prime numbers in AP. Were you successful in finding such sequences?

It is easy to find a **6-tuple** of prime numbers in AP. For example:

$$7, 37, 67, 97, 127, 157,$$

which we may describe as $7 + 30k$ for $k = 0, 1, 2, 3, 4, 5$. Please check for yourself that all these numbers are prime. The list cannot be extended further because the next term, 187, is not prime: $187 = 11 \times 17$.

Finding a **7-tuple** of prime numbers in AP is harder. After a great deal of searching we arrive at this example:

$$7, 157, 307, 457, 607, 757, 907,$$

which we may describe as $7 + 150k$ for $k = 0, 1, 2, 3, 4, 5, 6$. Please check for

yourself that all these numbers are prime. The list cannot be extended further because the next term, 1057, is not prime: $1057 = 7 \times 151$.

Finding an **8-tuple** of prime numbers in AP is harder still. It turns out that the smallest such example also yields a **10-tuple** of prime numbers in AP — much to our relief! Here are the numbers:

199, 409, 619, 829, 1039, 1249, 1459, 1669, 1879, 2089.

We may describe this as $199 + 210k$ for $k = 0, 1, 2, 3, 4, 5, 6, 7, 8, 9$. Please check for yourself that all these numbers are prime. The list cannot be extended further because the next term, 2299, is not prime: $2299 = 11^2 \times 19$.

If we start looking for longer sequences of prime numbers in AP, we begin to encounter much larger primes. Say we're looking for an **11-tuple**. It turns out (like earlier) that the smallest such example also yields a **12-tuple** of prime numbers in AP. Here it is:

110437, 124297, 138157, 152017, 165877, 179737,
193597, 207457, 221317, 235177, 249037, 262897.

We may describe this list as $110437 + 13860k$ for $k = 0, 1, 2, \dots, 10, 11$. So the smallest such example involves six-digit primes.

The smallest example of 15 primes in arithmetic progression is this list: $115453391 + 4144140k$ for $k = 0, 1, 2, \dots, 13, 14$. So the smallest such **15-tuple** involves nine-digit primes!

As we look for longer such sequences, the primes grow in size rapidly. You may enjoy looking at references [2], [4] and [5] for more details.

Further reading

- [1] Wikipedia, "List of prime numbers" from https://en.wikipedia.org/wiki/List_of_prime_numbers
- [2] Wikipedia, "Primes in arithmetic progression" from https://en.wikipedia.org/wiki/Primes_in_arithmetic_progression
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- [4] Chris Caldwell, "The PrimePages: prime number research & records" from <https://t5k.org/>
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Pattern Language — an Approach to Teaching Algebraic Expressions – 1

Article contributed by Tanuj Shah

A topic that children in middle school struggle with because of its abstract nature is understanding and simplifying algebraic expressions. There are many reasons for this, including terminology — variables, coefficients, like and unlike terms which do not make sense to them.

Algebraic expressions do not have a closure

Another reason for the difficulty is that *algebraic expressions do not have a closure*. For example, nothing more can be done with something like $5m + 8$. Almost certainly, they would have solved equations before being introduced to algebraic expressions. Therefore, they would have got used

to the idea of a letter representing an unknown number and would struggle with understanding the idea of a variable.

This may also get more confusing if the teacher is not careful in introducing the concept properly. Many teachers try introducing the letters in terms of a “fruit bowl approach.” $5a + 6b$ would be taught as 5 apples + 6 bananas which can not be combined together and altogether missing the point that $5a$ represents 5 multiplied by some number which is being represented by the letter a .

The late P.K Srinivasan, who devoted his life to making school mathematics interesting and aesthetically pleasing, had a very novel way of approaching algebraic expressions through pattern language. He would start by writing some arithmetic statements and ask questions like what stays the same, and what is changing.

$$3 + 7$$

$$3 + 5$$

$$3 + 9$$

$$3 + 12$$

Students will observe that the 3 does not change but the number being added to it is changing. So now the teacher can introduce the idea of something not changing as being a *constant*, and the thing that is changing as a *variable*. The variable can be expressed using a letter of the alphabet. Thus, the above can be stated as $3 + d$ which is an algebraic expression.

This can then be extended to multiplication:

Product	In reverse
3×7	7×3
3×5	5×3
3×9	9×3
3×12	12×3

which can be written as $3 \times d$, and the general convention of writing it is $3d$.

The above multiplications can be written in reverse and would yield $d \times 3$, but the convention to write it as $3d$ would be introduced to the students.

The complexity of the expressions could slowly increase through these patterns by introducing more variables and a mix of operations within the same expression:

$$2 \times 3 + 7 + 6 \times 5,$$

$$2 \times 5 + 8 + 9 \times 5,$$

$$2 \times 7 + 13 + 12 \times 5,$$

and so on; these give the expression $2a + b + 5c$.

This can be followed by expressions that require simplification:

$$7 + 7 \times 5 = 7 \times 6,$$

$$9 + 9 \times 5 = 9 \times 6,$$

$$3 + 3 \times 5 = 3 \times 6,$$

which is written as $a + 5a$ and would simplify to $6a$.

Second and third degree expressions could be introduced in a similar way:

$$7 \times 7 + 2 \times 7$$

$$6 \times 6 + 2 \times 6$$

$$9 \times 9 + 2 \times 9$$

which would be written as $c^2 + 2c$.

Students could then be asked to generate their own set of arithmetic operations when given an algebraic expression. For example, $3d - 4b + a^2$.

Questions which involve collecting like terms and simplifying can be shown as the distributive law.

$$2 \times 9 + 8 + 5 \times 9$$

$$2 \times 3 + 6 + 5 \times 3$$

$$2 \times 4 + 9 + 5 \times 4$$

This can be shown as $2a + b + 5a$ which simplifies to $7a + b$.

The students through this approach would be well equipped to understand the concept of like and unlike terms and what simplification entails.

In the next issue, we will look at how the area model can be used to introduce algebraic expressions.

About the author

Tanuj Shah did his teacher training at Nottingham University and taught in various schools in England before joining Rishi Valley School. He taught Mathematics in Rishi Valley School for over two decades, where he developed hands-on self learning-modules for the Junior and middle School. He has a deep passion for making mathematics accessible and interesting for all. He may be contacted at shahtanuj@hotmail.com

Did You Know ... ? – 1

Primes less than 100

Did you know that there are exactly 25 primes less than 100? (The smallest of these is, of course, 2, and the largest is 97.) This means that if we place 100 pieces of paper in a jar numbered from 1 to 100, mix up the pieces thoroughly and draw a piece out at random, the probability that it bears a prime number is exactly $\frac{1}{4}$.

Perhaps this fact could be made use of in sports tournaments?

Primes less than 1000

Similarly, there are 168 primes less than 1000. (The last few of these are 977, 983, 991, 997.)

Do you notice that the fraction $\frac{168}{1000}$ is very close to $\frac{1}{6}$? So, picking a number at random from the set $\{1, 2, 3, \dots, 999, 1000\}$ and checking whether it is prime is very much like rolling a fair die and checking whether we get 6.

Perhaps this fact too could be made use of in sports tournaments?

Perhaps the reader will find some nice uses of the following! (a) The number of primes less than 10^4 is 1229. (b) The number of primes less than 10^5 is 9592. (c) The number of primes less than 10^6 is 78498.

The year before us, 2025

The year that we've just entered is a special year — it is a perfect square:

$$2025 = 45^2.$$

The number can be written in many ways: $2025 = 27^2 + 36^2$, and also:

- $2025 = (1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9)^2$
- $2025 = 1^3 + 2^3 + 3^3 + 4^3 + 5^3 + 6^3 + 7^3 + 8^3 + 9^3$
- $2025 = (20 + 25)^2$
- As a sum of consecutive positive integers in fourteen different ways!
E.g., $1012 + 1013$, $674 + 675 + 676$, $403 + 404 + 405 + 406 + 407$,
 $335 + 336 + 337 + 338 + 339 + 340$, and there are ten other ways; see
if you can find them all!

Looking into the past, the perfect square year closest to us is $1936 = 44^2$, the year of the infamous 1936 Berlin Olympics, made famous by the memorable feats of Jesse Owens. Authoritarians were calling the shots across Europe that year.

After 2025 the next perfect square year is $2116 = 46^2$. Oh dear, that's a long way off; 91 years to go! What's that year going to be remembered for? And who among us will be around to say "Happy New Year!" that year?

The Global Hunt for Mega Primes

Adapted from: “New Prime Number, 41 Million Digits Long, Breaks Math Records,” <https://www.scientificamerican.com/article/new-prime-number-41-million-digits-long-breaks-math-records/>. Author: Jack Murtagh

Prime numbers, divisible only by 1 and themselves, are central to mathematics and encryption. Enthusiasts and researchers around the world use vast computing power to discover ever-larger primes.

Recently, Luke Durant from San Jose broke a six-year record with a prime of 41,024,320 digits. This new prime is far beyond human comprehension, with digits vastly exceeding even the number of atoms in the universe.

Durant’s achievement was facilitated by the Great Internet Mersenne Prime Search (GIMPS), a collaborative project using advanced software and distributed computing power. By creating a “cloud supercomputer” across

17 countries, Durant ended a tradition of personal computers discovering large primes.

Primes are foundational in mathematics, as every number greater than 1 can be expressed uniquely as a product of primes. The fascination with primes dates back to the ancient Greeks, with Euclid proving their infinite existence in 300 BCE. As numbers grow, finding primes becomes increasingly challenging, necessitating sophisticated algorithms and computing methods.

For instance, to determine if a number like 99,400,891 is prime, one would in principle have to test all smaller numbers for divisibility. However, optimizations allow checking only prime numbers up to the square root of the target, drastically reducing the workload.

Durant's prime is a **Mersenne prime**, a special form defined by the form $2^n - 1$, named after the 17th-century theologian Marin Mersenne. The first few primes of this family are $3 = 2^2 - 1$, $7 = 2^3 - 1$, $31 = 2^5 - 1$, $127 = 2^7 - 1$ and $8191 = 2^{13} - 1$. It is easy to show that if n is composite, then $2^n - 1$ is composite as well. Indeed, if $n = ab$ where $a > 1$ and $b > 1$, then both $2^a - 1$ and $2^b - 1$ are divisors of $2^n - 1$. But it is not true that if p is prime, then $2^p - 1$ is prime! (For example, $2^{11} - 1$ is not prime; we have $2^{11} - 1 = 23 \times 89$.) Identifying primes of this family thus becomes a non-trivial task. But fortunately for us, such primes are easier to identify using a test known as the *Lucas-Lehmer primality test*, which is faster than general methods for other large numbers.

GIMPS, launched in 1996, enables volunteers to search for Mersenne primes using this efficient test. This crowdsourced approach has successfully identified the seven largest known primes. Although smaller unknown primes certainly exist (and in very large numbers), their primal-

ity status is harder to verify as they are typically not of any special form, and this leads to less efficient testing methods.

Durant, a former Nvidia engineer, leveraged GPUs, originally designed for graphics but now used for tasks requiring parallel computation, to search for primes. His global supercomputer processed twelve times more data than all other computers in the Mersenne search combined.

Alongside hardware advancements, the Mersenne search software also received upgrades. A new “probable prime” test, faster than the Lucas-Lehmer test, identifies likely primes. Confirmed by additional testing, Durant’s prime, $2^{136,279,841} - 1$, became the largest known prime, exceeding the previous record by 16 million digits.

Durant’s discovery also led to the identification of the largest known “perfect number,” linked to Mersenne primes. Perfect numbers equal the sum of their divisors, excluding themselves. For example, 6 and 28 are perfect numbers. Euler showed that all even perfect numbers arise from Mersenne primes. For example, $6 = 2 \times (2^2 - 1)$ and $28 = 2^2 \times (2^3 - 1)$. Here, both $2^2 - 1$ and $2^3 - 1$ are Mersenne primes. The next Mersenne prime is $2^5 - 1$, and this leads to the perfect number $2^4 \times (2^5 - 1) = 496$. Euler showed that there is an exact one-to-one correspondence between even perfect numbers and Mersenne primes. The existence of odd perfect numbers remains one of the oldest unsolved mathematical problems. We do not know if even one such number exists!

The search for Mersenne primes could stall anytime, as it’s unknown if an infinite number of them exist.

Despite the high cost of Durant’s project, estimated under \$2 million, he plans to donate the \$3,000 prize to his Alma Mater. The pursuit of primes,

with no immediate practical application, reflects human curiosity and computational progress. As George Woltman, founder of GIMPS, stated, the hunt for primes is ultimately about fun.

Further reading

- [1] Wikipedia. "Mersenne prime." [Internet]. From: https://en.wikipedia.org/w/index.php?title=Mersenne_prime
- [2] Weisstein, Eric W. "Mersenne Prime." From MathWorld—A Wolfram Web Resource.
<https://mathworld.wolfram.com/MersennePrime.html>

Why India's latest Sun mission finding is crucial for the world

Adapted from "Why India's latest Sun mission finding is crucial for the world." <https://www.bbc.com/news/articles/c0qdy5dg7v7o>. Author: Geeta Pandey

India's Aditya-L1 Mission Scores First Win: Early Warning for Solar Storms

India's first solar observation mission, Aditya-L1, has achieved its first major milestone, providing valuable insights into solar phenomena that could help safeguard Earth's infrastructure from solar storms. The mission's primary instrument, the Visible Emission Line Coronagraph (Velc), collected

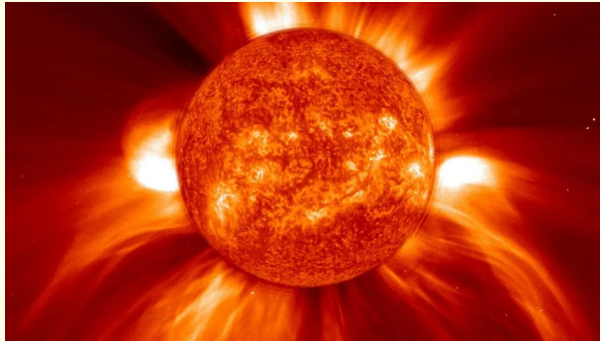


Figure 22: The Sun is the largest object in our solar system and essential to our survival

crucial data on 16 July 2023, allowing scientists to estimate the exact time a coronal mass ejection (CME) began. This breakthrough emphasizes the mission's potential to predict solar events and mitigate their impact on power grids, satellites, and communication systems.

CMEs: A Threat to Earth's Infrastructure

A CME is a massive burst of energy particles ejected from the Sun's outermost layer, the corona. These fireballs can weigh up to a trillion kilograms and travel at speeds of up to 3,000 km/s, potentially reaching Earth in just 15 hours. Such events can disrupt Earth's magnetic field, causing geomagnetic storms with diverse effects, from the creation of auroras to widespread infrastructure damage. The CME observed by Aditya-L1 on 16 July started at 13:08 GMT but was deflected away from Earth within 30 minutes, averting any significant impact.

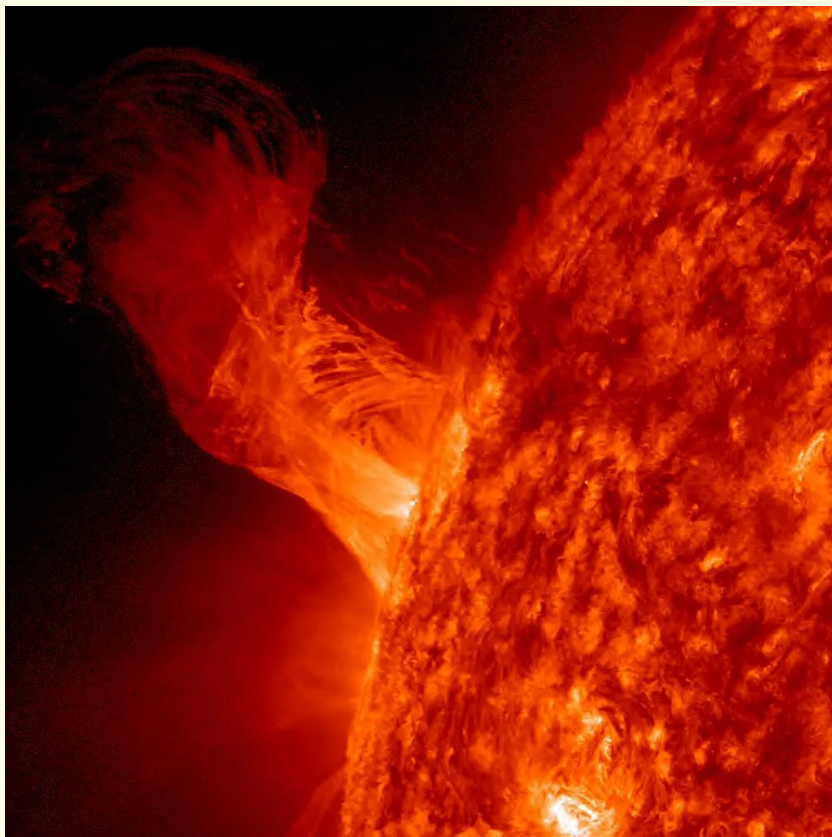


Figure 23: A coronal mass ejection can be several times larger than Earth

Solar activity, including CMEs, routinely affects both Earth and space weather. While these events rarely threaten human life directly, they can interfere with satellite electronics, disrupt power grids, and impair communication systems such as the internet, phone lines, and radio networks.

For instance, in 1989, a CME caused a nine-hour power outage in Quebec, affecting six million people. Similarly, in 2012, a powerful solar storm narrowly missed Earth, striking NASA's solar observatory instead. Such incidents underscore the importance of real-time solar monitoring to protect critical infrastructure.

Aditya-L1's Advantage: Uninterrupted Corona View

India's Aditya-L1, launched in early 2023, joins an elite group of space missions from agencies like NASA, the European Space Agency (ESA), Japan, and China that have been observing the Sun for decades. Positioned in space, Aditya-L1 has a unique advantage: it can continuously monitor the Sun, even during eclipses and occultations. Its coronagraph, Velc, plays a pivotal role by mimicking the Moon's function during a total solar eclipse. It artificially blocks the Sun's photosphere, providing an uninterrupted view of the corona. This capability surpasses the coronagraph on NASA-ESA's Solar and Heliospheric Observatory, which is larger and obscures parts of the corona along with the photosphere, limiting its ability to detect the genesis of CMEs.

Early Warning System for Future Threats

By precisely estimating the origins and trajectories of CMEs, Velc enhances our ability to predict their impact. This information could serve as an early warning system, allowing authorities to temporarily switch off power grids or satellites to prevent damage. Such foresight is critical in a world increasingly reliant on satellite-based communication and infrastructure.

India Joins the League of Solar Observers

India's solar research capabilities are further bolstered by three ground-based observatories in Kodaikanal, Gauribidanur, and Udaipur, which complement the findings from Aditya-L1. Together, these facilities aim to improve our understanding of solar dynamics and their implications for Earth.

The Sun's corona, visible only during total solar eclipses from Earth, is a focus of global scientific study due to its role in solar storms and CMEs. Aditya-L1's uninterrupted observations of the corona provide unprecedented insights, enabling scientists to better understand and predict solar phenomena. The mission's success not only marks a significant achievement for India's space program but also contributes to global efforts to mitigate the risks posed by solar activity.

Exoplanet plate tectonics and the hunt for alien life

Adapted from “Exoplanet plate tectonics: A new frontier in the hunt for alien life,” <https://www.newscientist.com/article/mg26435201-700-exoplanet-plate-tectonics-a-new-frontier-in-the-hunt-for-alien-life/>. Author: Elise Cutts

The Search for Earth-like Tectonics

“There is something strange about Earth. A few billion years ago, a process started here that we have never seen anywhere else. It completely reshaped the planet’s surface and its carbon cycle, sculpted new landscapes and has kept our home temperate and habitable for billions of years.” Thus begins this beautifully written article from the New Scientist.

Earth is unique among known planets due to its plate tectonics, a geologic

process that has reshaped the planet's surface, regulated its climate, and contributed to its habitability for billions of years. Plate tectonics involves the movement of Earth's outer rocky shell, where tectonic plates drift, collide, and recycle material back into the mantle. This recycling process plays a critical role in maintaining Earth's "natural thermostat," regulating carbon dioxide (CO₂) levels through the carbonate-silicate cycle, which helps stabilize the planet's temperature. This process, along with the creation of new land and mountain ranges, provides essential nutrients that sustain life in oceans and on land. Without plate tectonics, Earth's climate could have become uninhabitable, and water might not have remained in liquid form.

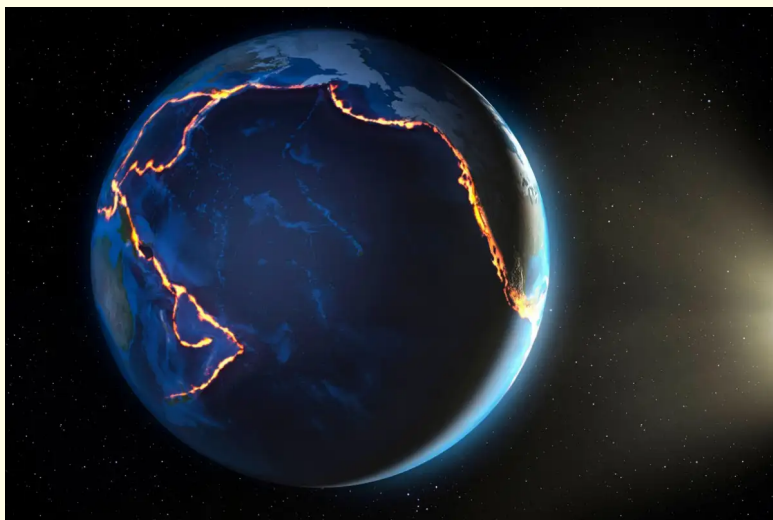


Figure 24: The Earth's tectonic plate system

Although plate tectonics is central to Earth's habitability, scientists are

unsure if it is a universal prerequisite for life. Earth is the only planet in the solar system exhibiting this process, and no evidence of it has been observed beyond our solar system. However, recent advancements, particularly with the James Webb Space Telescope (JWST), have enabled scientists to begin exploring the geology of exoplanets—planets outside our solar system. The study of exoplanet geology could help us understand the origins of plate tectonics on Earth and where to search for habitable worlds in the cosmos.

Plate Tectonics and Its Role in Earth's Habitability

On Earth, plate tectonics acts as a combined life-support system and thermostat. It continuously reshapes the planet's crust by forming new land through volcanic eruptions and recycling old crust into the mantle. This process drives the carbonate-silicate cycle, which regulates CO₂ levels and stabilizes Earth's climate over millions of years. This cycle has protected Earth from extreme climate events like "Snowball Earth" episodes (periods of global freezing) and hothouse periods (extreme warming). If human-induced climate change were to cause a catastrophe, this process might allow Earth to recover over millions of years.

Plate tectonics also ensures the delivery of nutrients to oceans and continents. Erosion of mountain ranges, created by tectonic processes, supplies nutrients vital for oceanic and terrestrial life. Without tectonics, land would flatten over time, and nutrient flow to oceans could slow, threatening ecosystems. Additionally, plate tectonics may have been crucial for the persistence of liquid water on Earth, further emphasizing its role in maintaining habitability.

Searching for Plate Tectonics on Exoplanets

The search for plate tectonics on exoplanets is challenging but crucial for understanding planetary habitability and the possibility of life beyond Earth. Although direct detection of plate tectonics on exoplanets is currently impossible, indirect clues, such as patterns of volcanism, surface composition, and atmospheric signatures, could provide evidence.

LHS 3844 b: A Potential Candidate

One promising exoplanet for studying tectonic processes is LHS 3844 b, a rocky planet located 49 light-years away. Discovered in 2018, it is slightly larger than Earth and orbits its star in just 12 hours. The planet is tidally locked, meaning one side always faces its star while the other remains in perpetual darkness, resulting in extreme temperature differences between the two hemispheres. Modeling suggests LHS 3844 b could exhibit “hemispheric tectonics,” where hot material rises on the day side, and cold material sinks on the night side. This form of tectonics, while different from Earth’s, could still involve a mobile crust broken into plates.

LHS 3844 b represents a breakthrough in exoplanet geology because it allows researchers to study an alien tectonic regime. While hemispheric tectonics would be less uniform than Earth’s, it could still qualify as a tectonic process, unlike the stagnant crusts of Venus, Mercury, or Mars.

Volcanism as a Clue

Volcanism, a potential by-product of tectonics, offers another avenue for studying exoplanet geology. On Earth, volcanic activity is closely tied to plate boundaries. Similarly, patterns of volcanism on exoplanets could hint

at tectonic processes. For instance, if one hemisphere of LHS 3844 b is more volcanically active than the other, it could support the idea of hemispheric tectonics.



Figure 25: Volcanic activity on the Earth has been going on for a long

However, detecting volcanic activity on exoplanets is challenging due to their distance and dimness. Scientists use indirect methods, such as analyzing light emitted by the planet or gases in its atmosphere. Volcanic activity produces heat and releases gases like sulphur dioxide, which could leave detectable signatures in an exoplanet's atmosphere. For example, researchers are studying the exoplanet LP 791-18 d, which may have a CO₂-rich atmosphere fuelled by volcanic activity. This planet, located 87 light-years away, experiences intense tidal heating from gravitational interactions with a neighbouring planet, driving volcanism similar to Jupiter's moon Io, the most volcanically active object in our solar system.

Atmospheric Signatures of Volcanism

Detecting volcanic gases in an exoplanet's atmosphere could help identify tectonic activity. For example, sulphur dioxide is a strong indicator of recent volcanic eruptions, though its signal can be masked by other atmospheric gases like ozone. Researchers are also exploring the possibility of detecting volcanic hazes or fluctuations in atmospheric composition over time, which could indirectly point to active volcanism.

Surface Composition and Tectonics

When exoplanets lack atmospheres, their exposed surfaces provide another way to study tectonic activity. The thermal glow from an airless planet's surface can reveal its rock composition and texture. For instance, basalt, a volcanic rock, is common on Earth's surface and could indicate recent volcanic activity. Granite, which forms through repeated recycling of crust, could suggest plate tectonics. On LHS 3844 b, scientists are analyzing whether the surface is smooth (indicating recent volcanic activity) or covered in rubble-like regolith (indicating an old, inactive surface). These studies mark the beginning of exoplanet geology, allowing researchers to probe the deep interiors of alien worlds.

The Future of Exoplanet Geology

The study of exoplanet geology is still in its infancy, but new tools and techniques are opening up possibilities for understanding the geologic processes of distant worlds. Future telescopes, such as NASA's proposed Habitable Worlds Observatory, could detect atmospheric signatures of volcanic and tectonic activity on Earth-like exoplanets. These advancements

could help determine whether plate tectonics is a universal phenomenon or unique to Earth.

Understanding plate tectonics on exoplanets could also shed light on Earth's own geologic history. Despite decades of research, the origins of plate tectonics on Earth remain a mystery. Studying other planets might reveal whether tectonics is common or rare in planetary systems, and whether it is essential for the development of life.

The Importance of Plate Tectonics in the Search for Life

Plate tectonics appears to be a crucial component of Earth's habitability, but its role in supporting life elsewhere remains uncertain. Finding another planet with plate tectonics could provide insights into the conditions necessary for life and the processes that sustain it. However, detecting plate tectonics on exoplanets is as difficult as finding signs of alien life, if not harder, due to the subtle and ambiguous nature of geologic signatures.

The search for plate tectonics on exoplanets is a long-term endeavour that will require future generations of telescopes and researchers. While the task is daunting, it is essential for understanding why Earth is uniquely habitable and whether similar conditions exist elsewhere in the universe. By studying exoplanets, scientists hope to unravel the mysteries of Earth's geology and the origins of life itself.

Astronauts Could Hitch a Ride on Asteroids to Get to Venus or Mars

Adapted from “Astronauts could hitch a ride on asteroids to get to Venus or Mars,” <https://www.newscientist.com/article/2453853-astronauts-could-hitch-a-ride-on-asteroids-to-get-to-venus-or-mars/>. Author: Jonathan O’Callaghan

Asteroids that regularly fly between Earth, Venus and Mars could provide radiation shielding for human missions to explore neighbouring planets.

Researchers have proposed an innovative idea for interplanetary travel: using asteroids as “space trains” to transport astronauts to Mars or Venus while providing protection from harmful radiation. Space radiation, including solar and galactic cosmic rays, poses a significant risk to astronauts



Figure 26: Asteroids that pass close to Earth could carry a spaceship to other worlds

on long missions beyond Earth's protective magnetic field. Prolonged exposure increases the likelihood of health issues like cancer, which makes shielding a critical concern for deep-space journeys.

While conventional solutions involve building spacecraft with heavy radiation shielding, Arsenii Kasianchuk and Volodymyr Reshetnyk from Ukraine's Taras Shevchenko National University suggest a more cost-effective alternative: travelling inside asteroids. These naturally occurring space rocks could offer sufficient protection against radiation without the need for extensive shielding materials. The researchers analyzed 35,000 near-Earth asteroids and identified 120 that met the criteria for interplanetary missions between now and 2120. These asteroids pass close enough to

Earth, Mars, or Venus to allow spacecraft to land on and depart from them, and they are large enough to house a spacecraft. The concept involves astronauts hitching a ride on one asteroid to a destination, such as Earth to Mars, and then waiting for another asteroid travelling in the opposite direction for the return trip. Kasianchuk compares the system to a train network, with journey times to Mars or Venus taking up to 180 days. For instance, in 2079, one asteroid will travel from Earth to Venus to Mars in 230 days, and another will return the following year, enabling a potential dual-planet mission.

However, there are significant technical challenges to this approach. Matching the high velocity of a passing asteroid—potentially as fast as 30 kilometres per second (67,000 mph)—requires an enormous amount of fuel, though some asteroids travel more slowly at around 2 kilometres per second (4,500 mph). Additionally, creating a safe space within the asteroid for astronauts to shelter is no small feat. Colin Snodgrass from the University of Edinburgh highlights the difficulty of landing on an asteroid and excavating a tunnel into its surface. Kasianchuk suggests using an uncrewed device to drill into the asteroid in advance, creating a cylindrical tunnel for the spacecraft to enter.

Despite these obstacles, Kasianchuk remains optimistic, noting that more near-Earth asteroids are discovered every year, increasing the pool of candidates for future interplanetary travel. This concept could pave the way for safer and more cost-effective space exploration.

The Starry Night

Adapted from “Beneath the brushstrokes, van Gogh’s sky is alive with real-world physics.” https://phys.org/news/2024-09-beneath-brushstrokes-van-gogh-sky.html?utm_source=nwletter&utm_medium=email&utm_campaign=daily-nwletter#google_vignette. Author: American Institute of Physics

Beneath the brushstrokes, van Gogh’s sky is alive with real-world physics

Researchers analyzed Vincent van Gogh’s iconic painting *The Starry Night* to explore its depiction of atmospheric dynamics and turbulence. By studying the swirling brushstrokes and variations in luminance, they sought to determine whether the painting aligns with physical laws governing real atmospheric motion. Their findings, published in *Physics of Fluids*, suggest that van Gogh intuitively captured complex atmospheric phenomena with remarkable accuracy.

The study, conducted by scientists specializing in fluid dynamics, used

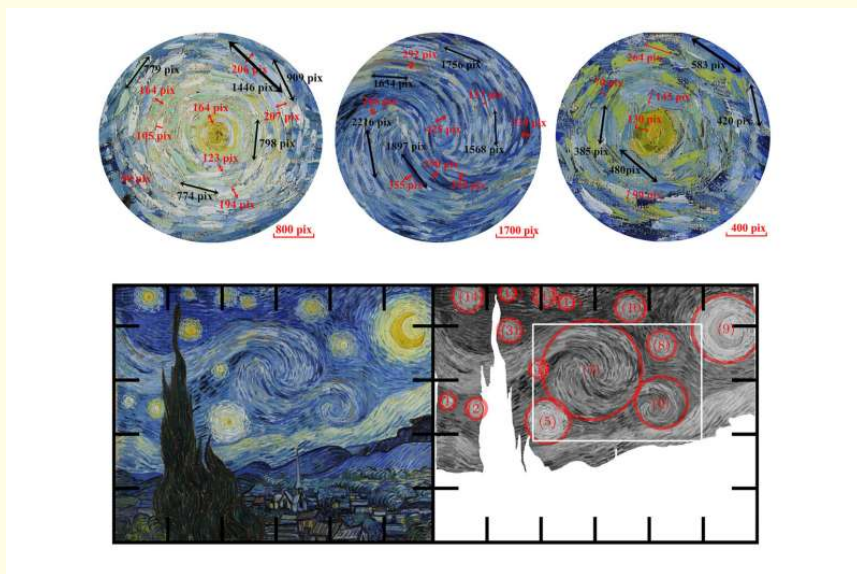


Figure 27: Study of Vincent van Gogh's "The Starry Night"

high-resolution digital images of the painting to measure the scale and spacing of van Gogh's brushstrokes. These measurements were compared to turbulence theories, revealing striking parallels between the artwork and natural atmospheric behaviour. The researchers treated the luminance of the paint as a proxy for kinetic energy, enabling them to analyze the dynamics and energy transfer within the depicted sky.

Key findings of the research include the discovery that van Gogh's brushstrokes align with Kolmogorov's law, which describes how energy cascades from large to small scales in turbulent flows. Additionally, the smaller-scale brushwork exhibited characteristics of Batchelor's scaling, which

governs passive scalar turbulence in atmospheric systems. This dual alignment of large- and small-scale turbulence is rare, making the painting an exceptional representation of natural phenomena.



Figure 28: Vincent van Gogh's "The Starry Night." Google Arts & Culture, Public Domain, <https://commons.wikimedia.org/w/index.php?curid=25498286>

The study highlights van Gogh's possible innate understanding of atmospheric motion or his keen observation of natural patterns, such as the

movement of clouds. The researchers suggest that turbulence, traditionally associated with high Reynolds number flows dominated by inertia, might need a broader definition to encompass phenomena observed in low Reynolds number systems, like those van Gogh appears to have captured.

The research underscores the depth of van Gogh's artistic vision, portraying his depiction of the sky not just as creative expression but as a potential reflection of real-world atmospheric physics.

The Rebellious Maths Teacher

Adapted from “The Rebellious Maths Teacher.” <https://www.thehindu.com/opinion/open-page/the-rebellious-maths-teacher/article68825889.ece>. Author: Ko Sesha

P.K. Srinivasan took his ideals across the world, from Indian village classrooms to American lecture halls and African colleges, embodying a lifelong rebellion against a system that suppressed creativity

P.K. Srinivasan (1924–2005), a celebrated Indian mathematics educator, dedicated his life to transforming the way mathematics was taught and perceived in schools. Born in Madras Province, he displayed an early aptitude for teaching, tutoring classmates even as a student. After earning a mathematics degree from Loyola College in 1947, Srinivasan, inspired by Mahatma Gandhi’s call to “Go Back to the villages,” rejected bureaucratic aspirations to work as a village teacher. At just 23, he moved to rural Lakshmi Naickenpalayam (a small village in Coimbatore district) with minimal possessions: khadi clothes and a trunk of books. Over the next five years, he introduced innovative, hands-on teaching methods to rural

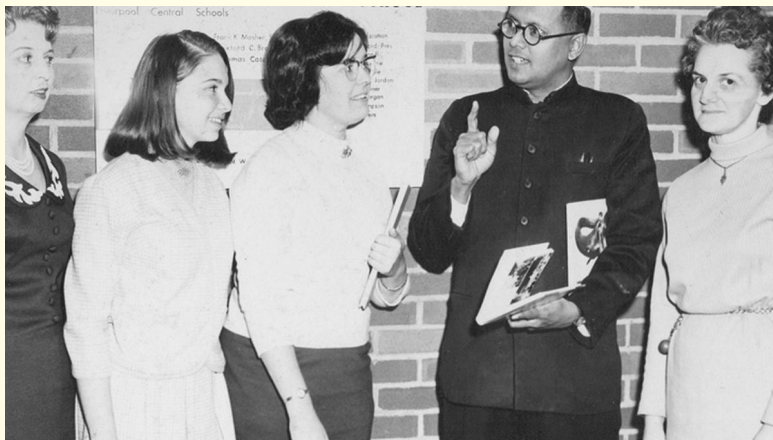


Figure 29: P.K. Srinivasan in New York with American maths teachers

students, aiming to eliminate their fear of mathematics. He also organized small math exhibitions and served as the first headmaster of Gandhi Kala Nilayam School before returning to Madras in 1952.

Back in Madras, Srinivasan joined Muthialpet High School, where he taught mathematics to students from marginalized communities, often visiting their homes to ensure they stayed in school. He firmly believed in the potential of every student, challenging colonial rote-learning methods and advocating for creativity and critical thinking. Among his students was R.G. Chandramogan, founder of Hatsun Foods, who credited Srinivasan's teachings for inspiring him to become a leader and entrepreneur.

Srinivasan was deeply inspired by the life of mathematical genius Srinivasa Ramanujan. In 1962, when the Indian government commemorated Ramanujan's 75th birth anniversary with a postal stamp, Srinivasan led ef-

forts to honour the mathematician. He organized a procession of students and teachers to purchase first-day covers and worked to unearth personal memorabilia, portraying Ramanujan as a role model for aspiring mathematicians. This tribute gained international recognition, being featured in Robert Kanigel's book *The Man Who Knew Infinity*.

In 1965, Srinivasan received a Fulbright Fellowship to the U.S., where his innovative teaching methods gained acclaim. He addressed the National Council of Teachers in New York, presenting Ramanujan as a "model student." Despite offers of a permanent job and relocation, Srinivasan chose to return to India, believing his country needed him more. Back home, he authored the first biography of Ramanujan, which was released by statesman C. Rajagopalachari.

Srinivasan later worked in Nigeria for seven years, training teachers in his capacity as Senior Federal Education Officer. Returning to India, he founded the Ramanujan Museum in Royapuram in 1993 and spearheaded initiatives like "Math Expos" and "Math Labs" to make math accessible to all. By the end of his life, Srinivasan had authored 30 books and developed numerous low-cost teaching aids. He passed away on June 20, 2005. His centenary, celebrated in 2024, highlighted his enduring legacy, with calls to designate his birthday, November 4, as "Maths Teachers' Day."

Ko Sessa is a Tamil film lyricist and screenwriter, and the grandson of PK Srinivasan.

Danger! — Counting Going on Ahead! – 1

It is well known that the topic of permutations and combinations is found to be tricky by large numbers of students. The reasons for this are varied and complex. We will devote a later article in this series to a discussion of these issues.

A particularly common kind of error is that of over-counting or under-counting. The former typically arises when we do not see that some configurations we consider to be different are actually the same. The latter typically arises when we miss some possibilities altogether. We discuss two examples of this. Example 1 is from [1].

Example 1

I have a class of 24 students, of which 16 are men and 8 are women. I want to choose a team of 8 that has 4 men and 3 women; the eighth member can be anyone (either man or woman). In how many ways can I choose the team?

One answer that comes forth from students is this:

$$\text{Number of ways} = \binom{16}{4} \times \binom{8}{3} \times \binom{17}{1}.$$

Why this answer?

- Stage I, selecting the men: $\binom{16}{4}$ represents the number of ways of choosing 4 men out of 16;
- Stage II, selecting the women: $\binom{8}{3}$ represents the number of ways of choosing 3 women out of 8; and
- Stage III, selecting the remaining member: $\binom{17}{1}$ represents the number of ways of choosing the remaining one member from 17 potential members.

The above product is equal to

$$1820 \times 56 \times 17 = 1732640.$$

But this answer cannot possibly be correct, surely, since the total number of ways of selecting 8 people from 24 people, with no constraints of any kind, is

$$\binom{24}{8} = 735471.$$

The desired answer must be less than this number. But we have obtained a figure that is more than twice this number!

Where is the error in reasoning? And what is the correct answer?

It is not difficult to spot the error.

Label the women as A, B, C, D, E, F, G, H (eight in all). Suppose that in stage II, when we are choosing 3 women out of 8, we happen to choose A, B, C ,

and in stage III, when we are selecting the last member from 17 people (keep in mind that there are both men and women left in this subset), we happen to choose woman D . So the women in our selected group are A, B, C, D . The same subset gets selected if in stage II we happen to choose women B, C, D , and in stage III we happen to choose woman A . *But these two selections are being counted separately in our scheme of reasoning.* It is clear, therefore, that there is a great deal of over-counting going on. No wonder we got an answer that is more than twice the largest possible figure!

The correct approach involves addition as well as multiplication. We must consider two possibilities separately: we select 4 men and 4 women; or we select 5 men and 3 women. The correct answer is therefore

$$\begin{aligned} \binom{16}{4} \times \binom{8}{4} + \binom{16}{5} \times \binom{8}{3} &= (1820 \times 70) + (4368 \times 56) \\ &= 127400 + 244608 = 372008. \end{aligned}$$

Example 2

Using the digits 0, 1, 2, 4, 5, 6 no more than once each, how many four-digit numbers can we make that are multiples of 5?

A student might answer as follows. The units digit of the number must be 0 or 5 for the number to be a multiple of 5; hence 2 choices. The remaining three digits can be chosen in any manner from the remaining five digits; hence $5 \times 4 \times 3 = 60$ choices. Therefore the required answer is $2 \times 60 = 120$.

But this answer is clearly wrong. Though there are indeed two choices for the units digit, the number of ways of selecting the remaining three digits is going to depend on whether the units digit is 0 or 5. (This is so

because the leading digit of a number cannot be 0.) The correct approach must consider both these possibilities. Here is how we can proceed.

- Suppose the units digit is 0. The remaining three digits can then be freely chosen from the remaining five digits; hence $5 \times 4 \times 3 = 60$ possibilities.
- Suppose the units digit is 5. The leading digit of the number can be freely chosen from the digit set $\{1, 2, 4, 6\}$; hence 4 choices. (The digit 0 is not available because the leading digit of a number cannot be 0.) The remaining two digits of the number can be freely chosen from the remaining four digits; hence $4 \times 3 = 12$ choices. So the number of possibilities in this case is $4 \times 12 = 48$.

The required answer is therefore $60 + 48 = 108$.

We see that the wrong answer is in excess by 12. What exactly is this number (i.e., 12) counting? We leave this for you to work out!

Further reading

[1] Math Stack Exchange, “Why/how am I over counting here?” from <https://math.stackexchange.com/questions/4984783/why-how-am-i-over-counting-here>

Article contributed by Shailesh Shirali

I-MoD Numbers

Article contributed by Hara Gopal R.

In this article we introduce four properties of numbers which we have called **I-MoD**, **Di-MoD**, **C-MoD** and **E-MoD**. We describe these below and explore how to find numbers with these properties.

We first need to recall some important facts about divisors of numbers. Given any positive integer n , let it be expressed in terms of its prime factorization as

$$n = p^a \times q^b \times r^c \times \dots, \quad (18)$$

where $p, q, r, \dots$ are prime numbers with $p < q < r < \dots$, and $a, b, c, \dots$ are positive integers. Then the number of divisors of n , denoted by $d(n)$, is given by

$$d(n) = (a + 1) \times (b + 1) \times (c + 1) \times \dots. \quad (19)$$

The sum of the divisors of n , denoted by $\sigma(n)$, is given by

$$\sigma(n) = \frac{p^{a+1} - 1}{p - 1} \times \frac{q^{b+1} - 1}{q - 1} \times \frac{r^{c+1} - 1}{r - 1} \times \dots. \quad (20)$$

To illustrate the use of these formulas, consider the number 24. Its divisors are 1, 2, 3, 4, 6, 8, 12 & 24 (8 divisors in all), and their sum is

$$1 + 2 + 3 + 4 + 6 + 8 + 12 + 24 = 60.$$

If we apply the formulas displayed above we get, since $24 = 2^3 \times 3^1$,

$$d(24) = (3 + 1) \times (1 + 1) = 4 \times 2 = 8,$$

which is correct; and

$$\sigma(24) = \frac{2^4 - 1}{2 - 1} \times \frac{3^2 - 1}{3 - 1} = 15 \times 4 = 60,$$

which too is correct.

If we divide the sum of the divisors with the number of divisors, the result will be called “Mean of Divisors” or MoD, for short. We shall denote the Mean of Divisors of a number n as $\text{MoD}(n)$. For example,

$$\text{MoD}(24) = \frac{60}{8} = 7.5.$$

This leads to our first definition.

Definition of an I-MoD number

We define an **I-MoD number** to be “a number whose mean of divisors is an integer.” Here, I-Mod stands for *Integral Mean of Divisors*.

For example, consider the number $14 = 2^1 \times 7^1$. We have:

$$d(14) = (1 + 1) \times (1 + 1) = 2 \times 2 = 4,$$

$$\sigma(14) = \frac{2^2 - 1}{2 - 1} \times \frac{7^2 - 1}{7 - 1} = 3 \times \frac{48}{6} = 3 \times 8 = 24,$$

$$\therefore \text{MoD}(14) = \frac{24}{4} = 6,$$

which is an integer. So 14 has the I-MoD property.

Definition of a Di-MoD number

If an I-MoD number is divisible by its MoD, we call it a **Di-MoD number**. Here, Di-MoD stands for *Divisible by Integral Mean of Divisors*.

For example, consider the number 6. We have:

$$d(6) = 4,$$

$$\sigma(6) = 12,$$

$$\text{MoD}(6) = \frac{12}{4} = 3.$$

As $\text{MoD}(6)$ is also a divisor of 6, we say that 6 is a Di-MoD number.

Comment We note that:

- A Di-MoD number must also be an I-MoD number. But an I-MoD number need not be a Di-MoD number.
- For an odd prime p , we have $d(p) = 2$, $\sigma(p) = p + 1$, which is even, so all odd primes have the I-MoD property.

Tables 1–3 list numbers between 1 and 100 having the I-MoD and Di-MoD properties.

n	$\sigma(n)$	$d(n)$	MoD(n)	Di-Mod number?
6	12	4	3	Yes
14	24	4	6	No
15	24	4	6	No
20	42	6	7	No
21	32	4	8	No
22	36	4	9	No
27	40	10	4	No
30	72	8	9	No
33	48	4	12	No
35	48	4	12	No
38	60	4	15	No
39	56	4	14	No
42	96	8	16	No
44	84	6	14	No
45	78	6	13	No
46	72	4	18	No
49	57	3	19	No

Table 1: Numbers from 1 to 50 having the I-MoD and Di-MoD properties

n	$\sigma(n)$	$d(n)$	MoD(n)	Di-Mod number?
51	72	4	18	No
54	120	8	15	No
55	72	4	18	No
56	120	8	15	No
57	80	4	20	No
60	168	12	14	No
62	96	4	24	No
65	84	4	21	No
66	144	8	18	No
68	126	6	21	No
69	96	4	24	No
70	144	8	18	No
77	96	4	24	No
78	168	8	21	No
85	108	4	27	No
86	132	4	33	No
87	120	4	30	No

Table 2: Numbers from 51 to 90 having the I-MoD and Di-MoD properties

n	$\sigma(n)$	$d(n)$	$\text{MoD}(n)$	Di-Mod number?
91	112	4	28	No
92	168	6	28	No
93	128	4	32	No
94	144	4	36	No
95	120	4	30	No
96	252	12	21	No
99	156	6	26	No

Table 3: Numbers from 91 to 100 having the I-MoD and Di-MoD properties

Very few Di-MoD numbers

If we study the above data, we see that there exist many I-MoD numbers up to 100 but only one Di-MoD number — the number 6.

If we extend the search to larger numbers we find a few more numbers with the Di-MoD property. Here is the list of all such numbers below 10000:

$$6, 140, 270, 672, 1638, 2970, 6200, 8190.$$

Some interesting features of the I-MoD numbers

The data also reveal some interesting properties of I-MoD numbers. For example, consider the consecutive numbers 6 and 7. We find:

$$\text{MoD}(6) = 3, \quad \text{MoD}(7) = 4.$$

So their MoD values are consecutive numbers too.

Similarly consider the consecutive numbers 20,21,22. Their MoD values are, respectively, 7,8,9. Observe that their MoD values are consecutive numbers too.

We shall refer to such number collections as **C-MoD numbers** (short for “Consecutive Mean of Divisors”). Can we find more such collections?

Again, consider the consecutive numbers 5 and 6. We find:

$$\text{MoD}(5) = 3, \quad \text{MoD}(6) = 3.$$

Their MoD values are equal. Consider another pair of consecutive integers 14 and 15. We find:

$$\text{MoD}(14) = 6, \quad \text{MoD}(15) = 6.$$

Their MoD values too are equal. Here we have instances of consecutive numbers with equal MoD values.

We shall call them **E-MoD numbers** (“Equal Mean of Divisors” numbers). Can we find more such number pairs?

We can similarly search for other interesting facts among these numbers.

About the author

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Heron's Formula for Area of a Triangle – 3

We have already encountered two different proofs of Heron's beautiful formula for the area (K) of a triangle with sides a, b, c ,

$$K = \sqrt{s(s-a)(s-b)(s-c)}.$$

The first proof was based on Pythagoras Theorem and was algebraic in nature, and the second proof used basic trigonometry and some circle geometry.

In this issue we shall study a third proof — the one given originally by Heron. It is adapted from the proof given in [2], which in turn is adapted from the proof given in [3]. As we shall see, it is a very beautiful proof — perhaps the nicest of them all!

We also mention Brahmagupta's extension of the formula to cyclic quadrilaterals.

Heron's proof

We work with the construction shown in Figure 30. It shows a triangle ABC and its incircle, centre I , that touches the sides at D, E, F .

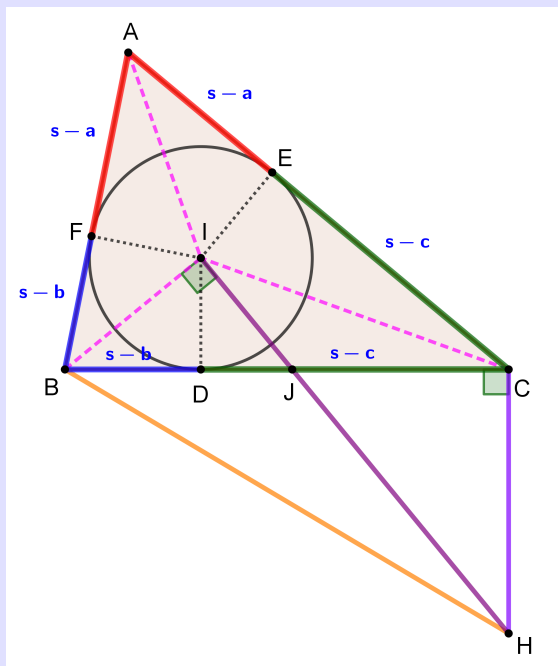


Figure 30: Heron's proof

Draw a line through I perpendicular to BI , and a line through C perpendicular to BC ; let them intersect at H . Let IH intersect BC at J . Let the lengths of the sides of the triangle be a, b, c , and let $s = (a + b + c)/2$ be the semi-perimeter. We now reason as follows.

- $AE = s - a = AF$, $BD = s - b = BF$, $CD = s - c = CE$.
- We take it as known that the area of $\triangle ABC$ is rs , where r is the radius of the incircle.
- $\angle BIC = 90^\circ + A/2$ (well-known). Next, quadrilateral $BICH$ is cyclic, since $\angle BIC = 90^\circ = \angle BCH$. Hence $\angle BHC = 90^\circ - A/2$.
- Hence $\triangle AEI \sim \triangle BCH$ (both have angles $A/2$, $90^\circ - A/2$ and 90°).
Therefore,

$$\frac{IE}{AE} = \frac{HC}{BC}, \quad \therefore HC = \frac{ra}{s-a}.$$

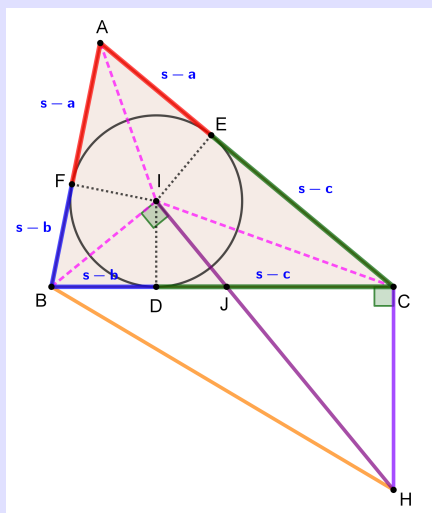


Figure 31: Figure duplicated for convenience

- Again, we have $\triangle IDJ \sim \triangle HCJ$, so

$$\frac{DJ}{CJ} = \frac{ID}{HC}, \quad \therefore \frac{DJ}{CJ} = \frac{r}{ra/(s-a)} = \frac{s-a}{a}.$$

Hence

$$\frac{DJ}{DJ+CJ} = \frac{s-a}{(s-a)+a}, \quad \therefore \frac{DJ}{DC} = \frac{s-a}{a},$$

giving

$$DJ = \frac{(s-a) \cdot (s-c)}{s}.$$

- Next, observe that $\triangle IDJ \sim \triangle BDI$ (to see why, note that ID is an altitude of $\triangle BIJ$ which is right-angled at I). Hence

$$\frac{DJ}{ID} = \frac{DI}{BD}, \quad \therefore DJ = \frac{DI^2}{BD} = \frac{r^2}{s-b}.$$

- From the previous two findings, we obtain the equality

$$\frac{(s-a) \cdot (s-c)}{s} = \frac{r^2}{s-b},$$

which yields $r^2s = (s-a)(s-b)(s-c)$ and hence

$$r^2s^2 = s(s-a)(s-b)(s-c).$$

Heron's formula for the area of the triangle follows. ■

Brahmagupta's formula for area of a cyclic quadrilateral

The formula found by Brahmagupta in the 6th century CE for the area of a cyclic quadrilateral is striking and has a beautiful symmetry about it.

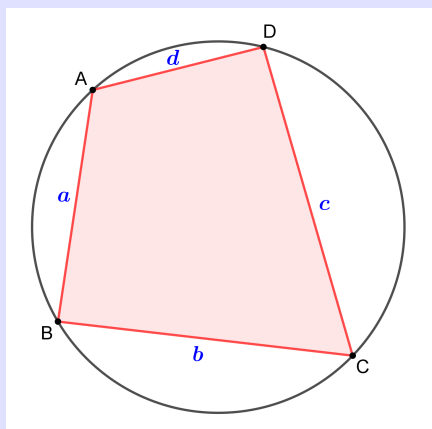


Figure 32

Theorem (Brahmagupta). *A cyclic quadrilateral with sides a, b, c, d and semi-perimeter $s = (a + b + c + d)/2$ has area K given by*

$$K = \sqrt{(s - a)(s - b)(s - c)(s - d)}.$$

It is important to note that *the formula holds only for a cyclic quadrilateral*. Indeed, if we know only the lengths of the sides of a quadrilateral and nothing else, we cannot (in general) say what the area of the quadrilateral will be; there will be infinitely many quadrilaterals having the same lengths for the four sides, and infinitely many possible values for the area. Further information would be needed before we can compute the area.

A striking aspect of Brahmagupta's theorem is that *it yields Heron's formula as a corollary*. For, consider a quadrilateral with sides $a, b, c, 0$. Note that here the fourth side has shrunk to a point (with $d = 0$). In this extreme situation

the quadrilateral has become a triangle! Now a triangle can always be circumscribed by a circle; no additional conditions are needed. So this peculiar ‘quadrilateral’ is cyclic and therefore Brahmagupta’s formula applies. The above formula with $d = 0$ yields

$$K = \sqrt{s(s-a)(s-b)(s-c)},$$

and this is just Heron’s formula.

We will take up the proof of Brahmagupta’s theorem in the closing article of this series (April 2025 issue).

Further reading

[1] Wikipedia, “Heron’s formula.”

https://en.wikipedia.org/wiki/Heron's_formula

[2] Shannon Umberger, “A Geometric Proof of Heron’s Formula.”

<https://jwilson.coe.uga.edu/EMT668/EMAT6680.2000/Umberger/MATH7200/HeronFormulaProject/GeometricProof/geoproof.html>

[3] Howard Eves, “An Introduction to the History of Mathematics” (1969). Holt, Rinehart and Winston. (Chapter 6, page 172)

Article contributed by Shailesh Shirali

Divisibility Properties of Sets of Consecutive Integers

Article contributed by Anand Prakash

We are familiar with the notion of *twin primes* (e.g., $\{5, 7\}$ or $\{11, 13\}$; they are of the form $\{p, p + 2\}$, with p odd) and *prime triplets* (e.g., $\{5, 7, 11\}$ or $\{17, 19, 23\}$; they are of the form $\{p, p + 2, p + 6\}$, with p odd). (Observe that we have not defined a prime triplet to be of the form $\{p, p + 2, p + 4\}$ where all three of $p, p + 2, p + 4$ are prime. Indeed, there is only one triplet of that form; namely, $\{3, 5, 7\}$. Please prove this for yourself! So, defining a prime triplet that way would turn out to be most uninteresting.)

In this article, we shall look for sets of consecutive integers that are divisible by twin primes or prime triplets. That is, we look for values of n (a positive integer greater than 1) and p (an odd prime number) such that:

Twin primes case: n is divisible by p , and $n + 1$ is divisible by $p + 2$;

OR

Prime triplets case: n is divisible by p , $n + 1$ is divisible by $p + 2$, and $n + 2$ is divisible by $p + 6$.

The case of twin primes

Let $\{p, p + 2\}$ be a pair of twin primes, and let $\{n, n + 1\}$ be a pair of consecutive integers such that

$$p(m + k) = n, \quad (p + 2)m = n + 1, \quad (21)$$

where m, k are positive integers. Subtraction yields

$$2m - pk = 1, \quad \therefore m = \frac{pk + 1}{2}. \quad (22)$$

Relationship (22) allows us to find many solutions of the required type. For example, let $p = 11$. (This will do since 11 and 13 are primes.) Then we have

$$m = \frac{11k + 1}{2}.$$

We may now assign any odd value to k and get a solution. E.g.,

- $k = 3$ gives $m = 17$, and we have the pair $\{220, 221\}$; here, 220 is divisible by 11, and 221 is divisible by 13.
- $k = 5$ gives $m = 28$, and we have the pair $\{363, 364\}$; here, 363 is divisible by 11, and 364 is divisible by 13.
- $k = 7$ gives $m = 39$, and we have the pair $\{506, 507\}$; here, 506 is divisible by 11, and 507 is divisible by 13.

We can take other values of p such that $p + 2$ is prime and generate many such consecutive number pairs (indeed, an unlimited number of them).

The case of prime triplets

Let $\{p, p + 2, p + 6\}$ be prime triplet, and let $\{n, n + 1, n + 2\}$ be consecutive integers such that

$$p(m + k + l) = n, \quad (p + 2)(m + k) = n + 1, \quad (p + 6)m = n + 2, \quad (23)$$

where m, k, l are positive integers. Subtraction yields

$$2(m + k) - pl = 1, \quad \therefore 2(m + k) = pl + 1,$$

$$6m - pk - 2(m + k) = 1, \quad \therefore 2(m + k) = 6m - pk - 1.$$

Therefore we have:

$$pl + 1 = 6m - pk - 1, \quad \therefore 6m = pl + pk + 2,$$

giving

$$p(l + k) = 2(3m - 1). \quad (24)$$

We split the analysis here by considering two cases.

Suppose that $3m - 1$ is prime: Then equation (24) implies that $p = 3m - 1$, $l + k = 2$, giving $l = 1 = k$. Since $2(m + k) = pl + 1$, this now leads to

$$2(m + 1) = p + 1, \quad \therefore p = 2m + 1.$$

So $3m - 1 = 2m + 1$, giving $m = 2$, $p = 5$. Hence the three integers are

$$n = 5(2 + 1 + 1) = 20, \quad n + 1 = (5 + 2)(2 + 1) = 21, \quad n + 2 = (5 + 6)2 = 22.$$

This is the only solution if we assume that $3m - 1$ is prime. We may check that 20, 21, 22 are divisible (respectively) by 5, 7, 11.

Suppose that $3m - 1$ is not prime: From $p(l + k) = 2(3m - 1)$ we get

$$l + k = \frac{2(3m - 1)}{p}. \quad (25)$$

We also have

$$6m - pk - 1 = 2(m + k), \quad \therefore 4m - 2k - 1 = pk,$$

giving

$$k = \frac{4m - 2k - 1}{p}. \quad (26)$$

From (26) and (25), we get, by subtraction,

$$l = \frac{2(3m - 1)}{p} - \frac{4m - 2k - 1}{p} = \frac{2m + 2k - 1}{p}. \quad (27)$$

We also have, from (26),

$$k = \frac{4m - 1}{p + 2}. \quad (28)$$

Let's see if $p = 5$ yields a solution. We should try this because $\{5, 7, 11\}$ is indeed a prime triplet. We get, from (27)

$$2m + 2k - 1 = 5l,$$

$$\therefore 2m + \frac{8m - 2}{5 + 2} = 5l + 1,$$

$$\therefore 22m - 2 = 35l + 7,$$

giving

$$m = \frac{35l + 9}{22}, \quad l = \frac{22m - 9}{35}.$$

By looking for values of l such that $35l + 9$ is a multiple of 22, we find values of m, k .

In this manner, we can find solutions to the stated problem. Here are a few solutions corresponding to the prime triplet $\{5, 7, 11\}$:

- $(n, n + 1, n + 2) = (405, 406, 407)$, obtained from $m = 37, k = 21, l = 23$; note that 405, 406, 407 are divisible by 5, 7, 11 respectively.
- $(n, n + 1, n + 2) = (790, 791, 792)$, obtained from $m = 72, k = 41, l = 45$; note that 790, 791, 792 are divisible by 5, 7, 11 respectively.
- $(n, n + 1, n + 2) = (1175, 1176, 1177)$, obtained from $m = 107, k = 61, l = 67$; note that 1175, 1176, 1177 are divisible by 5, 7, 11 respectively.

About the author

Anand Prakash runs a small garment shop at Kesariya village in the state of Bihar. He has a keen interest in number theory and recreational mathematics and has published many papers in international journals in these fields. He also has a deep interest in classical Indian music as well as cooking. In addition, he has written a large number of poems in Hindi. He may be contacted at prakashanand805@gmail.com

Surprising limits – 1

We are familiar with the fact that as $x \rightarrow 0$, we also have $\sin x \rightarrow 0$.

What can be said about the behaviour of $\frac{1}{\sin x} - \frac{1}{x}$ as $x \rightarrow 0$?

This is quite non-obvious because both quantities become infinitely large as $x \rightarrow 0$.

Let's see what the numbers tell us. Here is a table of values.

x	$\frac{1}{x}$	$\sin x$	$\frac{1}{\sin x}$	$\frac{1}{\sin x} - \frac{1}{x}$
0.1	10	0.0998	10.0167	0.0167
0.01	100	0.0099998	100.00167	0.00167
0.001	1000	0.0009999998	1000.000167	0.000167

The table is full of tantalising patterns! But the evidence seems clear. It appears that $\frac{1}{\sin x} - \frac{1}{x} \rightarrow 0$ as $x \rightarrow 0$.

Let's try to justify this. The standard way, with which you are surely familiar, is by using L'Hopital's Rule. The rule states that for functions f

and g and a point c in the domain of f and g , if $f(c) = 0 = g(c)$, then

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}, \quad (29)$$

provided the second limit exists. (A few other conditions are included to avoid 'bad situations.' We do not list them here.)

We have, now,

$$\begin{aligned} \lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right) &= \lim_{x \rightarrow 0} \left(\frac{x - \sin x}{x \cdot \sin x} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{1 - \cos x}{x \cdot \cos x + \sin x} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{\sin x}{2 \cos x - x \cdot \sin x} \right) = \frac{0}{2} = 0, \end{aligned}$$

as expected. But this is not the only way of finding the limit. Another approach is to use the series expansion for the sine function:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots,$$

which we approximate to

$$\sin x \approx x - \frac{x^3}{6} \quad \text{for } x \approx 0. \quad (30)$$

Hence we have:

$$\begin{aligned} \frac{1}{\sin x} - \frac{1}{x} &\approx \frac{1}{x - \frac{1}{6}x^3} - \frac{1}{x} \\ &= \frac{1}{x} \left(\frac{1}{1 - \frac{1}{6}x^2} - 1 \right) \\ &\approx \frac{1}{x} \left(1 + \frac{x^2}{6} - 1 \right) = \frac{x}{6}. \end{aligned}$$

This approach gives us a stronger result (i.e., stronger than that given by L'Hopital's Rule) than what we might have expected:

$$\frac{1}{\sin x} - \frac{1}{x} \approx \frac{x}{6} \quad \text{for } x \approx 0. \quad (31)$$

From (31) it immediately follows that

$$\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right) = 0.$$

Relation (31) exactly matches the data in the table displayed earlier (just look at the last column). The following graph is of interest.

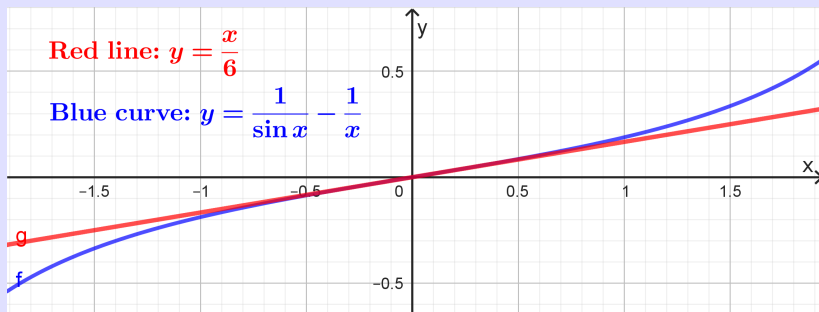


Figure 33: Graphs of $y = \frac{1}{\sin x} - \frac{1}{x}$ and $y = \frac{x}{6}$

The visual evidence in the graph suggests that the graphs of $y = \frac{1}{\sin x} - \frac{1}{x}$ and $y = \frac{x}{6}$ are tangent to each other at $x = 0$. This is indeed the case. The reader should verify this claim using differentiation.

Divergence between the graphs begins to be noticeable beyond $|x| \approx 0.6$. For $|x| < 0.5$, the graphs are practically indistinguishable.

An unexpected limit

Relation (31) also yields the following result which is quite non-obvious:

$$\lim_{x \rightarrow 0} \left(\frac{1}{x \cdot \sin x} - \frac{1}{x^2} \right) = \frac{1}{6}. \quad (32)$$

Why do we call it non-obvious? Because the limit is of the type $\infty - \infty$, and such limits can never be predicted in advance!

Obviously, we could also have evaluated the limit using L'Hopital's Rule. The author's preference, however, is to use power series approximations! This approach seems more elegant and also more efficient.

Another surprise

There is a second surprise to be found in this problem. Another victory for the non-obvious!

Considering that $\sin x \approx x$ for $x \approx 0$, we might expect that

$$\lim_{x \rightarrow 0} \left(\frac{1}{\sin^2 x} - \frac{1}{x^2} \right) = \lim_{x \rightarrow 0} \left(\frac{1}{x \cdot \sin x} - \frac{1}{x^2} \right).$$

But this turns out to be not the case! For example,

$$\text{for } x = 0.001, \quad \frac{1}{\sin^2 x} - \frac{1}{x^2} \approx 0.3333,$$

which strongly suggests that the limit is $\frac{1}{3}$ and not $\frac{1}{6}$. Let's work out why.

To evaluate the first limit, we again use the power series approximation $\sin x \approx x - \frac{1}{6}x^3$. We have,

$$\frac{1}{\sin^2 x} \approx \frac{1}{\left(x - \frac{1}{6}x^3\right)^2} = \frac{1}{x^2} \cdot \frac{1}{\left(1 - \frac{1}{6}x^2\right)^2}.$$

Now we make use of another approximation (this one comes from the binomial theorem): for $|a| < 1$:

$$\frac{1}{(1-a)^2} = 1 + 2a + 3a^2 + \dots,$$

and therefore,

$$\frac{1}{(1-a)^2} \approx 1 + 2a \quad \text{for } a \approx 0.$$

Invoking this approximation:

$$\begin{aligned} \frac{1}{\sin^2 x} &\approx \frac{1}{x^2} \cdot \left(1 + \frac{x^2}{3}\right) \\ &= \frac{1}{x^2} + \frac{1}{3}. \end{aligned} \tag{33}$$

From (33) we deduce:

$$\lim_{x \rightarrow 0} \left(\frac{1}{\sin^2 x} - \frac{1}{x^2} \right) = \frac{1}{3}. \tag{34}$$

So the limit in question is $\frac{1}{3}$ and not $\frac{1}{6}$.

Yet another surprise

There is a third surprise to be found in this problem! Let us evaluate the following limit:

$$\lim_{x \rightarrow 0} \left(\frac{x}{\sin^3 x} - \frac{1}{x^2} \right). \tag{35}$$

As earlier, we might expect that

$$\lim_{x \rightarrow 0} \left(\frac{x}{\sin^3 x} - \frac{1}{x^2} \right) = \lim_{x \rightarrow 0} \left(\frac{1}{\sin^2 x} - \frac{1}{x^2} \right).$$

Yet again this turns out to be not the case! For example,

$$\text{for } x = 0.01, \frac{x}{\sin^3 x} - \frac{1}{x^2} \approx 0.5000142251,$$

which suggests that the limit is $\frac{1}{2}$ and not $\frac{1}{3}$. Let's see if the limit is indeed equal to $\frac{1}{2}$. We argue in the same manner as earlier. We have,

$$\frac{x}{\sin^3 x} \approx \frac{x}{\left(x - \frac{1}{6}x^3\right)^3} = \frac{1}{x^2} \cdot \frac{1}{\left(1 - \frac{1}{6}x^2\right)^3}. \quad (36)$$

We again make use of the binomial theorem. For $|a| < 1$ we have,

$$\frac{1}{(1-a)^3} = 1 + 3a + 6a^2 + \cdots,$$

and therefore,

$$\frac{1}{(1-a)^3} \approx 1 + 3a \quad \text{for } a \approx 0. \quad (37)$$

Invoking this approximation, we see that for $x \approx 0$,

$$\begin{aligned} \frac{x}{\sin^3 x} &\approx \frac{1}{x^2} \cdot \frac{1}{\left(1 - \frac{1}{6}x^2\right)^3} \\ &\approx \frac{1}{x^2} \cdot \left(1 + \frac{1}{2}x^2\right) = \frac{1}{x^2} + \frac{1}{2}. \end{aligned} \quad (38)$$

It follows from this that

$$\lim_{x \rightarrow 0} \left(\frac{x}{\sin^3 x} - \frac{1}{x^2} \right) = \frac{1}{2}, \quad (39)$$

as suspected.

We anticipate from the above that for $x \approx 0$,

$$\frac{x^2}{\sin^3 x} - \frac{1}{x} \approx \frac{x}{2}, \quad (40)$$

so it may be of interest to compare the graphs of the above two expressions in the vicinity of the origin. Here they are:

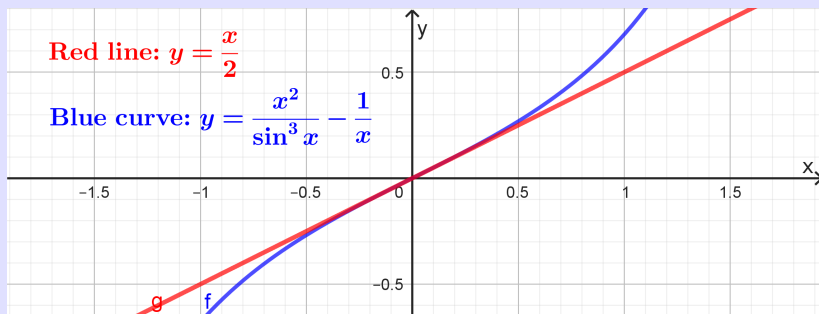


Figure 34: Graphs of $y = \frac{x^2}{\sin^3 x} - \frac{1}{x}$ and $y = \frac{x}{2}$

The visual evidence strongly suggests that the two graphs are tangent to each other at the origin.

Infinity ahead. Beware!

We see here the counter-intuitive nature of infinity. When we work with infinity, we should be prepared to expect anything!

Article contributed by Shailesh Shirali

A Tale of Two Parabolas – 1

We describe here an unexpected result associated with parabolas. We came across it in [1].

Consider a pair $\mathcal{P}_1$ and $\mathcal{P}_2$ of parabolas in the coordinate plane such that their axes are perpendicular to each other.

What can be said about their points of intersection? Various possibilities can arise.

For example, $\mathcal{P}_1$ and $\mathcal{P}_2$ may miss each other altogether (Figure 35). That is, there are *no points of intersection* between $\mathcal{P}_1$ and $\mathcal{P}_2$.

Or $\mathcal{P}_1$ and $\mathcal{P}_2$ may be *tangent to one another* (Figure 36). That is, they may touch each other at a single point.

Or $\mathcal{P}_1$ and $\mathcal{P}_2$ may cut each other in *two distinct points* (Figure 37). The second possibility is actually a special case of this one.

Or $\mathcal{P}_1$ and $\mathcal{P}_2$ may cut each other in *four distinct points* (Figure 38).

Our interest for now is only in the fourth situation. Let us label the four points of intersection as A, B, C, D (Figure 39).

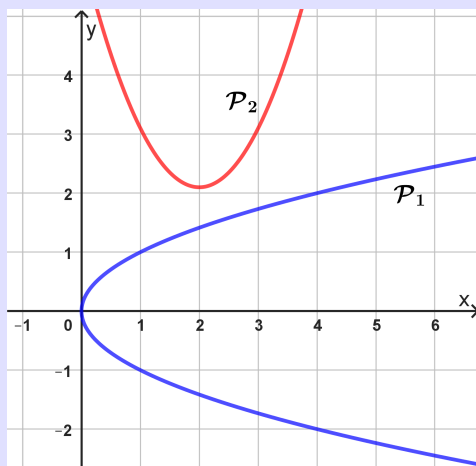


Figure 35: The case of no points of intersection

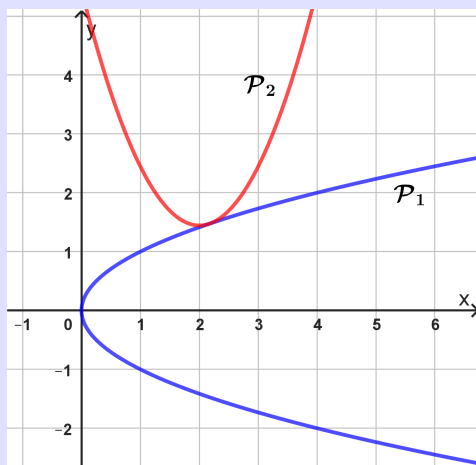


Figure 36: The case of tangency

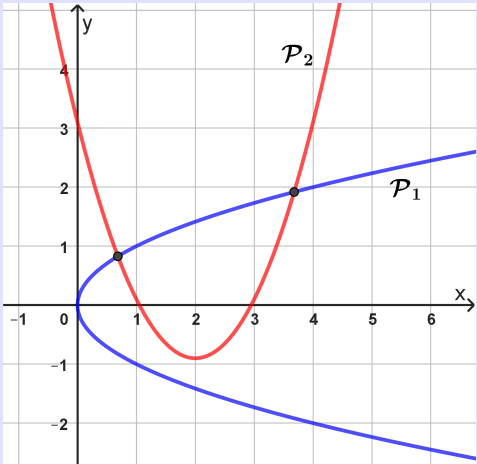


Figure 37: Two points of intersection

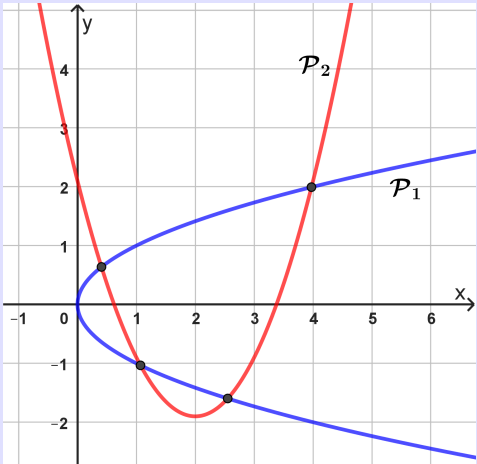


Figure 38: Four points of intersection

Claim

Points A, B, C, D lie on a circle (Figure 39). See [1].

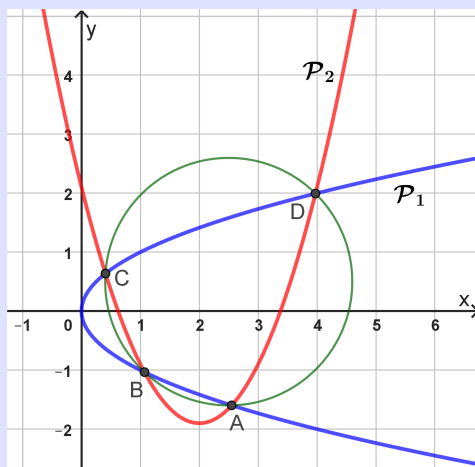


Figure 39: The four points of intersection of $\mathcal{P}_1$ and $\mathcal{P}_2$ lie on a circle

Inasmuch as this is a geometric claim, we expect that there must be a pure geometry proof of the assertion. This turns out to be quite difficult to find. But there is an elegant and compact proof using coordinates and the theory of equations. This is what we shall now present.

Algebraic proof of the claim

Adopt a system of coordinates in which the axis of $\mathcal{P}_1$ coincides with the x -axis and its vertex lies at the origin.

We may assume without any loss of generality that the equation of $\mathcal{P}_1$ is $y^2 = x$.

By assumption, the axis of $\mathcal{P}_2$ is parallel to the y -axis. Let the equation of $\mathcal{P}_2$ be $y = ax^2 + bx + c$. Here $a \neq 0$.

We write the equations of the two parabolas in the following form:

- $\mathcal{P}_1$: $f_1(x, y) = 0$ where $f_1(x, y) = y^2 - x$.
- $\mathcal{P}_2$: $f_2(x, y) = 0$ where $f_2(x, y) = ax^2 + bx + c - y$.

The points of intersection of $\mathcal{P}_1$ and $\mathcal{P}_2$ are, as mentioned earlier, A, B, C, D . This means, obviously, that points A, B, C, D lie on both the curves. Hence both f_1 and f_2 vanish (i.e., they are identically 0) at points A, B, C, D .

Let u, v be any real numbers. Let

$$f(x, y) = uf_1(x, y) + vf_2(x, y).$$

Since f_1 and f_2 vanish at points A, B, C, D , it follows that f too vanishes at points A, B, C, D .

The curve $\mathcal{C} : f(x, y) = 0$ is obviously a conic section as its equation is of the second degree. Depending on the values of u and v , it could be any of the conic sections. But whatever be this conic section, it must pass through the points A, B, C, D .

Let us assign these values to u, v :

$$u = a, \quad v = 1.$$

Then f has the form

$$f(x, y) = a(x^2 + y^2) + (b - a)x + y + c.$$

Hence $\mathcal{C}$ has the equation

$$a(x^2 + y^2) + (b - a)x + y + c = 0.$$

But this is the equation of a circle!

As noted earlier, points A, B, C, D lie on $\mathcal{C}$ for any choice of u, v .

It follows that points A, B, C, D lie on the circle with equation

$$a(x^2 + y^2) + (b - a)x + y + c = 0.$$

We have thus established the stated property, and we have even found the equation of the circumscribing circle. ■

Is there a “pure geometry” proof? We shall take up this question next time.

Further reading

- [1] Math Stack Exchange, “Intersection of two parabolae” from
<https://math.stackexchange.com/questions/134366/intersection-of-two-parabolae>

Article contributed by Shailesh Shirali

A New Proof of the Theorem of Pythagoras

This article has been adapted from the article [1] written on the same topic and by the same author.

The Pythagorean theorem ('PT' for short) is easily the best known result in all of mathematics. What is less well-known is the fact that among all theorems in mathematics, it holds the "world record" for the number of different proofs. There is no other theorem that even comes close! (See [3] and [4].) In the book *The Pythagorean Proposition* [2] (published in 1940), the author Elisha S. Loomis lists as many as 370 different proofs of the theorem. Since that time, close to a century back, still more proofs have appeared.

We present the most recent of these proofs in this article. It is by two high-school teenagers, Calcea Johnson and Ne'Kiya Jackson; they were students of St. Mary's Academy in New Orleans, USA, when they discovered this proof. (They are now in college.)

The proof raises a very interesting question. *Can there be a proof of the PT based on trigonometry?* In [2], the author claims definitively and strongly that there cannot be such a proof. It turns out that Loomis was wrong in making this claim, and the proof by Johnson & Jackson is itself a counterexample! We say more about this interesting debate below.

Proof by Calcea Johnson and Ne’Kiya Jackson

We start by asking, *Can there be a trigonometric proof of the PT?*

It is possible that by ‘trigonometric proof’ we have in mind the following ‘argument.’ Let $\triangle ABC$ be given, right-angled at C . Using the usual symbols we must prove that $c^2 = a^2 + b^2$. Observe that $\sin A = a/c$ and $\cos A = b/c$. Since $\sin^2 A + \cos^2 A = 1$, we obtain $a^2/c^2 + b^2/c^2 = 1$, or $c^2 = a^2 + b^2$. Hence proved!

But how do we know that $\sin^2 A + \cos^2 A = 1$? By using the Pythagorean Theorem, of course! So we have got trapped into a circular argument here, which means that we have not actually proved anything.

But is there a way of proving the basic identity $\sin^2 \theta + \cos^2 \theta = 1$ *without using the PT* and using only the definitions of sine and cosine? Yes, there is — all we need to do is mimic the proof-by-similar-triangles of the PT.

Here are the details (Figure 40). Let $\triangle ABC$ be right-angled at vertex C ; let its hypotenuse AB have unit length. Denote $\angle ABC$ by θ . Draw a perpendicular CD from C to AB . Then $\angle ACD = \theta$.

Since $AB = 1$, we get $CA = \sin \theta$ and $CB = \cos \theta$, from the definitions of the trigonometric ratios. Since $\triangle CBD$ is right-angled with hypotenuse CB ,

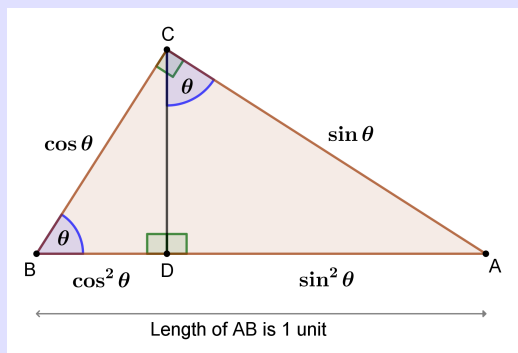


Figure 40: Proof of the identity $\sin^2 \theta + \cos^2 \theta = 1$ — without using the PT

we also have

$$\frac{BD}{CB} = \cos \theta, \quad \therefore BD = \cos^2 \theta, \quad (41)$$

and since $\triangle CAD$ is right-angled with hypotenuse AC , we have

$$\frac{AD}{CA} = \sin \theta, \quad \therefore AD = \sin^2 \theta. \quad (42)$$

Since $BD + AD = 1$, we get $\sin^2 \theta + \cos^2 \theta = 1$, as required.

And since $\cos \theta = BC / AB$ and $\sin \theta = AC / AB$, we get $CB^2 + CA^2 = AB^2$ or $a^2 + b^2 = c^2$, which is the PT.

This way, we arrive at the statements of the PT and the basic trigonometric identity ($\sin^2 \theta + \cos^2 \theta = 1$) at the same time. *Now there is no circularity of argument.*

With that preamble, let us study the proof given by Ms. Calcea Johnson and Ms. Ne’Kiya Jackson of New Orleans; see Figure 2.



📷 Ne'Kiya Jackson, left, and Calcea Johnson recently presented their findings at the American Mathematical Society's south-eastern chapter's semi-annual meeting. Photograph: WWL-TV

Figure 41: Calcea Johnson and Ne'Kiya Jackson of St. Mary's Academy, New Orleans, USA. Credit: https://www.theguardian.com/us-news/2023/mar/24/new-orleans-pythagoras-theorem-trigonometry-prove?CMP=oth_b-aplnews_d-1

Here are the prerequisites needed to understand the proof.

1. The basic theorems concerning similar triangles (the results are well-known so we do not list them here);
2. The formula for the limiting sum of an infinite geometric progression with first term a and common ratio r between -1 and 1 :

$$a + ar + ar^2 + ar^3 + \dots = \frac{a}{1-r} \quad (-1 < r < 1);$$

3. The double-angle trigonometric identity $\sin 2\theta = 2 \sin \theta \cdot \cos \theta$.

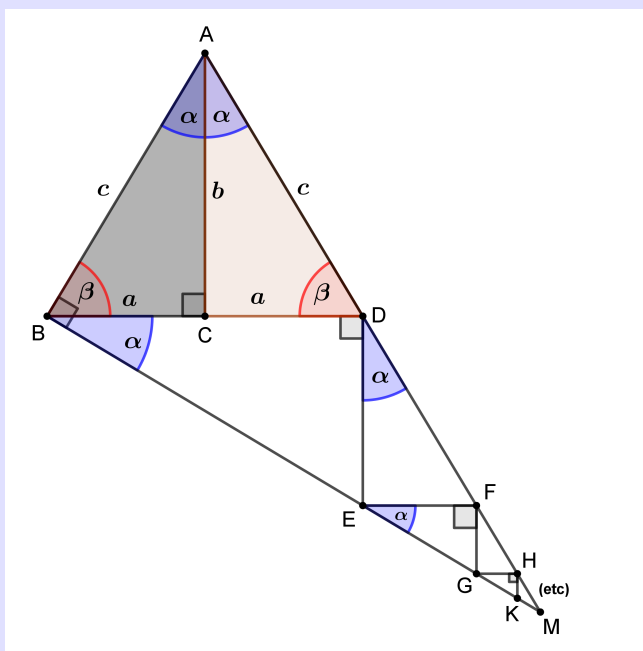


Figure 42: Proof of the PT by Calcea Johnson and Ne’Kiya Jackson

Details of the proof Figure 42 shows the given $\triangle ABC$ which is right-angled at C. We must show that $c^2 = a^2 + b^2$. We shall assume throughout that $a < b$; equivalently, that $\alpha < \beta$.

Reflect $\triangle ABC$ in AC to give $\triangle ADC$; then $\angle BAD = 2\alpha$. Since $\alpha < \beta$, it follows that $\alpha < 45^\circ$, so $\angle BAD < 90^\circ$. Draw ray AD. Draw a line through B perpendicular to AB. This line will meet ray AD at M, say. (The two lines meet since $\angle BAM < 90^\circ$.) Join BM. Now draw $DE \perp BD$, meeting

BM at E ; draw $EF \parallel BD$, meeting ray AM at F ; draw $FG \perp BD$, meeting BM at G ; draw $GH \parallel BD$, meeting ray AM at H ; and so on ... (see the 'etc' in Figure 3).

We shall now compute the lengths of BM and AM ; this yields the desired result. We shall feel free to make use of trigonometric identities — *provided that those identities do not require the PT for their proofs.*

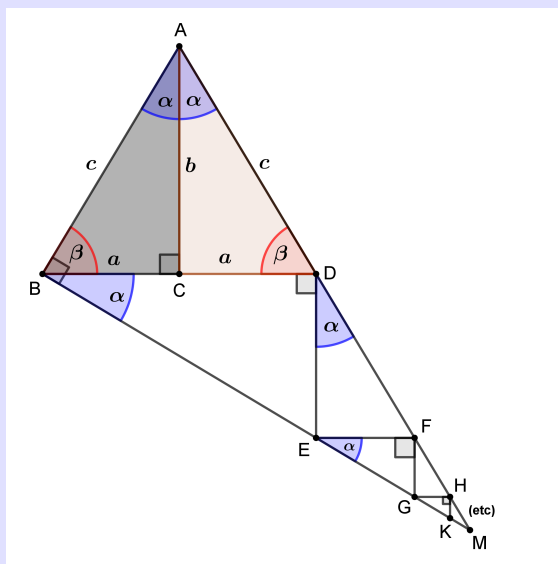


Figure 43: Proof of the PT (figure redrawn for convenience)

Note the many triangles that are similar to $\triangle ABC$ (see Figure 43, which is Figure 42 redrawn for convenience). We have:

$$\triangle ABC \sim \triangle BED \sim \triangle DFE \sim \triangle EGF \sim \triangle FGH \sim \triangle GKH \sim \dots, \quad (43)$$

as each of these triangles has angles α , β , 90° .

It follows that $DE : BD : BE = a : b : c$, so $DE : 2a : BE = a : b : c$, hence

$$BE = \frac{2ac}{b}, \quad DE = \frac{2a^2}{b}. \quad (44)$$

Next: $EF : DE : DF = a : b : c$, so $EF : 2a^2/b : DF = a : b : c$, hence

$$EF = \frac{2a^3}{b^2}, \quad DF = \frac{2a^2c}{b^2}. \quad (45)$$

Next: $FG : EF : EG = a : b : c$, so $FG : 2a^3/b^2 : EG = a : b : c$, hence

$$FG = \frac{2a^4}{b^3}, \quad EG = \frac{2a^3c}{b^3}. \quad (46)$$

Proceeding thus, we find that $DF, FH, \dots$ form a geometric progression with common ratio a^2/b^2 :

$$DF = c \cdot \frac{2a^2}{b^2}, \quad FH = c \cdot \frac{2a^4}{b^4}, \quad \dots \quad (47)$$

Similarly, $BE, EG, GK, \dots$ form a geometric progression with the same common ratio, a^2/b^2 :

$$BE = c \cdot \frac{2a}{b}, \quad EG = c \cdot \frac{2a^3}{b^3}, \quad GK = c \cdot \frac{2a^5}{b^5}, \quad \dots \quad (48)$$

We are now able to compute the total length of the segments $DF, FH, \dots$ by summing an infinite geometric progression with common ratio less than 1:

$$\begin{aligned} DF + FH + \dots &= c \cdot \frac{2a^2}{b^2} \cdot \left(1 + \frac{a^2}{b^2} + \frac{a^4}{b^4} + \frac{a^6}{b^6} + \dots \right) \\ &= c \cdot \frac{2a^2}{b^2} \cdot \frac{1}{1 - a^2/b^2} = \frac{2ca^2}{b^2 - a^2}, \end{aligned} \quad (49)$$

and therefore:

$$AM = c + \frac{2ca^2}{b^2 - a^2} = c \cdot \frac{a^2 + b^2}{b^2 - a^2}. \quad (50)$$

Similarly,

$$\begin{aligned}
 BM &= BE + EG + GK + \cdots \\
 &= \frac{2ac}{b} \cdot \left(1 + \frac{a^2}{b^2} + \frac{a^4}{b^4} + \frac{a^6}{b^6} + \cdots \right) \\
 &= \frac{2ac}{b} \cdot \frac{1}{1 - a^2/b^2} = \frac{2abc}{b^2 - a^2}.
 \end{aligned} \tag{51}$$

It follows that

$$\frac{BM}{AM} = \left(\frac{2abc}{b^2 - a^2} \right) \div c \left(\frac{a^2 + b^2}{b^2 - a^2} \right) = \frac{2ab}{a^2 + b^2}. \tag{52}$$

On the other hand, $BM/AM = \sin 2\alpha$, as may be seen by observing the angles of the right-angled triangle ABM . Hence:

$$\sin 2\alpha = \frac{2ab}{a^2 + b^2}. \tag{53}$$

We now make use of the trigonometric identity $\sin 2\alpha = 2 \sin \alpha \cos \alpha$. From $\triangle ABC$, we get

$$\sin \alpha = \frac{a}{c}, \quad \cos \alpha = \frac{b}{c}, \quad \therefore \sin 2\alpha = \frac{2ab}{c^2}. \tag{54}$$

From the two relations for $\sin 2\alpha$, we conclude that

$$c^2 = a^2 + b^2, \tag{55}$$

and there we have it: the Pythagorean theorem! ■

You may wonder about the double angle identity $\sin 2\alpha = 2 \sin \alpha \cos \alpha$. Does it require the PT for its proof? No, it does not. (We show this below, in the appendix.) So there is no circular reasoning in this proof.

You may also wonder what happens to the proof in the isosceles case when $a = b$ (or $\alpha = \beta$). We leave this part to the reader.

Closing remark

Though the proof by Calcea Johnson and Ne’Kiya Jackson is long and involved, it is also subtle and deep. A truly wonderful achievement by the two teenagers.

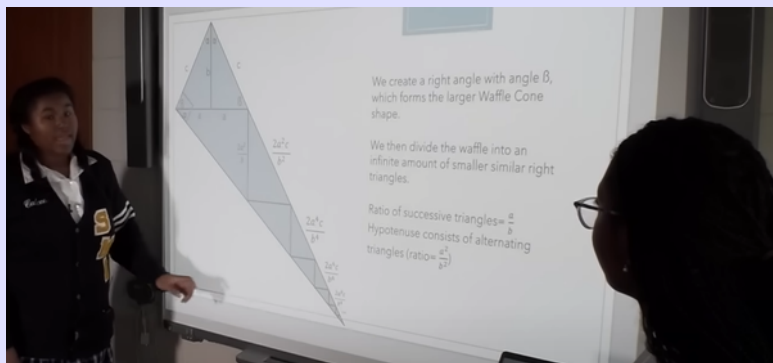


Figure 44: Calcea Johnson and Ne’Kiya Jackson presenting their proof

Further reading

- [1] Shirali, Shailesh (2023), “Two New Proofs of the Pythagorean Theorem - Part I.” *At Right Angles*, July 2023. pp. 7-12. <https://publications.azimpremjiuniversity.edu.in/id/eprint/4877>
- [2] Elisha Scott Loomis, *The Pythagorean Proposition*. From <https://files.eric.ed.gov/fulltext/ED037335.pdf>; see also <https://mathlair.allfunandgames.ca/pythprop.php>.

- [3] Wikipedia, "Pythagorean theorem." From
https://en.wikipedia.org/wiki/Pythagorean_theorem.
- [4] Cut-the-Knot, "Pythagorean Theorem." From
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- [5] Leila Sloman, "2 High School Students Prove Pythagorean Theorem. Here's What That Means." From
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- [6] Ramon Antonio, "US teens say they have new proof for 2,000-year-old mathematical theorem." From
https://www.theguardian.com/us-news/2023/mar/24/new-orleans-pythagoras-theorem-trigonometry-prove?CMP=oth_b-aplnews_d-1
- [7] MathTrain, "How High Schoolers Proved Pythagoras Using Just Trig! (and some other stuff)." From
<https://www.youtube.com/watch?v=nQD6lDwFmCc>
- [8] polymathematic, "Pythagoras Would Be Proud: High School Students' New Proof of the Pythagorean Theorem." From
<https://www.youtube.com/watch?v=p6j2nZKwf20>

Proof of a double angle identity

As promised we shall prove the double angle identity without making use of the PT at any stage.

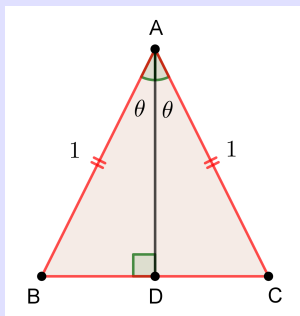


Figure 45: Proof of the double angle formula for sine

Consider an isosceles triangle ABC with $AB = 1 = AC$ and $\angle BAC = 2\theta$ (see Figure 45). Let D be the midpoint of BC . Then AD is the bisector of angle $\angle BAC$, and it is perpendicular to BC . Clearly, $AD = \cos \theta$ and $BD = \sin \theta$. Hence the area K of $\triangle ABC$ is

$$K = \frac{1}{2}(\text{base} \times \text{height}) = \sin \theta \cdot \cos \theta.$$

We may also compute the area using another well-known formula:

$$K = \frac{1}{2}(AB \cdot AC \cdot \sin \angle BAC) = \frac{1}{2} \sin 2\theta.$$

Equating the two expressions for K , we obtain the desired identity:

$$\sin 2\theta = 2 \sin \theta \cdot \cos \theta.$$

Note that we did not have to invoke the PT at any point in the proof. ■

Article contributed by Shailesh Shirali

A Striking Property of 2025

Article contributed by Prithwiji De

The year 2025 has arrived, and it is that time of the year when people find all sorts of nice (and at times crazy) ways of writing 2025 using the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 and the mathematical operations $+$, $-$, $\times$, $/$. Some might also use sophisticated mathematical functions.

The one that many find is

$$\begin{aligned}2025 &= (1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9)^2 \\ &= 1^3 + 2^3 + 3^3 + 4^3 + 5^3 + 6^3 + 7^3 + 8^3 + 9^3.\end{aligned}$$

But the relation that is easy to miss is this one:

$$(20 + 25)^2 = 2025.$$

Are there any other 4-digit numbers that have this striking property?

Answering this question fully requires solving the following problem: find all 4-tuples of decimal digits (a, b, c, d) with $a, c \neq 0$ such that

$$\left((10a + b) + (10c + d) \right)^2 = 100(10a + b) + 10c + d.$$

Expanding and simplifying the equation, we obtain

$$99(10c + d) = (100 - p - 10c - d)(p + 10c + d)$$

where $p = 10a + b$. Now observe that by the AM-GM inequality,

$$\begin{aligned} & (100 - p - 10c - d)(p + 10c + d) \\ & \leq \left(\frac{100 - p - 10c - d + p + 10c + d}{2} \right)^2 = \left(\frac{100}{2} \right)^2 = 2500, \end{aligned}$$

giving $99(10c + d) \leq 2500$, and so

$$10c + d \leq \lfloor 2500/99 \rfloor = 25.$$

Therefore $c \in \{1, 2\}$ and $d \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

Observe that $100 - (p + 10c + d)$ and $p + 10c + d$ have the same parity (i.e., they are both even or both odd).

If $c = 1$, we get

$$99(10 + d) = (90 - p - d)(10 + p + d),$$

and if both factors on the right hand side are even, then 4 divides the product on the right hand side and hence it divides the product on the left hand side too. This implies that $10 + d$ is divisible by 4, and hence d is either 2 or 6.

So p either satisfies

$$(99)(12) = (88 - p)(p + 12) \Rightarrow p^2 - 76p + 132 = 0,$$

or

$$(99)(16) = (84 - p)(16 + p) \Rightarrow p^2 - 68p + 240 = 0.$$

But neither of these equations has an integer root.

Therefore both $90 - p - d$ and $10 + p + d$ are odd, which shows that $10 + d$ is odd. So, $d \in \{1, 3, 5, 7, 9\}$.

If $d = 1$, then

$$1089 = (89 - p)(11 + p) \Rightarrow p^2 - 78p + 110 = 0$$

which does not have an integer root.

Similarly, when $d = 3, 5, 7, 9$ we obtain (respectively) the equations

$$p^2 - 74p + 156 = 0,$$

$$p^2 - 70p + 210 = 0,$$

$$p^2 - 66p + 272 = 0,$$

$$\text{and } p^2 - 62p + 342 = 0.$$

None of these equations has an integer root.

Thus $c = 2$, and we want $d \in \{0, 1, 2, 3, 4, 5\}$ and p such that

$$99(20 + d) = (80 - p - d)(20 + p + d).$$

The factors on the right side are either both even or both odd.

If the two factors are even, then 4 divides d , so $d \in \{0, 4\}$. If $d = 0$, then p satisfies

$$p^2 - 70p + 1180 = 0,$$

and if $d = 4$, then p is a root of

$$p^2 - 52p + 552 = 0.$$

But neither of these equations has an integer root.

Thus both factors are odd, and hence d is odd.

The possibilities $d = 1$ and $d = 3$ yield (respectively) the equations

$$p^2 - 58p + 420 = 0$$

and

$$p^2 - 54p + 506 = 0;$$

but neither equation has an integer root.

So we discard these two possibilities.

The only remaining possibility is $d = 5$. If $d = 5$ then

$$p^2 - 50p + 600 = 0,$$

and this equation does have integer roots (20 and 30); hence, $p \in \{20, 30\}$.

Both possibilities yield solutions to our problem.

It follows that there are two 4-digit numbers with the required property: 2025 and 3025. It is easy to check that

$$(20 + 25)^2 = 2025,$$

$$\text{and } (30 + 25)^2 = 3025.$$

We note in passing that if we had allowed $c = 0$, then we would have also obtained

$$(98 + 01)^2 = 9801.$$

About the author

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Problems for Solution – 3

Here are some problems for you to ponder over. The answers to the problems are given on page 179.

Solutions to the problems in this set will be given in the next issue of the magazine. Enjoy!

Problems for the January 2025 issue

1. For some two real numbers $x \neq y$, let

$$\frac{x^2 + y^2}{x^2 - y^2} + \frac{x^2 - y^2}{x^2 + y^2} = k.$$

Find the value of the following expression in terms of k :

$$\frac{x^8 + y^8}{x^8 - y^8} + \frac{x^8 - y^8}{x^8 + y^8}.$$

(1st Junior Balkan Mathematical Olympiad, Macedonia, June 1997)

2. Prove that for each prime $p > 2$, there exists exactly one positive integer n , such that $n^2 + np$ is a perfect square. (Sophie Fellowship, October 20, 2023)

3. For any triangle ABC , let a, b, c be the length of its sides, and let R be its circumradius. Deduce the angles of the triangle given that

$$R(b + c) = a\sqrt{bc}.$$

(1st Junior Balkan Mathematical Olympiad, Macedonia, June 1997)

4. Figure 46 shows a semicircle with diameter BC and centre O . A square $ODEM$ is drawn as shown, with one vertex at O and the opposite vertex on the arc BC . Chord AMB is drawn. Find the ratios $AM : MO$ and $MO : OB$.

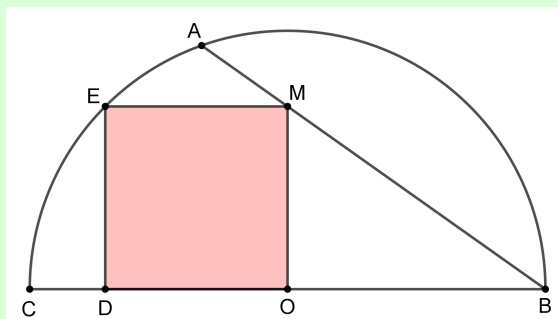


Figure 46: A semicircle, a chord and an inscribed square

5. Prove that the number

$$\underbrace{11 \dots 11}_{1997 \text{ 1's}} \underbrace{22 \dots 22}_{1998 \text{ 2's}} 5$$

(with 1997 1's and 1998 2's) is a perfect square. (2nd Junior Balkan Mathematical Olympiad, Macedonia, June 1998)

6. Find all the pairs (x, y) of positive integers such that

$$x^y = y^{x-y}$$

(2nd Junior Balkan Mathematical Olympiad, Macedonia, June 1998)

Solutions to problems from the October 2024 issue

Problem 1 (October 2024): (From IMO 1959.) Prove that the fraction

$$\frac{21n + 4}{14n + 3}$$

is irreducible (i.e., in its “lowest terms”) for every positive integer n .

Solution: We must show that $21n + 4$ and $14n + 3$ are coprime. For this, we only need to exhibit a pair a, b of integers such that

$$a(21n + 4) + b(14n + 3) = 1.$$

To see why, let $g = \text{GCD}(21n + 4, 14n + 3)$. Then g will divide

$$a(21n + 4) + b(14n + 3)$$

for any pair of integers a, b . So if we are able to find a pair a, b of integers such that $a(21n + 4) + b(14n + 3) = 1$, then it will force g to be equal to 1 (since the only positive divisor of 1 is 1 itself). But it is easy to find such a pair of integers just by inspection. Since $\frac{21}{14} = \frac{3}{2}$, we must have $\frac{a}{b} = -\frac{2}{3}$. The choice $a = -2, b = 3$ is immediately seen to work, for

$$(-2)(21n + 4) + 3(14n + 3) = 1.$$

The claim that $\text{GCD}(21n + 4, 14n + 3) = 1$ is thus proved. ■

Problem 2 (October 2024): Find all positive integers n such that $2^n + 1$ is a perfect square.

Solution: Let $2^n + 1 = m^2$ where m is a positive integer. Then we have:

$$2^n + 1 = m^2,$$

$$\therefore 2^n = m^2 - 1,$$

$$\therefore 2^n = (m - 1)(m + 1).$$

The only proper divisors of a power of 2 are smaller powers of 2. Hence $m - 1$ and $m + 1$ are both powers of 2, say $m - 1 = 2^a$ and $m + 1 = 2^b$ where a, b are non-negative integers. By subtraction, we get

$$2^b - 2^a = 2.$$

The only way to write 2 as a difference of two powers of 2 is

$$2 = 4 - 2 = 2^2 - 2^1.$$

Hence $b = 2$ and $a = 1$, giving $m = 2^2 - 1 = 3$. So there is just one solution in positive integers to the given equation: $2^3 + 1 = 3^2$. Therefore, the only positive integer n such that $2^n + 1$ is a perfect square is $n = 3$. ■

Problem 3 (October 2024):

- (a) Are there any positive integers n such that $n(n + 1)$ is a perfect square?
- (b) Are there any positive integers n such that $n(n + 1)(n + 2)$ is a perfect square?

Solution: (a) We start by noting that if the product of two coprime positive integers a, b is a perfect square, then both a, b are perfect squares. To see why this is so, observe that as there are no prime factors common to a, b , each prime factor must occur to an *even power* in both a, b separately. Hence both a, b must be perfect squares.

Next, note that the consecutive integers $n, n + 1$ differ by 1 and therefore have no prime factors in common (i.e., they are coprime). Therefore, if the product $n(n + 1)$ is a perfect square, then both n and $n + 1$ must be perfect squares. But there can be no two consecutive positive integers that are both perfect squares. (Please prove this for yourself.)

So, there is no positive integer n such that $n(n + 1)$ is a perfect square. ■

(b) We use the same principle as the one used above — that if the product of two coprime positive integers is a perfect square, then both of them must be perfect squares. But the reasoning is more subtle now, as there are *three* factors here and it is not true that they are all coprime to each other. Here we observe that since $n + 1, n$ are coprime, and so too are $n + 1, n + 2$, it must be that $n + 1$ and $n(n + 2)$ are coprime. Therefore if the product of $n + 1$ and $n(n + 2)$ is a perfect square, then both $n + 1$ and $n(n + 2)$ must be perfect squares. This means, in particular, that $n(n + 2)$ is a perfect square. But that can never be true for any positive integer n , since $n^2 + 2n$ and the perfect square $(n + 1)^2 = n^2 + 2n + 1$ are consecutive integers.

So, there is no positive integer n such that $n(n + 1)(n + 2)$ is a perfect square. ■

Problem 4 (October 2024): Trapezium $ABCD$ has $AB \parallel DC$ and is such that the bisector of $\angle D$ is perpendicular to side BC (Figure 47). The lengths

of some of the segments are shown in the figure ($u = BF$, $v = CF$, $w = DF$). Find the area of trapezium $ABCD$ in terms of u, v, w .

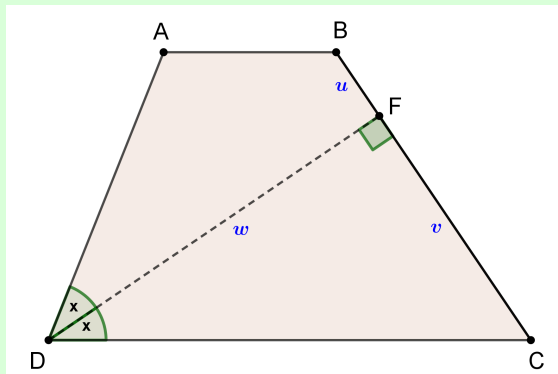


Figure 47

Solution: We ‘complete’ the trapezium into a triangle by extending the sides DA beyond A and CB beyond B so that they meet at E (Figure 48). We also draw $AG \perp EB$, which means that $AG \parallel DF$.

We are going to compute the area of the trapezium by subtracting the area of $\triangle ABE$ from the area of $\triangle DCE$.

Clearly, $\triangle EDF \cong \triangle CDF$, so $EF = CF = v$.

Hence the area of $\triangle DCE$ is $\frac{1}{2}(2v)w = vw$.

Since $EF = v$ and $BF = u$, we get $EB = v - u$ and $EG = \frac{1}{2}(v - u)$.

By similarity, $AG : DF = EB : EC$, hence

$$AG = \frac{w(v - u)}{2v}.$$

Hence the area of $\triangle ABE$ is

$$\frac{1}{2}(v-u) \cdot \frac{w(v-u)}{2v} = \frac{w(v-u)^2}{4v}.$$

It follows that the required area of trapezium $ABCD$ is

$$vw - \frac{w(v-u)^2}{4v} = \frac{w(u+v)(3v-u)}{4v}.$$

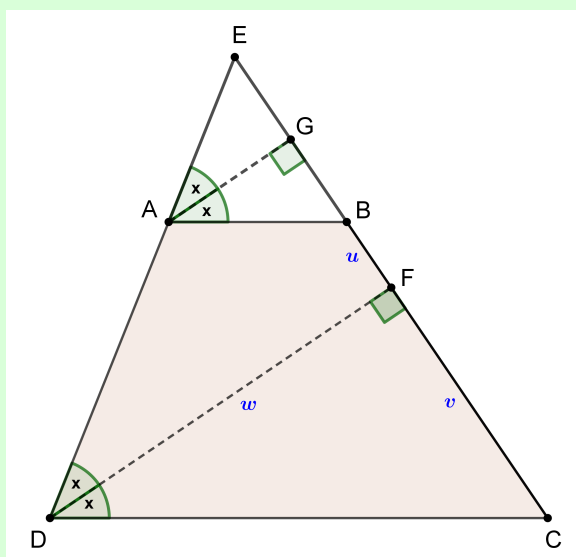


Figure 48

Solution by Shri Durgesh Bannu.

Problem 5 (October 2024): Find all positive integers a, b, c, d such that

$$a + \frac{1}{b + \frac{1}{c + \frac{1}{d}}} = \frac{421}{130}.$$

Solution: Observe that for any positive integers b, c, d , we have

$$\frac{1}{b + \frac{1}{c + \frac{1}{d}}} < 1.$$

Since $\frac{421}{130} = 3 + \frac{31}{130}$ lies strictly between 3 and 4, it follows that $a = 3$. Hence

$$\frac{1}{b + \frac{1}{c + \frac{1}{d}}} = \frac{31}{130},$$

and therefore,

$$b + \frac{1}{c + \frac{1}{d}} = \frac{130}{31} = 4 + \frac{6}{31}.$$

Using the same reasoning, we conclude that $b = 4$. This implies that

$$c + \frac{1}{d} = \frac{31}{6} = 5 + \frac{1}{6},$$

and by using the same reasoning yet again, we conclude that $c = 5, d = 6$.
Hence the required answer is $a = 3, b = 4, c = 5, d = 6$. ■

Problem 6 (October 2024): Find all the positive integers n such that $n^2 + 3^n$ is a perfect square (4th Junior Balkan Mathematical Olympiad, Macedonia, June 2000).

Solution: Some solutions are easily found by inspection: $n = 1$ yields $1^2 + 3^1 = 2^2$ and $n = 3$ yields $3^2 + 3^3 = 36 = 6^2$. We must now explore whether there are any solutions $n > 3$.

Many problems of this type are best approached using divisibility and congruence arguments. But we will solve this problem using inequalities!

Suppose that for some positive integer n , we have $n^2 + 3^n = m^2$ where m is a positive integer; then $m > n$, and we have

$$3^n = m^2 - n^2 = (m - n)(m + n).$$

Since the only proper factors of 3^n are smaller powers of 3, it must be that

$$m - n = 3^a, \quad m + n = 3^{n-a},$$

for some integer $a \geq 0$. Since $m + n > m - n$, we get $n - a > a$, so $2a < n$. Hence $2a \leq n - 1$. We now consider two possibilities.

Suppose that $2a = n - 1$, i.e., $n - 2a = 1$. Then we have:

$$\begin{aligned} 2n &= 3^{n-a} - 3^a = 3^a \cdot (3^{n-2a} - 1) \\ &= 3^a \cdot (3^1 - 1) = 2 \cdot 3^a, \end{aligned}$$

so $n = 3^a$. Since $n = 2a + 1$, this yields

$$3^a = 2a + 1.$$

But it is easy to prove that $3^a > 2a + 1$ for all integers $a > 1$. We will not prove this here; it is easy to do so using induction, and we ask you to fill in the details. Equality holds precisely when $a \in \{0, 1\}$. Both these values yield solutions to the given problem: $a = 0$ leads to $n = 1$, and $a = 1$ leads to $n = 3$. These are all the solutions when $n - 2a = 1$.

Next, suppose that $2a < n - 1$, i.e., $n - 2a > 1$. Then we have $n - 2a \geq 2$, so $a \leq n - a - 2$. Hence,

$$\begin{aligned} 2n &= 3^{n-a} - 3^a \\ &\geq 3^{n-a} - 3^{n-a-2} \\ &= 3^{n-a-2} \cdot (3^2 - 1) = 8 \cdot 3^{n-a-2}. \end{aligned}$$

We already know that $3^k \geq 2k + 1$ for all integers $k \geq 0$. Hence

$$3^{n-a-2} \geq 2(n - a - 2) + 1 = 2n - 2a - 3.$$

This implies that $2n \geq 8 \cdot (2n - 2a - 3) = 16n - 16a - 24$, i.e.,

$$7n \leq 8a + 12.$$

At the same time we have $n - 2a \geq 2$, i.e., $n \geq 2a + 2$. Substituting this into the above relation we get

$$14a + 14 \leq 8a + 12,$$

which is clearly not possible for $a \geq 0$.

This shows that the given equation has no solution beyond $n = 3$. Therefore the only solutions are $n = 1$ and $n = 3$. ■

Problem 7 (October 2024): Find all fractions which can be written simultaneously in the forms

$$\frac{7a-5}{5a-3} \quad \text{and} \quad \frac{6b-1}{4b-3}$$

for some integers a, b (Indian Regional Mathematics Olympiad 2015).

Solution: We have (there is no danger in setting out the equivalence in the first line below, as the denominators of the two fractions can never be 0),

$$\frac{7a-5}{5a-3} = \frac{6b-1}{4b-3} \iff (7a-5)(4b-3) = (6b-1)(5a-3)$$

$$\iff 2ab + 16a + 2b = 12$$

$$\iff ab + 8a + b = 6$$

$$\iff (a+1)(b+8) = 14.$$

Therefore, $a+1, b+8$ are a pair of integers (positive or negative) whose product is 14. Since $14 = 2 \times 7$, we can write 14 as a product of two integers in the following ways:

$$\pm 1 \times \pm 14, \quad \pm 14 \times \pm 1, \quad \pm 2 \times \pm 7, \quad \pm 7 \times \pm 2$$

(same sign to be used in each case for the two factors).

By setting a and b equal to the respective numbers in these ordered pairs and solving the resulting equations, we obtain a total of 8 different integer pairs (a, b) for which the two given fractions are equal.

For example, from $(a + 1, b + 8) = (-1, -14)$ we get $(a, b) = (-2, -22)$; and so on. The 8 pairs are:

$$(-15, -9), \quad (-8, -10), \quad (-3, -15), \quad (-2, -22), \\ (0, 6), \quad (1, -1), \quad (6, -6), \quad (13, -7).$$

The 8 fractions that can be represented by the given expressions are:

$$\frac{55}{39}, \quad \frac{61}{43}, \quad \frac{13}{9}, \quad \frac{19}{13}, \quad \frac{5}{3}, \quad 1, \quad \frac{37}{27}, \quad \frac{43}{31}.$$

For example, from $a = -15$ we get the fraction

$$\frac{7(-15) - 5}{5(-15) - 3} = \frac{-120}{-48} = \frac{5}{3}.$$

Similarly for the other fractions. ■

A Problem from National Mathematics Talents Contest Finals 2024

Article contributed by Rishon Fernandes

Problem 1. During a break, 128 children at a school sit in a circle to play a game. The teacher walks clockwise around the circle to hand out chocolates to the children according to the following scheme: he selects one child and hands over a chocolate to the child, then he skips the next child and hands over a chocolate to the next one, then he skips 2 children and gives a chocolate to the next one, then he skips 3 children, and so on. Show that after many rounds, each child will have received at least one chocolate.

If there were 129 children to start with, will the same conclusion hold?

Solution

We use modular arithmetic to solve the problem. The 128 children are seated in a circle. So we use modulo 128 to define positions. Each position of a child uniquely corresponds to a remainder modulo 128. Label the first child $0 \pmod{128}$, the next child $1 \pmod{128}$, the child after that $2 \pmod{128}$, and so on.

Hereafter, when we refer to the position of a child, we shall mean the corresponding remainder modulo 128.

Let $f(k)$ be the position of the child who gets the k -th chocolate.

Claim: $f(k) \equiv \frac{k(k+1)}{2} \pmod{128}$.

Proof. We use induction.

Base case: $f(1) \equiv 1 \pmod{128}$. This is trivially true.

Inductive step: Suppose $f(m) \equiv \frac{m(m+1)}{2} \pmod{128}$.

Since the teacher skips m children after giving the m -th chocolate, we get

$$f(m+1) \equiv f(m) + m \equiv \frac{(m+1)(m+2)}{2} \pmod{128}.$$

This establishes the inductive step.

As the base claim and the inductive step have been established, it follows that the claim is true. ■

Claim: Every child gets a chocolate.

The claim essentially asserts that for any $r \in \{0, 1, 2, \dots, 126, 127\}$, there exists a positive integer k such that

$$\frac{k(k+1)}{2} \equiv r \pmod{128}.$$

Proof. The statement

$$\frac{k(k+1)}{2} \equiv r \pmod{128}.$$

is equivalent to

$$k(k+1) \equiv 2r \pmod{256}.$$

Let R be the set $\{0, 2, 4, 6, \dots, 250, 252, 254\}$ modulo 256. This is the set of all *even remainders* modulo 256.

Consider the function $f : R \rightarrow R$ defined by

$$f(a) = a(a+1) = a^2 + a.$$

The map is well-defined because if a is even, then $a^2 + a$ too is even, and so is $a^2 + a \pmod{256}$.

Claim: f is injective.

Proof. Suppose $f(a) \equiv f(b) \pmod{256}$.

Then $a^2 + a \equiv b^2 + b \pmod{256}$, hence

$$(a-b)(a+b+1) \equiv 0 \pmod{256}.$$

As a and b are even, $a+b+1$ is odd, hence coprime to 256. Therefore, it can be divided out of the above congruence.

So we get $a - b \equiv 0 \pmod{256}$, i.e.,

$$a \equiv b \pmod{256}.$$

Therefore f is injective. ■

Since f is injective and $f : R \rightarrow R$, it follows that the range of f must be R .

Therefore, for all even residues $2r \pmod{256}$ there exists k such that

$$k(k+1) \equiv 2r \pmod{256},$$

which is equivalent to existence of k such that

$$\frac{k(k+1)}{2} \equiv r \pmod{128}$$

for all residues $r \pmod{128}$.

Therefore every child gets a chocolate. ■

What happens when there are 129 children?

Claim: The conclusion does not hold for 129 children. Specifically, the third child (2 modulo 129) does not get a chocolate.

Proof. Suppose the third child gets the k -th chocolate. Then

$$f(k) \equiv \frac{k(k+1)}{2} \equiv 2 \pmod{129}.$$

As 3 divides 129, the above implies that

$$\frac{k(k+1)}{2} \equiv 2 \pmod{3}.$$

As 8 and 3 are coprime, we may multiply both sides by 8 without destroying the congruence. We get:

$$4k^2 + 4k \equiv 16 \pmod{3},$$

$$\therefore 4k^2 + 4k \equiv 1 \pmod{3},$$

$$\therefore 4k^2 + 4k + 1 \equiv 2 \pmod{3},$$

$$\therefore (2k + 1)^2 \equiv 2 \pmod{3},$$

This is a contradiction as 2 is not a quadratic residue mod 3. (That is, they do not exist any squares that are of the form $2 \pmod{3}$.)

Therefore the third child never gets a chocolate. ■

Source of the problem: Problem 8 from the 56th National Mathematics Talent Contest (Senior finals) held on 19 October 2024. The contest is organised by the Association of Mathematics Teachers of India (AMTI).

About the author

Rishon Fernandes is currently a student of Class 11 at The Learning Center PU College, Mangalore, Karnataka. He has a deep love for Mathematics. He participated in INMO 2024 and took part in the PROMYS Residential Summer Program held at Indian Institute of Science, Bangalore (2024). He loves playing basketball. He enjoys meeting and working with strong minded and focused people. He hopes to pursue a career in mathematics or computer science. He may be contacted at rishon86f@gmail.com.

An Altitude Problem

The fact that the altitudes of any triangle meet in a point is well-known (see Figure 49). The point of concurrence is called the *orthocentre*.

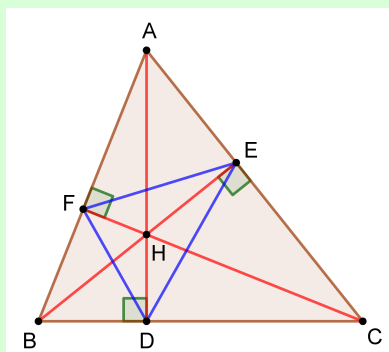


Figure 49: AD , BE , CF are altitudes. They concur at the orthocentre H

What is much less well-known is the following: AD , BE , CF bisect the angles of $\triangle DEF$. This means that the point H is the incentre of $\triangle DEF$. We invite the reader to prove this before proceeding forward.

Proof of claim It suffices if we prove that AD bisects $\angle EDF$. That is, we must prove that $\angle HDE = \angle HDF$. We do so by making use of the cyclic quadrilaterals present in this configuration. Observe that:

- Quadrilateral $HDCE$ is cyclic, because $\angle HDC = 90^\circ = \angle HEC$.
- Quadrilateral $HFBD$ is cyclic, because $\angle HDB = 90^\circ = \angle HFB$.
- Quadrilateral $BCEF$ is cyclic, because $\angle BEC = 90^\circ = \angle BFC$.

From these, we deduce the following (respectively):

- $\angle HDE = \angle HCE$
- $\angle HDF = \angle HBF$
- $\angle HCE = \angle HBF$

The three statements together imply that $\angle HDE = \angle HDF$, as desired. ■

So, the point H is the incentre of $\triangle DEF$.

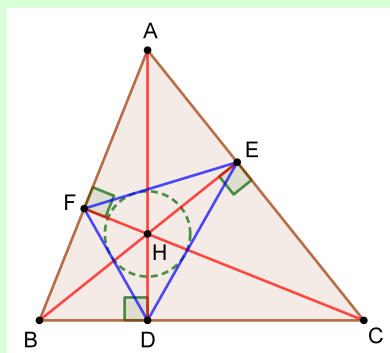


Figure 50: H is the incentre of $\triangle DEF$

The reader may want to explore other ways of proving the above result.

The result we now describe will surely come as a surprise. Given any triangle ABC , draw its altitude AD from A to side BC . Let K be any point on AD . Draw rays BK and CK , and let them intersect sides AC and AB at points E and F , respectively. (See Figure 51.)

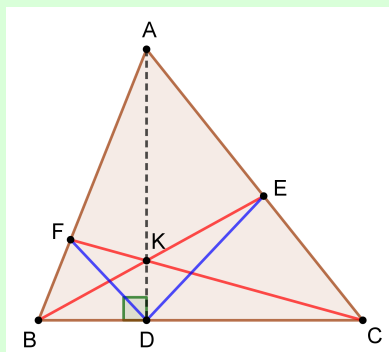


Figure 51: AD bisects $\angle EDF$

Claim

AD bisects $\angle EDF$, and this is true for any position of K on AD .

Proof of claim: We offer two proofs of this beautiful result. The first is a “pure geometry” proof based on *Ceva’s theorem*. This theorem is important enough for us to state it here in full but we postpone for now the proof of the theorem.

The second proof makes use of elementary coordinate geometry.

Here is the statement of Ceva’s theorem (see Figure 52).

Theorem (Ceva). *Let P be an arbitrary point within triangle ABC . Join A, B, C to P and extend the lines to meet the opposite sides at points D, E, F . Then we have:*

$$\frac{BD}{DC} \times \frac{CE}{EA} \times \frac{AF}{FB} = 1.$$

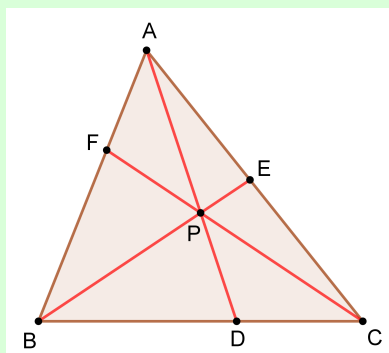


Figure 52: Ceva's theorem

Remarkably, the theorem has a converse:

Theorem (Converse of Ceva). *Given a $\triangle ABC$, let points D, E, F lie on sides BC, CA, AB respectively, and suppose that*

$$\frac{BD}{DC} \times \frac{CE}{EA} \times \frac{AF}{FB} = 1.$$

Then lines AD, BE, CF meet in a point.

The reader may enjoy looking for a proof of Ceva's theorem.

Let us now show how these two results prove the angle bisector property.

Through vertex A draw a line parallel to BC . Next, extend BE and CF to meet the line just drawn at points G and H , respectively (see Figure 53).

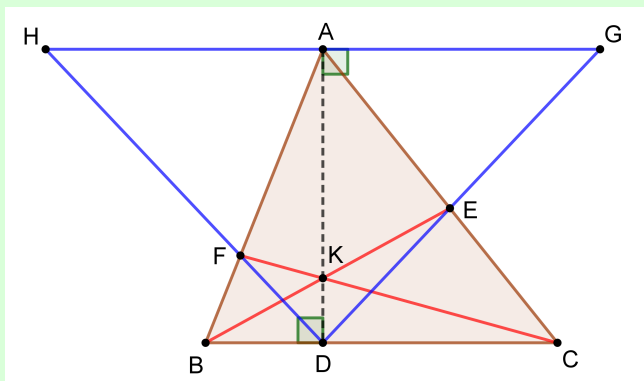


Figure 53: $GH \parallel BC$

Observe that showing that AD bisects $\angle EDF$ is equivalent to showing that A is the midpoint of GH , i.e., that $HA = AG$. We now have:

$$\frac{HA}{AG} = \frac{HA}{BD} \times \frac{DG}{AG} \times \frac{BD}{DG} \quad (56)$$

$$= \frac{AF}{FB} \times \frac{CE}{EA} \times \frac{BD}{DG} = 1, \quad (57)$$

by Ceva's theorem. So $HA = AG$, and the stated result follows. ■

Comment

What a short proof! Ceva's theorem is a very powerful result, and we should be prepared to make full use of it.

Proof using coordinates

In opting to prove a geometric result using coordinates, the first decision to be made is the choice of coordinates. Here we opt to locate the origin of the coordinate axes at D , and the x -axis along line BC .

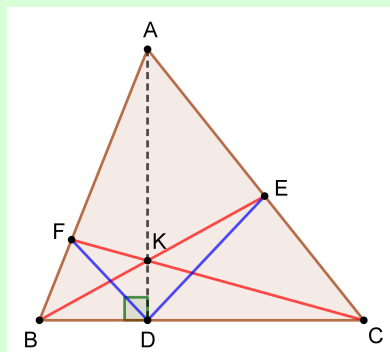


Figure 54: Let BC be the x -axis, with D as the origin

We start by assigning symbols for the coordinates of the points. Let $B = (b, 0)$, $C = (c, 0)$, $A = (0, a)$, $K = (0, k)$. Next, we obtain the equations of various lines:

$$BK: y = -\frac{k}{b}x + k,$$

$$CK: y = -\frac{k}{c}x + k,$$

$$AB: y = -\frac{a}{b}x + a,$$

$$AC: y = -\frac{a}{c}x + a.$$

These equations enable us to find the coordinates of the points E and F .

For E , we must solve the equations

$$y = -\frac{k}{b}x + k,$$

$$y = -\frac{a}{c}x + a.$$

The solution of this pair of equations is

$$x = \frac{bc(a-k)}{ab-ck}, \quad y = \frac{ak(b-c)}{ab-ck},$$

giving

$$E = \left(\frac{bc(a-k)}{ab-ck}, \frac{ak(b-c)}{ab-ck} \right).$$

Similarly, we find the coordinates of F :

$$F = \left(\frac{bc(a-k)}{ac-bk}, \frac{ak(c-b)}{ac-bk} \right).$$

From these we determine the slopes of DE and DF :

$$\text{Slope of } DE = \frac{ak(b-c)}{bc(a-k)},$$

$$\text{Slope of } DF = \frac{ak(c-b)}{bc(a-k)}.$$

We see immediately that

$$\text{Slope of } DF = -\text{Slope of } DE,$$

showing that $\angle KDE = \angle KDF$. This proves the claim. ■

Article contributed by Shailesh Shirali

Answers to “Problems for Solution” – 3

We give here the answers to the problems on page 155. Note that these are only the answers; we will give the solutions in the next issue.

1. The required value is $2 \cdot \frac{(k+2)^4 + (k-2)^4}{(k+2)^4 - (k-2)^4}$.

2. The value of n which ‘works’ is $n = \left(\frac{p-1}{2}\right)^2$

3. The triangle is right-angled isosceles, so the angles are $90^\circ, 45^\circ, 45^\circ$.

4. $AM : MO = 1 : \sqrt{3}$ and $MO : OB = 1 : \sqrt{2}$

5. The given number is the square of the following number:

$$\underbrace{33 \dots 33}_{1997 \text{ 3's}}5.$$

6. The required pairs are $(1,1), (9,3), (8,2)$.

Call for Articles

We invite mathematics teachers and practitioners to contribute short articles to The Compass.

Articles could deal with mathematical content or the teaching of mathematics (classes 6 to 12). They could feature real-life classroom experiences with projects and explorations.

They could be about the use of audiovisual material or social media in the mathematics classroom.

They could feature reflections on the teaching of mathematics, or on the nature of mathematics.

Articles about the use of software are particularly welcome, as are articles dealing with non-routine problems and classroom experiences with problem-solving.

In short, anything at all to do with school mathematics!